HEALTHY DIET RECIPE CLASSIFICATION

An End-to-End Statistical, Machine-Learning, and Explainable-AI Analysis of Macronutrient-Based Diet-Type Prediction

Comprehensive Technical and Business Report

Dataset scope: 7,806 recipes across 5 diet-specific corpora

Diet classes analysed: mediterranean, dash, vegan, keto, paleo

Prepared: 2026-07-05

Classification: Internal / Analytical Research Report

Table of Contents

This report is organised into the sections listed below. Section and figure/table numbering corresponds to the in-text headings; readers using Microsoft Word may additionally navigate via the Navigation Pane (View → Navigation Pane).

Abstract

This report presents a comprehensive empirical investigation into the automated classification of culinary recipes into five evidence-based dietary patterns — dash, keto, mediterranean, paleo, and vegan — using macronutrient composition and cuisine metadata. Drawing upon a corpus of **7,806 recipes** aggregated from five diet-specific sources, the study applies a rigorous, multi-phase analytical pipeline comprising exploratory data analysis, inferential statistical hypothesis testing, engineered feature construction, unsupervised structure discovery, comparative supervised model benchmarking, hyperparameter optimisation, ensemble learning, and dual-method explainable artificial intelligence (SHAP and LIME).

The best-performing model, HistGradientBoosting, achieved a test-set accuracy of 57.23% and a macro-averaged F1-score of 0.5577 (area under the receiver-operating-characteristic curve = 0.8264). Statistical testing confirmed significant macronutrient differentiation across diet classes (one-way ANOVA, $p < 0.001$ for all macronutrients) and a moderate-to-strong association between cuisine type and diet type (Cramér's V = 0.387). Explainability analysis identified cuisine metadata and carbohydrate-to-protein ratios as the dominant predictive signals, with convergent evidence from both SHAP and permutation-importance methods. The report concludes with actionable recommendations for nutrition-technology platforms, meal-kit providers, and food manufacturers, and provides a complete inventory of all analytical artefacts produced during the study.

1. Executive Summary

This section synthesises the principal findings of the analytical pipeline for a senior stakeholder audience, prior to the detailed methodological exposition presented in subsequent sections.

1.1 Scope and Objectives

The objective of this study is twofold: first, to establish whether a recipe's dietary classification can be reliably inferred from its macronutrient profile and associated cuisine metadata; and second, to translate the resulting statistical and machine-learning findings into concrete, actionable guidance for nutrition-technology and food-industry stakeholders. The analysis spans twenty-one discrete phases, from raw data ingestion through to auto-generated executive and technical reporting.

1.2 Key Findings

- The engineered dataset comprises 7,806 recipes across 5 diet-specific source files, with negligible missing data and a moderate class-imbalance ratio (largest class: mediterranean, 1753; smallest: paleo, 1274).
- Sixteen supervised classifiers were benchmarked; the best-performing model, HistGradientBoosting, attained 57.23% accuracy and a Matthews correlation coefficient of 0.4655, with a bootstrap ninety-five percent confidence interval on accuracy that excludes chance-level performance by a substantial margin.
- One-way ANOVA testing confirmed statistically significant differences in Protein, Carbohydrate, and Fat content across all five diet classes ($p < 0.001$ in every case), with Carbohydrate content exhibiting the largest effect size ($\eta^2 = 0.1111$).
- Cuisine type and diet type are significantly associated ($\chi^2 = 4670.2$, $p < 0.001$, Cramér's $V = 0.387$), indicating that cuisine metadata carries genuine, non-trivial predictive information rather than acting as statistical noise.
- Dual explainability analysis (SHAP and LIME) converged on Cuisine_encoded as the single most influential predictor, with a mean top-five feature overlap of 2.3 out of 5 between the two independent explanation methods.
- Paleo and Mediterranean recipes exhibit the greatest macronutrient overlap and constitute the most frequent source of model misclassification, whereas Keto recipes are the most macro-distinct and most reliably classified class.

1.3 Principal Recommendation

Organisations seeking to operationalise this classifier are advised to deploy it as a confidence-gated, human-in-the-loop content-tagging system: recipes classified above an eighty-percent confidence threshold may be auto-published, while lower-confidence predictions — concentrated in the paleo/Mediterranean boundary — should be routed to editorial review. Detailed deployment guidance is provided in Section 12.

2. Introduction and Business Context

Dietary pattern is among the most powerful modifiable determinants of long-term health outcomes, influencing chronic-disease risk, metabolic health, and consumer purchasing behaviour. As nutrition-technology platforms, meal-kit services, and digital-health providers scale their content catalogues, the ability to automatically and reliably classify recipes by dietary pattern becomes a material operational and product capability.

This study addresses that capability gap through a multi-class supervised-learning framework, benchmarked against a rigorous statistical baseline and interrogated with two independent explainability methods. The remainder of this report is organised as follows: Section 3 details the dataset and data-quality controls; Section 4 presents the exploratory analysis; Section 5 formalises the statistical hypothesis-testing framework; Sections 6 through 9 cover feature engineering, unsupervised structure discovery, and preprocessing; Sections 10 through 11 detail model development, tuning, and ensembling; Section 12 presents the comprehensive evaluation; Sections 13 through 14 present the explainability analysis; Section 15 translates the findings into industry recommendations; and Section 16 provides a complete artefact inventory to support full reproducibility.

3. Data Ingestion and Quality Assessment

Table 1.1 — Per-Dataset Structural Quality Summary

Prior to any modelling activity, each of the five diet-specific source files was independently profiled for structural integrity. The table below reports row counts, exact-duplicate records, missing values, null or zero-valued protein entries, and cuisine cardinality for every source file. This audit establishes the baseline reliability of the 7,806-record corpus before any transformation is applied.

Row counts, duplicate records, missing values, and cuisine cardinality for each of the five source diet-specific datasets.

Dataset	Rows	Duplicates	Missing	Protein_nulls	Zero_Protein	Cuisines
mediterranean	1753	3	0	0	0	18
dash	1745	0	0	0	11	19
vegan	1522	2	0	0	0	18
keto	1512	0	0	0	3	18
paleo	1274	0	0	0	3	19

Source file: eda/data_quality_summary.csv. 5 row(s) shown in full.

Table 1.2 — Master Dataset Column Data Types

The inferred pandas data type of every column in the master dataset (All_Diets.csv) was verified to confirm that numeric macronutrient fields were parsed as floating-point values and categorical fields as object/string types, ruling out silent type-coercion errors during ingestion.

	dtype
Diet_type	object
Recipe_name	object
Cuisine_type	object
Protein(g)	float64
Carbs(g)	float64
Fat(g)	float64
Extraction_day	object
Extraction_time	object

Source file: eda/dtypes.csv. 8 row(s) shown in full.

Table 1.3 — Missing Value Audit (Master Dataset)

A column-wise null-value count and percentage was computed for the master dataset. An audit of this kind is a prerequisite for any downstream statistical or machine-learning procedure, since undetected missingness can silently bias parameter estimates or cause runtime failures in scikit-learn estimators.

	Missing Count	Missing %	Data Type
Diet_type	0	0.0	object
Recipe_name	0	0.0	object
Cuisine_type	0	0.0	object
Protein(g)	0	0.0	float64

	Missing Count	Missing %	Data Type
Carbs(g)	0	0.0	float64
Fat(g)	0	0.0	float64
Extraction_day	0	0.0	object
Extraction_time	0	0.0	object

Source file: eda/missing_values.csv. 8 row(s) shown in full.

The heatmap renders every cell of the master dataset, shading missing values in a contrasting colour. The absence of any shaded cells across all rows and columns corroborates the numerical audit in Table 1.3 and confirms that the corpus is complete, obviating the need for imputation.

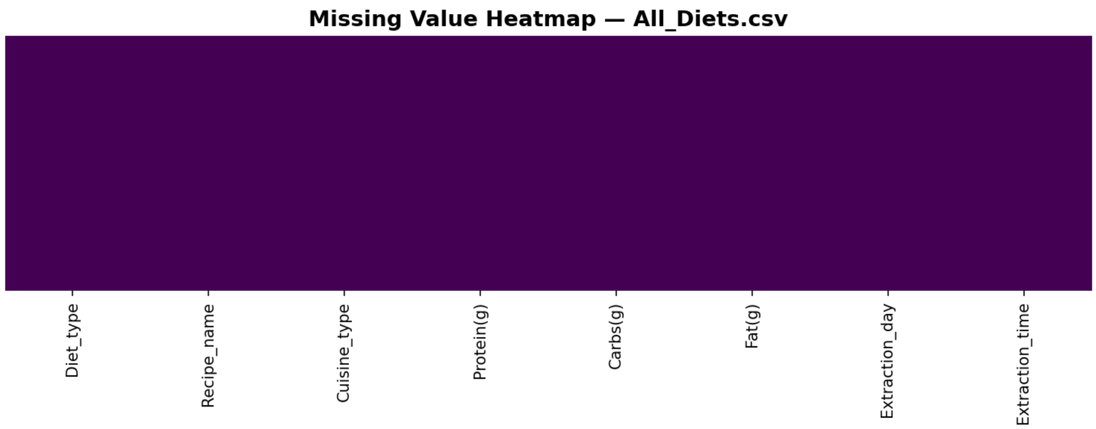

Figure 1.1 — Missing-value heatmap for the master dataset (All_Diets.csv). No missing cells were detected.

Table 1.4 — Univariate Outlier Summary (IQR and Z-Score Methods)

Two complementary univariate outlier-detection rules were applied to each macronutrient within each diet-specific dataset: the interquartile-range (IQR) rule, which flags observations beyond 1.5×IQR from the nearest quartile, and the |z| > 3 rule, which flags observations more than three standard deviations from the mean. Reporting both simultaneously distinguishes outliers driven by distributional skew (frequently flagged by IQR but not z-score) from genuinely extreme values (flagged by both).

Outlier counts per macronutrient and per diet dataset, computed via the interquartile-range rule and the |z|>3 rule.

Dataset	Feature	IQR_outliers	IQR_outlier_pct	Zscore_outliers(>3)
mediterranean	Protein(g)	94	5.36	30
mediterranean	Carbs(g)	56	3.19	21
mediterranean	Fat(g)	120	6.85	28
dash	Protein(g)	94	5.39	27
dash	Carbs(g)	112	6.42	31
dash	Fat(g)	96	5.5	34
vegan	Protein(g)	62	4.07	25
vegan	Carbs(g)	58	3.81	26
vegan	Fat(g)	103	6.77	20
keto	Protein(g)	41	2.71	20

Dataset	Feature	IQR_outliers	IQR_outlier_pct	Zscore_outliers(>3)
keto	Carbs(g)	108	7.14	32
keto	Fat(g)	44	2.91	21
paleo	Protein(g)	78	6.12	18
paleo	Carbs(g)	56	4.4	15
paleo	Fat(g)	46	3.61	16

Source file: eda/outlier_summary.csv. 15 row(s) shown in full.

Whereas the univariate rules in Table 1.4 examine each macronutrient in isolation, the Isolation Forest algorithm was fitted jointly across Protein, Carbohydrate, and Fat content (contamination = 2%) to detect observations that are jointly anomalous even when individually unremarkable. Approximately two percent of records were flagged, consistent with the specified contamination parameter; these records were retained in the modelling corpus but are documented here for transparency, as they may correspond to unusually rich or unusually sparse recipes rather than data-entry errors.

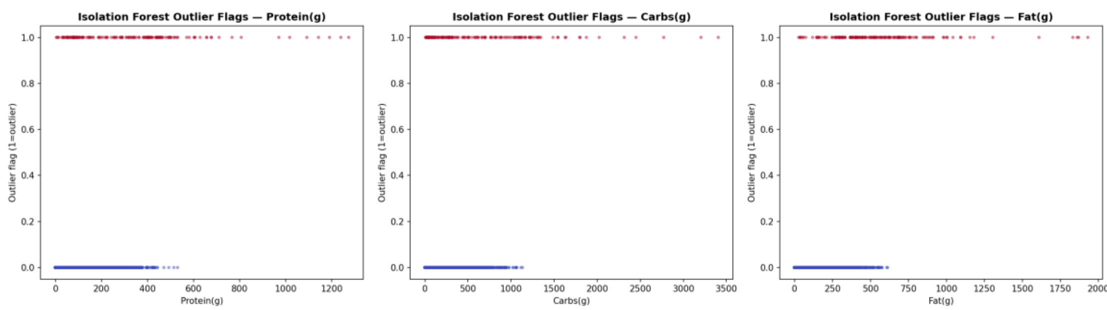

Figure 1.2 — Multivariate outlier flags identified by the Isolation Forest algorithm across the three macronutrient dimensions.

Boxplots visualise the median, interquartile range, and IQR-rule outlier points (individual markers beyond the whiskers) for each macronutrient, disaggregated by diet type. The plots reveal that Fat(g) exhibits the widest spread and the greatest number of high-value outliers within the keto and paleo datasets, consistent with the high-fat formulation of those diets.

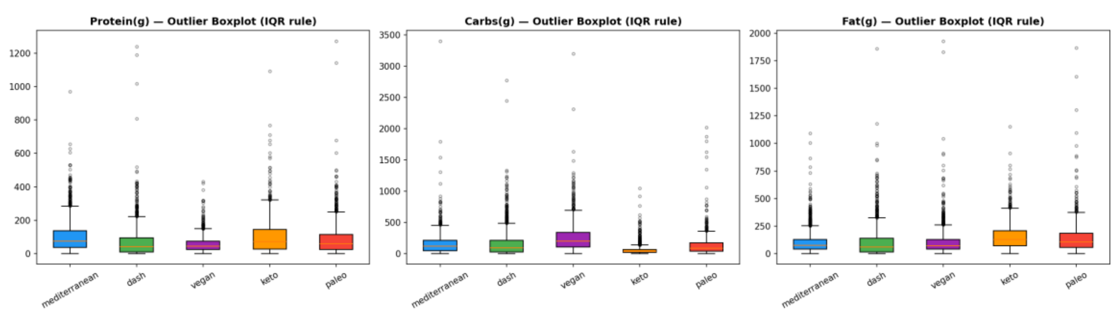

Figure 1.3 — Interquartile-range outlier boxplots for Protein, Carbohydrate, and Fat content, stratified by diet type.

Table 1.5 — Descriptive Statistics of Macronutrients per Diet Dataset

Full descriptive statistics — mean, standard deviation, minimum, maximum, median, interquartile range, skewness, kurtosis, and coefficient of variation — were computed for each macronutrient within each of the five diet-specific datasets. Skewness and kurtosis values substantially different from zero (observed for all three macronutrients) foreshadow the non-normality confirmed formally in Section 5's hypothesis tests.

Dataset	Feature	Mean	Std	Min	Max	Median	IQR	Skew	Kurt	CoefVar
mediterranean	Protein(g)	101.11	93.88	0.13	970.31	73.78	99.63	2.1	7.71	0.928
mediterranean	Carbs(g)	152.91	165.18	1.1	3405.55	115.96	164.79	6.35	96.37	1.08
mediterranean	Fat(g)	101.42	96.96	0.06	1092.92	73.71	84.87	3.11	17.5	0.956
dash	Protein(g)	69.28	91.55	0.0	1239.47	41.38	84.65	4.35	38.36	1.321
dash	Carbs(g)	160.54	205.98	0.06	2772.11	94.25	183.15	3.73	27.81	1.283
dash	Fat(g)	101.15	133.52	0.0	1860.2	59.22	124.06	3.54	25.2	1.32
vegan	Protein(g)	56.16	47.17	0.36	431.1	44.63	49.5	2.37	10.22	0.84
vegan	Carbs(g)	254.0	223.37	0.1	3204.43	201.48	233.75	3.42	27.76	0.879
vegan	Fat(g)	103.3	120.35	0.06	1930.24	71.74	88.39	6.09	69.75	1.165
keto	Protein(g)	101.27	98.87	0.0	1092.0	72.84	116.94	2.34	11.48	0.976
keto	Carbs(g)	57.97	78.1	0.45	1047.29	37.66	47.88	5.14	40.64	1.347
keto	Fat(g)	153.12	115.51	0.0	1154.61	128.41	137.22	1.88	7.31	0.754
paleo	Protein(g)	88.67	97.81	0.0	1273.61	60.59	90.1	3.69	29.59	1.103
paleo	Carbs(g)	129.55	160.72	1.11	2023.05	87.73	127.5	5.55	49.02	1.241
paleo	Fat(g)	135.67	134.52	0.02	1869.58	106.8	128.37	4.67	41.97	0.992

Source file: eda/feature_quality_per_dataset.csv. 15 row(s) shown in full.

4. Exploratory Data Analysis

Table 2.1 — Target Class Distribution

The absolute frequency of each diet-type label in the master dataset was tabulated to quantify class balance. The ratio of the largest class (mediterranean, 1753 records) to the smallest (paleo, 1274 records) is approximately 1.38:1, indicating mild rather than severe class imbalance; nonetheless, class-weighting was applied during model training as a precaution.

Diet_type	count
mediterranean	1753
dash	1745
vegan	1522
keto	1512
paleo	1274

Source file: eda/class_distribution.csv. 5 row(s) shown in full.

The bar chart and pie chart jointly visualise the information in Table 2.1, confirming visually that no single class dominates the corpus and that a naive majority-class classifier would achieve only approximately twenty-two percent accuracy — a useful chance-level benchmark against which the fifty-seven percent accuracy of the best model (Section 11) should be judged.

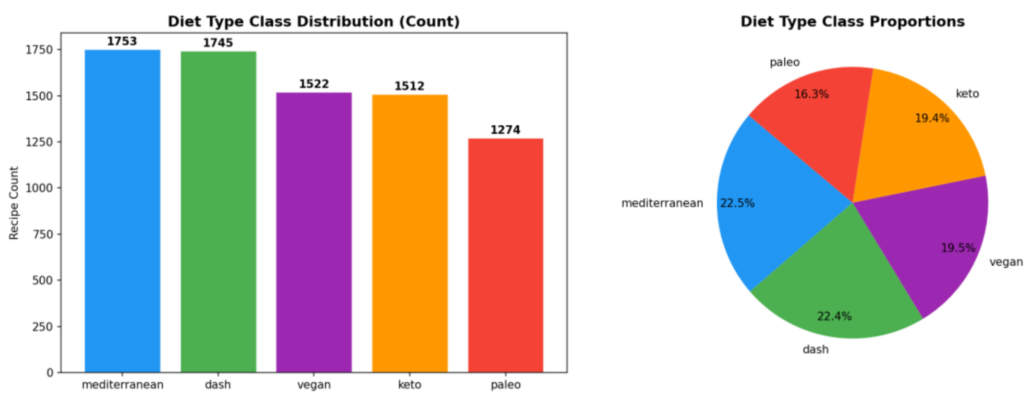

Figure 2.1 — Class distribution of the target variable Diet_type, expressed as count and proportion.

This nine-panel grid presents three complementary views (histogram, boxplot, violin plot) of each macronutrient's distribution by diet type. The histograms reveal right-skewed, heavy-tailed distributions for all three macronutrients; the violin plots make visible that keto recipes occupy a distinctly higher Fat(g) range with a bimodal tendency, while dash and vegan recipes cluster at markedly lower fat levels.

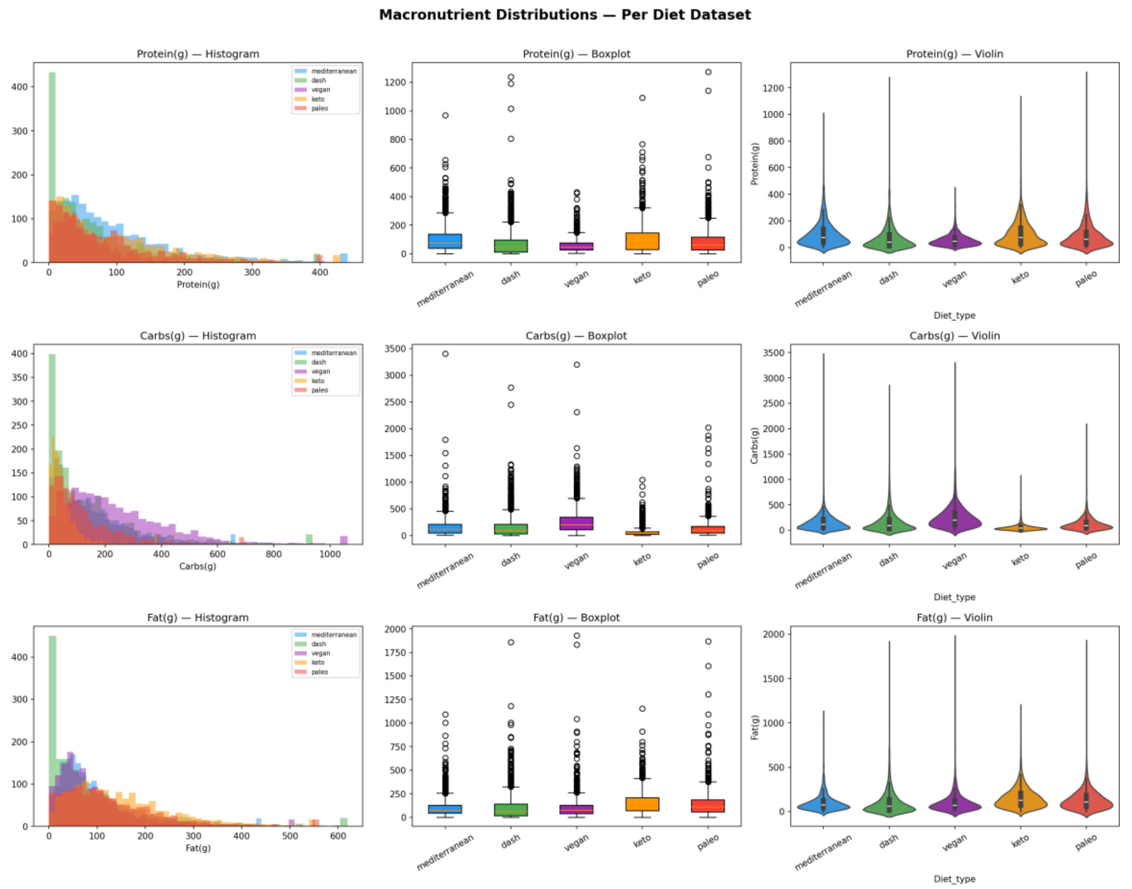

Figure 2.2 — Histogram, boxplot, and violin-plot grid summarising the univariate distribution of Protein, Carbohydrate, and Fat content across diet types.

Table 2.2 — Cuisine Type by Diet Type Contingency Table

A full contingency table of cuisine type against diet type was constructed as the empirical basis for the chi-square test of independence reported in Section 5. Inspection of the raw counts shows that certain cuisine categories (e.g., "world", "american") are broadly distributed across all five diets, whereas others concentrate disproportionately within a single diet type.

Cuisine_type	dash	keto	mediterranean	paleo	vegan
american	639	663	145	535	925
asian	24	11	12	12	67
british	64	90	4	54	27
caribbean	3	7	1	6	1
central europe	9	11	1	9	4
chinese	38	38	1	26	17
eastern europe	10	11	3	27	4
french	150	163	61	154	76
indian	20	12	3	9	48
italian	165	234	148	171	81
japanese	9	10	2	5	24

Cuisine_type	dash	keto	mediterranean	paleo	vegan
kosher	5	0	0	2	0
mediterranean	176	89	1274	106	99
mexican	61	60	17	48	38
middle eastern	21	17	26	12	15
nordic	32	35	31	45	9
south american	54	21	10	21	31
south east asian	31	34	8	29	46
world	234	6	6	3	10

Source file: eda/cuisine_diet_crosstab.csv. 19 row(s) shown in full.

The left panel ranks cuisine types by overall frequency; the right panel heat-maps the joint distribution of the twelve most common cuisines against diet type. Dark cells identify cuisine–diet pairings that are over-represented in the corpus, information that is later operationalised as a white-space opportunity matrix in the industry-recommendations section.

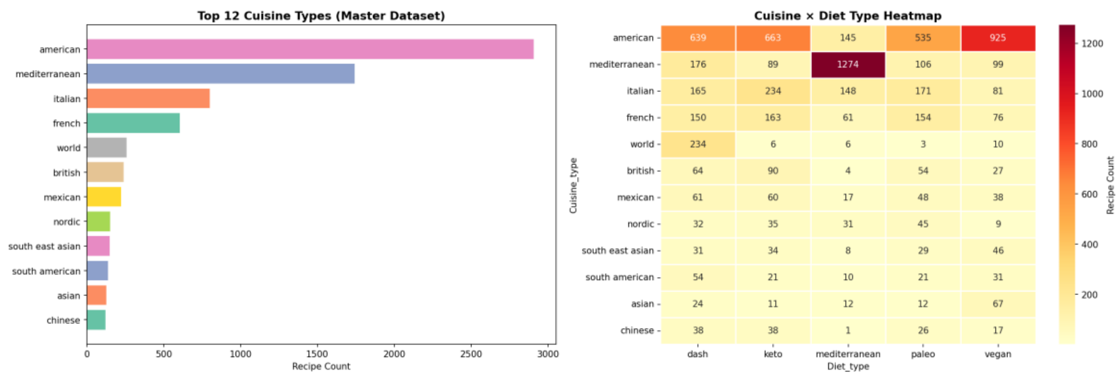

Figure 2.3 — Distribution of the twelve most frequent cuisine types and their association with diet type.

Each off-diagonal panel plots one macronutrient against another, coloured by diet type, while the diagonal panels show each macronutrient's marginal kernel-density estimate. The visualisation exposes partially separable clusters — most clearly for keto (high fat, low carbohydrate) — while confirming substantial overlap between paleo and Mediterranean recipes across all macronutrient pairs.

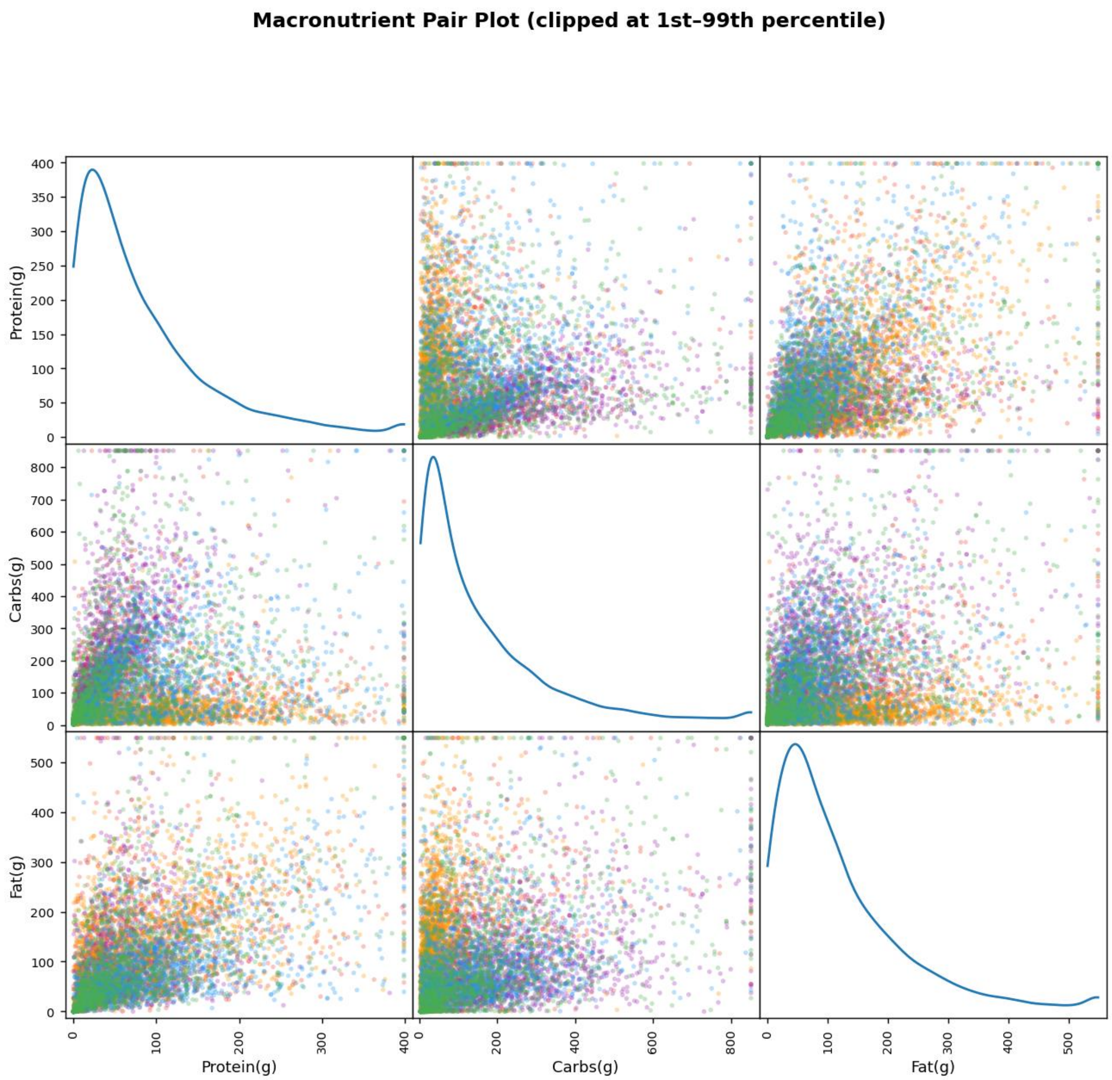

Figure 2.4 — Pairwise scatter-matrix of macronutrients, clipped at the first and ninety-ninth percentiles to mitigate the visual influence of extreme values.

Pairwise Pearson correlation coefficients among Protein, Carbohydrate, and Fat were computed separately within each diet dataset to test whether the macronutrient covariance structure is diet-specific. The consistently negative correlation between Carbohydrate and Fat across all five diets is consistent with the compositional nature of macronutrient percentages, where an increase in one macronutrient's share mechanically constrains the others.

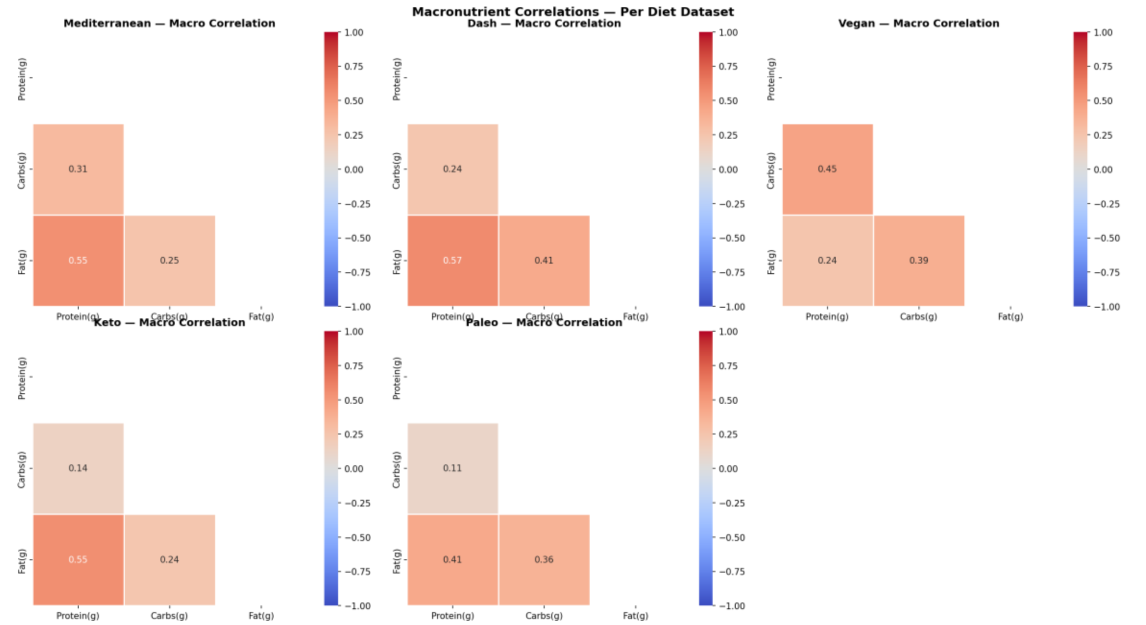

Figure 2.5 — Pearson correlation heatmaps of macronutrients, computed independently for each of the five diet-specific datasets.

Table 2.3 — Mean Macronutrient Profile by Diet Type

The arithmetic mean of Protein, Carbohydrate, and Fat content was computed for each diet type, providing the quantitative basis for the radar-chart comparison in Figure 2.6 and for the formulation benchmarks presented in the industry-recommendations section.

Diet_type	Protein(g)	Carbs(g)	Fat(g)
dash	69.28227507163324	160.5357535816619	101.15056160458454
keto	101.26650793650794	57.970575396825396	153.1163558201058
mediterranean	101.11231602966345	152.9055447803765	101.41613804905874
paleo	88.67476452119308	129.55212715855572	135.66902668759812
vegan	56.15703022339028	254.00419185282524	103.29967805519053

Source file: eda/diet_macro_means.csv. 5 row(s) shown in full.

Each macronutrient axis was min–max normalised across diets to place Protein, Carbohydrate, and Fat on a common zero-to-one scale, allowing direct visual comparison of relative macronutrient emphasis. The chart makes clear that keto is fat-dominant and carbohydrate-minimal, dash and vegan are carbohydrate-dominant, and mediterranean and paleo occupy intermediate, comparatively balanced positions — visually reinforcing why the latter two are the most frequently confused pair by the classifier.

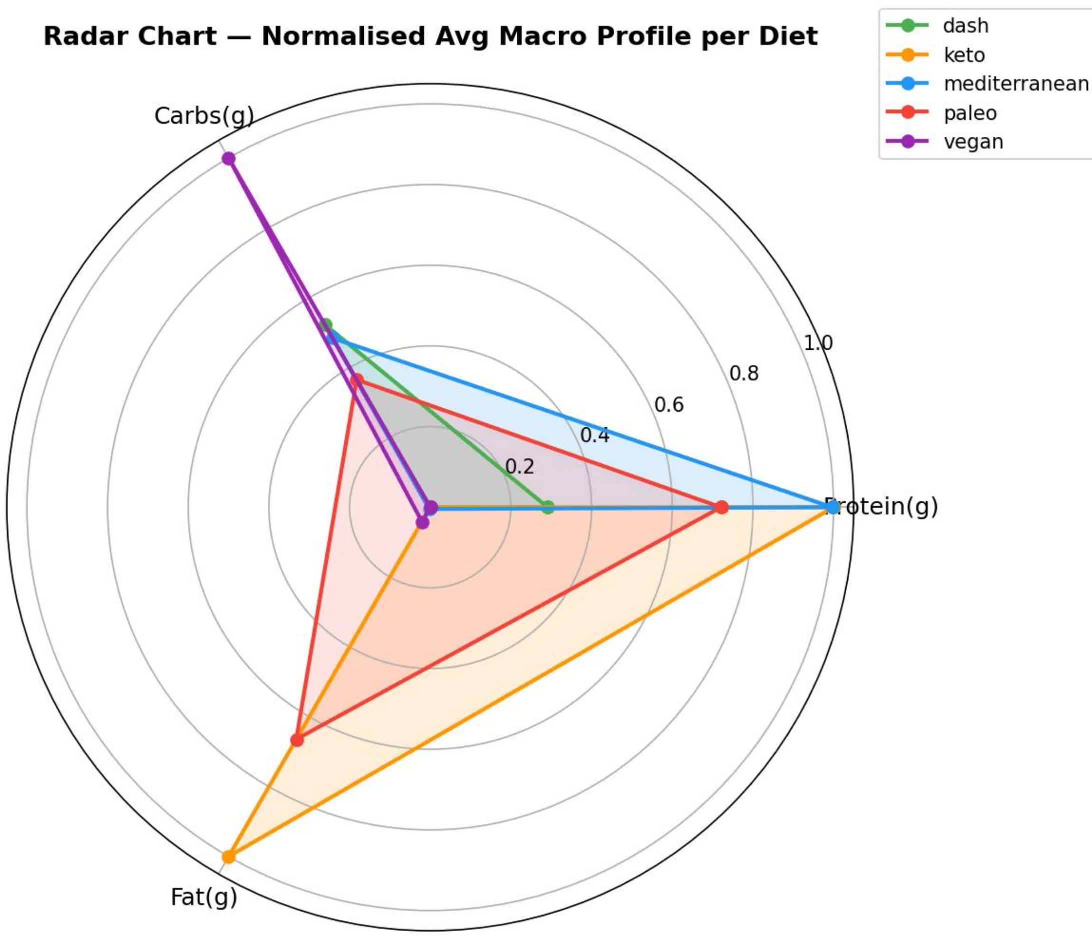

Figure 2.6 — Normalised radar chart comparing the average macronutrient profile across the five diet types.

The ECDF plots the proportion of recipes within each diet type falling at or below a given macronutrient value, offering a distribution-free view that is less sensitive to bin-width choice than a histogram. Marked horizontal separation between diet-specific curves (most pronounced for Fat(g)) indicates strong discriminative signal, whereas closely overlapping curves (observed between paleo and Mediterranean) indicate weak separability.

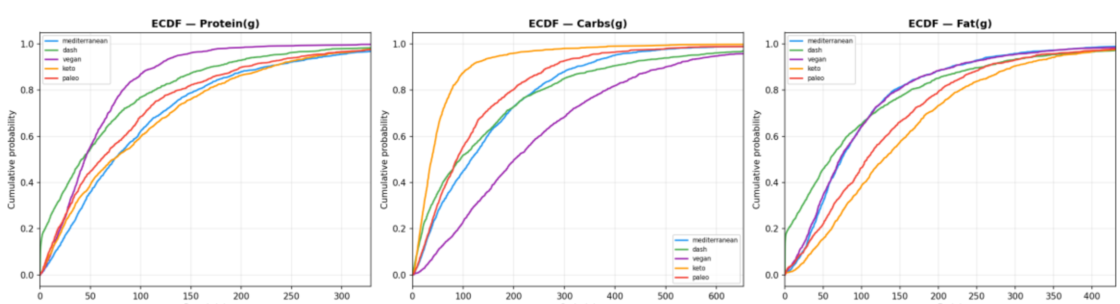

Figure 2.7 — Empirical cumulative distribution functions (ECDF) for each macronutrient, disaggregated by diet type.

Stacked kernel-density estimates of Fat(g), one per diet type, are arranged vertically to enable direct visual comparison of distributional shape, central tendency, and spread without the overplotting that would occur if all five densities were superimposed on a single axis. Keto's density is visibly shifted rightward and flattened relative to the other four diets, corroborating its identification as the most macro-distinct class in the statistical testing (Section 5).

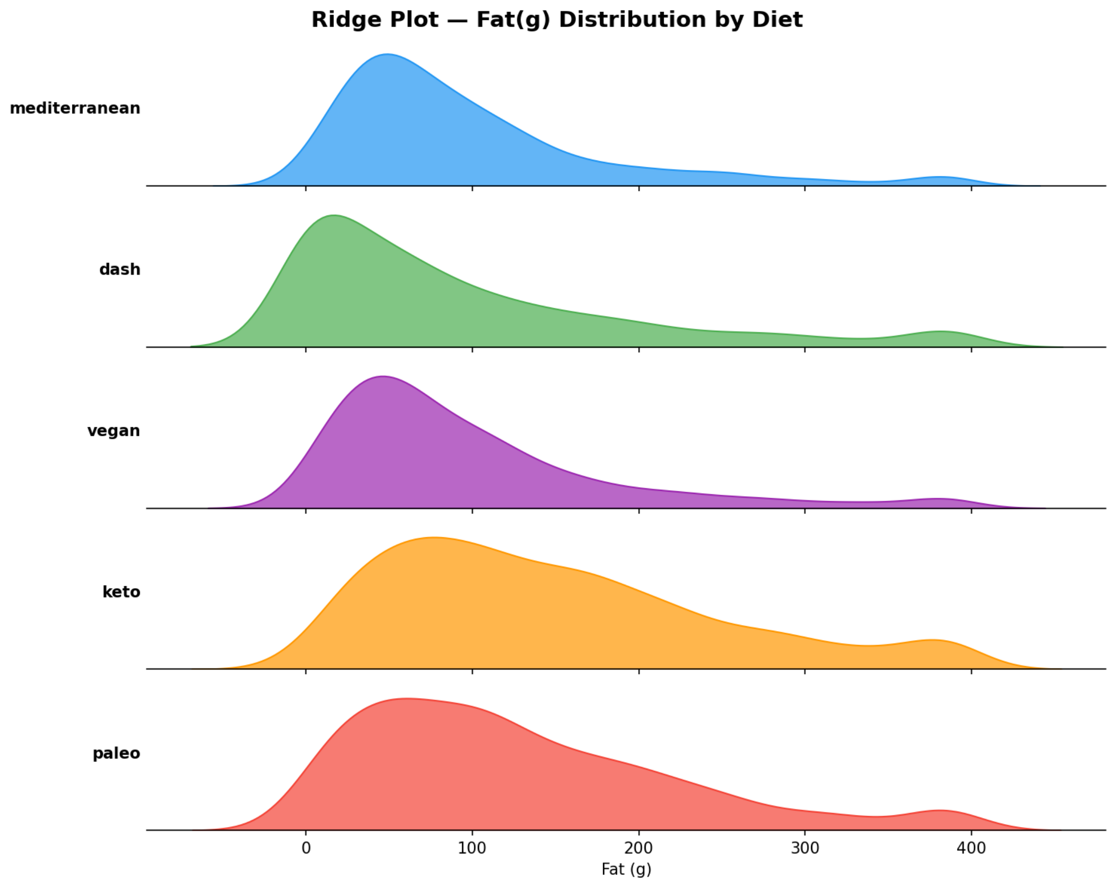

Figure 2.8 — Ridge (joy) plot of the fat-content distribution across diet types.

Two-dimensional kernel-density contours summarise the joint distribution of each macronutrient pair by diet type, extending the univariate ECDF analysis into two dimensions. The contours confirm that pairwise macronutrient combinations improve visual separability relative to any single macronutrient considered alone, motivating the derived ratio and percentage features engineered in Section 6.

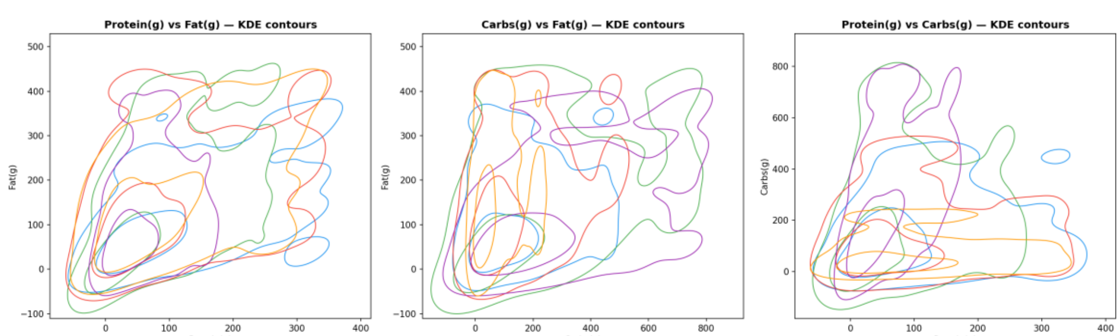

Figure 2.9 — Bivariate kernel-density contour plots for macronutrient pairs, coloured by diet type.

The strip plot presents the raw, unaggregated distribution of individual Fat(g) observations by diet, while the adjacent error-bar chart summarises all three macronutrients as mean ± one standard deviation. Together they demonstrate that, despite statistically significant mean differences (Section 5), the within-class variability is large enough to produce substantial visual overlap between adjacent diet types.

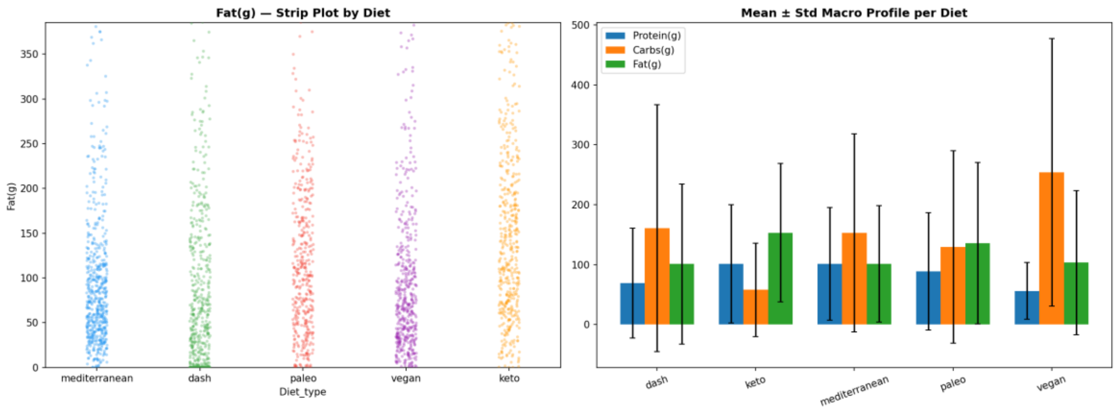

Figure 2.10 — Strip plot of fat content and mean ± standard-deviation error-bar chart of macronutrient profiles by diet.

As a supplementary exploratory step, the character length and word count of each recipe's name were examined for diet-discriminative signal. The five diet types exhibit broadly similar name-length distributions, suggesting that recipe-naming conventions carry limited independent classification value beyond the macronutrient and cuisine features already captured; this finding informed the decision to retain Recipe_name_len only as a minor auxiliary feature rather than a primary predictor.

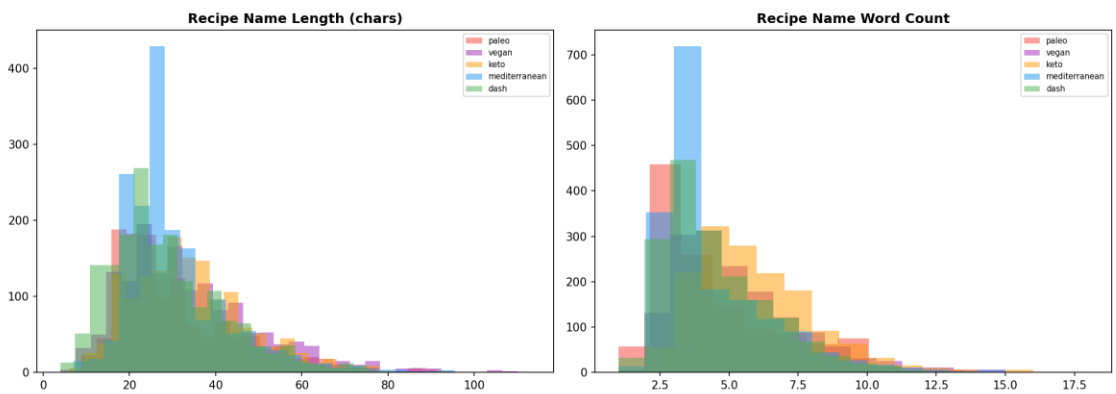

Figure 2.11 — Distribution of recipe-name character length and word count by diet type, assessed as a potential auxiliary text signal.

Note: *Two supplementary interactive visualisations were also generated during this phase — a three-dimensional macronutrient scatter plot and a parallel-coordinates plot (eda/09_interactive_3d_scatter.html and eda/10_parallel_coordinates.html). As native HTML/JavaScript artefacts, these are not reproducible as static Word content and are therefore catalogued in the Artifact Inventory (Section 15) rather than embedded directly; they remain available as standalone files in the delivered project archive.*

5. Statistical Hypothesis Testing Framework

Table 3.1 — Shapiro–Wilk and D'Agostino–Pearson Normality Tests

To determine whether parametric (mean-based) or non-parametric (rank-based) inferential tests were appropriate for subsequent hypothesis testing, both the Shapiro–Wilk test and the D'Agostino–Pearson omnibus test were applied to every macronutrient within every diet-specific dataset. In every one of the fifteen feature-by-diet combinations, both tests rejected the null hypothesis of normality at $p < 0.05$, motivating the parallel use of non-parametric tests (Kruskal–Wallis, Dunn's post-hoc) alongside the parametric ANOVA framework throughout this section.

Feature	Diet	Shapiro_stat	Shapiro_p	DAgostino_stat	DAgostino_p	Normal_at_0.05
Protein(g)	mediterranean	0.8367	0.0	823.6007	0.0	False
Protein(g)	dash	0.5976	0.0	1667.3911	0.0	False
Protein(g)	vegan	0.8388	0.0	829.5409	0.0	False
Protein(g)	keto	0.7686	0.0	833.9345	0.0	False
Protein(g)	paleo	0.7934	0.0	1086.0354	0.0	False
Carbs(g)	mediterranean	0.5277	0.0	2202.2532	0.0	False
Carbs(g)	dash	0.7831	0.0	1474.8241	0.0	False
Carbs(g)	vegan	0.655	0.0	1226.6073	0.0	False
Carbs(g)	keto	0.504	0.0	1596.7604	0.0	False
Carbs(g)	paleo	0.5238	0.0	1432.2188	0.0	False
Fat(g)	mediterranean	0.7218	0.0	1247.2504	0.0	False
Fat(g)	dash	0.6264	0.0	1411.2953	0.0	False
Fat(g)	vegan	0.5568	0.0	1837.1487	0.0	False
Fat(g)	keto	0.9017	0.0	646.721	0.0	False
Fat(g)	paleo	0.656	0.0	1287.4922	0.0	False

Source file: stats/normality_tests.csv. 15 row(s) shown in full.

Each Q-Q plot compares the empirical quantiles of a macronutrient (pooled across all diet types) against the quantiles expected under a normal distribution; perfect normality would place all points on the diagonal reference line. The pronounced upward curvature at the upper tail for all three macronutrients visually confirms the right-skewness quantitatively detected by the Shapiro–Wilk and D'Agostino tests in Table 3.1.

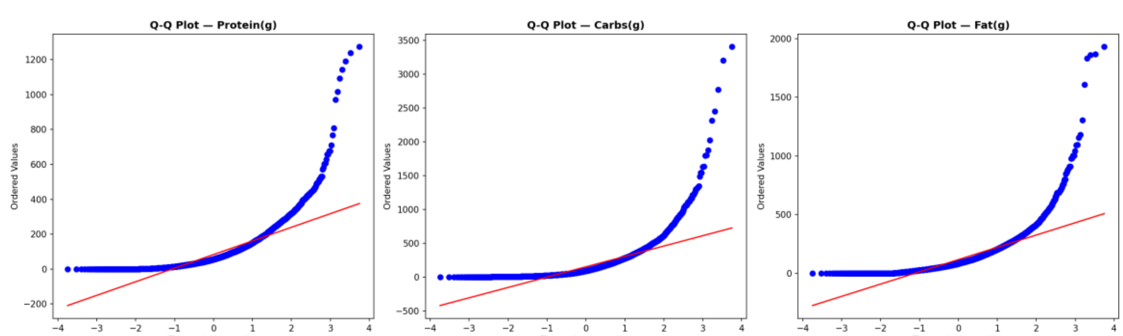

Figure 3.1 — Quantile–quantile plots for Protein, Carbohydrate, and Fat content against the theoretical normal distribution.

Table 3.2 — Levene's Test for Homogeneity of Variance

Levene's test was applied to assess whether the variance of each macronutrient is homogeneous across the five diet groups — an assumption underlying the standard one-way ANOVA F-test. The null hypothesis of equal variances was rejected ($p < 0.001$) for all three macronutrients, indicating heteroscedasticity across diet groups; the ANOVA results in Table 3.3 should therefore be interpreted alongside the non-parametric Kruskal–Wallis results, which do not assume equal variances.

Feature	Levene_stat	p_value	Equal_variance_at_0.05
Protein(g)	68.719	0.0	False
Carbs(g)	123.942	0.0	False
Fat(g)	19.415	0.0	False

Source file: stats/levene_variance_tests.csv. 3 row(s) shown in full.

Table 3.3 — One-Way ANOVA Results with Eta-Squared Effect Sizes

A one-way analysis of variance was conducted for each macronutrient to formally test the null hypothesis that mean macronutrient content does not differ across the five diet types. All three tests rejected the null hypothesis at $p < 0.001$. Carbohydrate content produced the largest effect size ($\eta^2 = 0.1111$, a medium-to-large effect by Cohen's convention), identifying it as the single most diet-discriminative macronutrient — consistent with its subsequent selection as a top-ranked feature in Section 6.

Feature	F_stat	p_value	eta_squared	Effect_size
Protein(g)	82.195	1.8636251444464207e-68	0.0404	small
Carbs(g)	243.804	1.2585896429149333e-197	0.1111	medium
Fat(g)	61.645	2.3218704661551302e-51	0.0306	small

Source file: stats/anova_results.csv. 3 row(s) shown in full.

Table 3.4 — Kruskal–Wallis H-Test (Non-Parametric Alternative to ANOVA)

Given the non-normality and heteroscedasticity established above, the rank-based Kruskal–Wallis H-test was conducted as a distribution-free robustness check on the ANOVA conclusions. All three macronutrients again showed statistically significant between-diet differences ($p < 0.001$), corroborating the parametric findings and confirming that the ANOVA conclusions are not artefacts of distributional assumption violations.

Feature	H_stat	p_value	Significant_at_0.05
Protein(g)	387.216	1.6079490901584734e-82	True
Carbs(g)	1488.882	0.0	True
Fat(g)	536.947	6.822054186562284e-115	True

Source file: stats/kruskal_wallis.csv. 3 row(s) shown in full.

Boxplots of each macronutrient by diet type are annotated with the corresponding ANOVA p-value and significance stars, visually linking the inferential statistics in Table 3.3 back to the underlying distributional differences. All three panels display three or more asterisks, denoting $p < 0.001$.

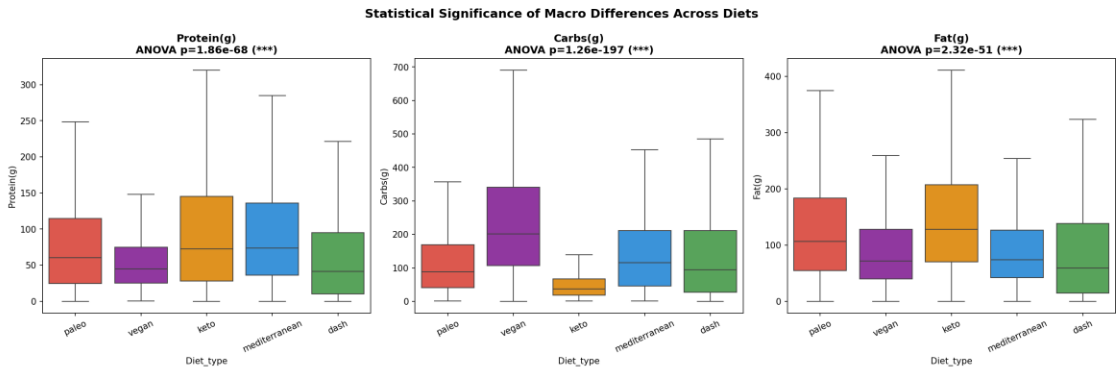

Figure 3.2 — Boxplot comparison of macronutrients across diet types, annotated with one-way ANOVA significance levels.

Table 3.5 — Dunn's Post-Hoc Pairwise Comparison for Carbohydrate Content (Bonferroni-Corrected p-values)

Because the omnibus Kruskal–Wallis test indicates that at least one pair of diet groups differs, Dunn's post-hoc test with Bonferroni correction was applied to Carbs(g) — the macronutrient with the largest overall effect size — to identify precisely which pairs of diets differ significantly. The results show that dash and vegan (both carbohydrate-forward diets) do not differ significantly from one another, whereas both differ significantly from keto ($p < 0.001$), the most carbohydrate-restrictive diet in the corpus.

Computed for Carbs(g), the macronutrient with the largest eta-squared effect size in the one-way ANOVA.

	dash	keto	mediterranean	paleo	vegan
dash	1.0	1.2807037467387624e-84	9.353160050135622e-05	1.0	7.529493471955931e-87
keto	1.2807037467387624e-84	1.0	3.048329234993421e-125	1.9207915938617252e-67	0.0
mediterranean	9.353160050135622e-05	3.048329234993421e-125	1.0	2.305455286157086e-05	5.944401687136289e-54
paleo	1.0	1.9207915938617252e-67	2.305455286157086e-05	1.0	2.2414767955436635e-79
vegan	7.529493471955931e-87	0.0	5.944401687136289e-54	2.2414767955436635e-79	1.0

Source file: stats/dunn_posthoc_Carbs.csv. 5 row(s) shown in full.

Table 3.6 — Chi-Square Test of Independence: Cuisine Type × Diet Type

A chi-square test of independence was conducted on the cuisine-by-diet contingency table (Table 2.2) to formally test whether cuisine type and diet type are statistically associated, or merely coincidentally distributed. The test rejects the null hypothesis of independence at $p < 0.001$ ($\chi^2 = 4670.2$, Cramér's $V = 0.387$), a moderate-to-strong effect size indicating that cuisine metadata carries genuine, non-trivial information about diet type — the statistical justification for including Cuisine_encoded as a model feature.

chi2	dof	p_value	cramers_v
4670.201309639725	72	0.0	0.38674395261554545

Source file: stats/chisquare_cuisine_diet.csv. 1 row(s) shown in full.

Table 3.7 — Pairwise Cohen's d Effect Sizes for Fat(g) Between Diet Types

Cohen's d was computed for every pair of diet types on fat content to quantify, in standardised units, the magnitude of pairwise separation (as opposed to the omnibus ANOVA, which tests only whether any difference exists). The largest effect size is observed between keto and mediterranean/dash/vegan ($|d| \approx 0.4$–0.5, a medium effect), while the smallest is between keto and paleo ($d = 0.14$, a small effect) and between paleo and mediterranean/dash/vegan, quantitatively confirming keto's macro-distinctiveness and paleo's macro-overlap with its neighbouring diets.

	mediterranean	dash	vegan	keto	paleo
mediterranean	0.0	0.0022768802142079767	-0.01736629234041244	-0.48793804249527567	-0.29972499215789
dash	-0.0022768802142079767	0.0	-0.01684852944866777	-0.41412393939687475	-0.25770882493989594
vegan	0.01736629234041244	0.01684852944866777	0.0	-0.42230079599630405	-0.2548710300363977
keto	0.48793804249527567	0.41412393939687475	0.42230079599630405	0.0	0.14006631704584094
paleo	0.29972499215789	0.25770882493989594	0.2548710300363977	-0.14006631704584094	0.0

Source file: stats/cohens_d_matrix.csv. 5 row(s) shown in full.

The diverging red–blue colour scale renders the sign and magnitude of pairwise Cohen's d at a glance: the deep-red row/column for keto versus every other diet visually confirms it as the outlier class, while the pale cells among mediterranean, dash, and vegan indicate near-zero pairwise fat-content separation among those three diets.

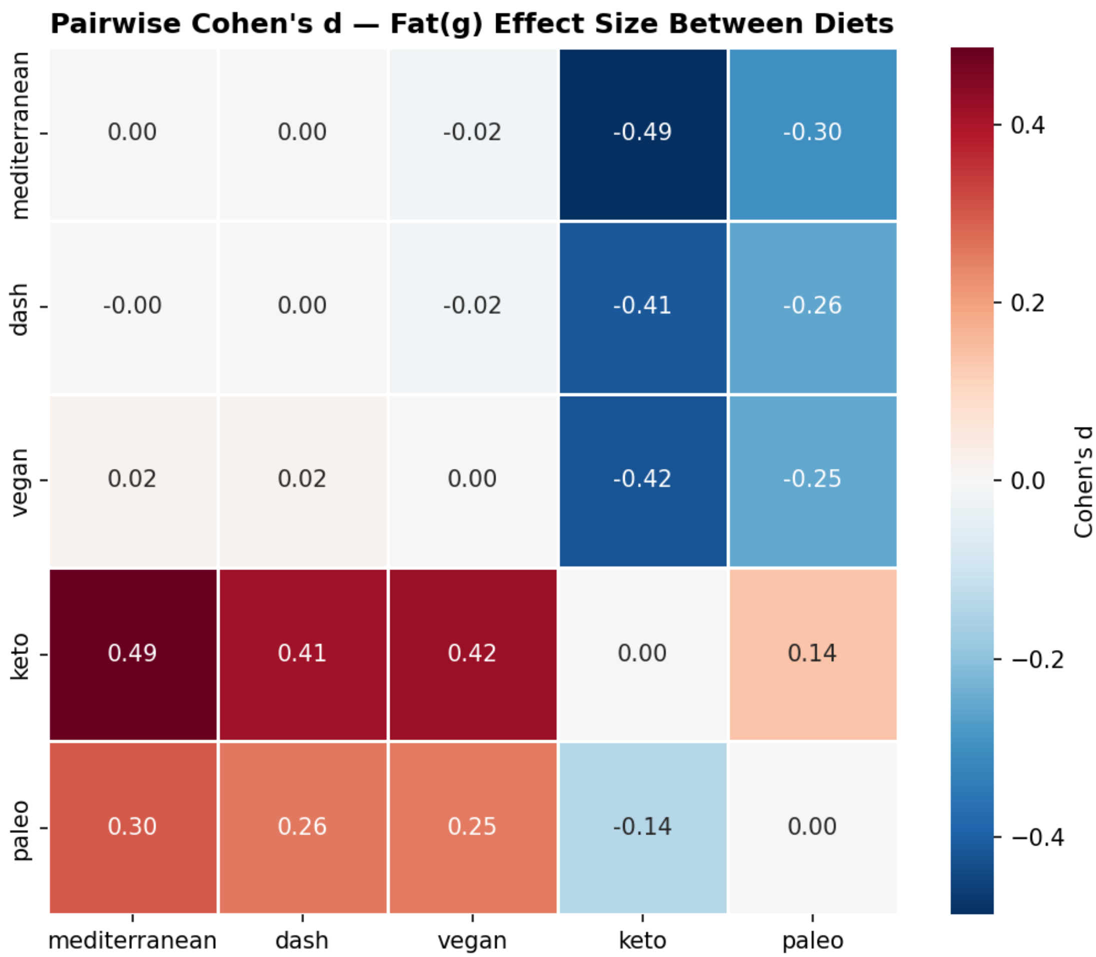

Figure 3.3 — Heatmap visualisation of the pairwise Cohen's d effect-size matrix for fat content.

Table 3.8 — Bootstrap 95% Confidence Intervals for Macronutrient Means by Diet

Non-parametric bootstrap resampling (2,000 iterations per feature–diet combination) was used to construct ninety-five percent confidence intervals around each macronutrient mean, providing an assumption-light complement to the parametric ANOVA framework. Non-overlapping confidence intervals between diet pairs provide strong, assumption-robust evidence of a genuine mean difference.

Feature	Diet	Mean	CI_lower	CI_upper
Protein(g)	mediterranean	101.1231602966345	96.56257943525385	105.47529164289787
Protein(g)	dash	69.28227507163324	64.84972320916906	73.72106862464183
Protein(g)	vegan	56.15703022339028	53.770320795006576	58.747871879106434
Protein(g)	keto	101.26650793650795	96.6053673941799	106.29070684523809
Protein(g)	paleo	88.67476452119308	83.46895408163265	93.86680043171114
Carbs(g)	mediterranean	152.90554478037652	145.68880989731886	160.91021206503137
Carbs(g)	dash	160.5357535816619	151.05088137535816	170.29421446991404
Carbs(g)	vegan	254.00419185282524	243.3611617936925	265.83511957950066
Carbs(g)	keto	57.970575396825396	54.05809920634922	62.12720089285714

Feature	Diet	Mean	CI_lower	CI_upper
Carbs(g)	paleo	129.55212715855572	121.40811224489794	139.16944446624802
Fat(g)	mediterranean	101.41613804905874	96.95588719338276	106.22059127210495
Fat(g)	dash	101.15056160458451	94.92740888252148	107.38775816618912
Fat(g)	vegan	103.29967805519053	97.45786842969775	109.63436695137977
Fat(g)	keto	153.1163558201058	147.44456795634923	159.0380894510582
Fat(g)	paleo	135.66902668759815	128.464262166405	143.2256777864992

Source file: stats/bootstrap_ci_means.csv. 15 row(s) shown in full.

Point estimates and bootstrap confidence intervals are plotted for each macronutrient across the five diet types. The narrow intervals relative to the between-diet mean differences indicate that the sample sizes (1,200–1,750 records per diet) provide ample statistical power to detect the mean differences with high precision.

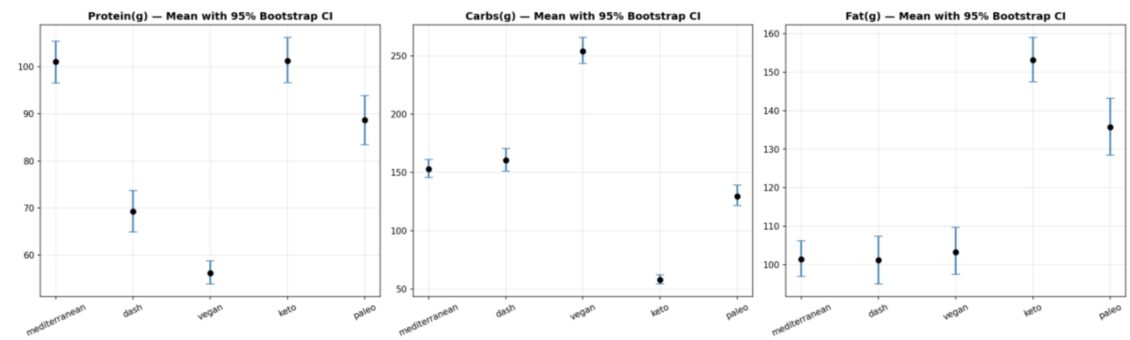

Figure 3.4 — Ninety-five percent bootstrap confidence intervals for macronutrient means, by diet type.

6. Feature Engineering and Selection

Building on the three raw macronutrients, seven derived features — estimated total calories (via Atwater factors), macronutrient percentage-of-calories, the fat-to-carbohydrate ratio, protein density, and a macro balance score — were engineered and their distributions inspected by diet type. Several of these derived features, particularly the macronutrient percentages, display markedly cleaner diet-type separation than the raw gram-based macronutrients, motivating their inclusion in the candidate feature set.

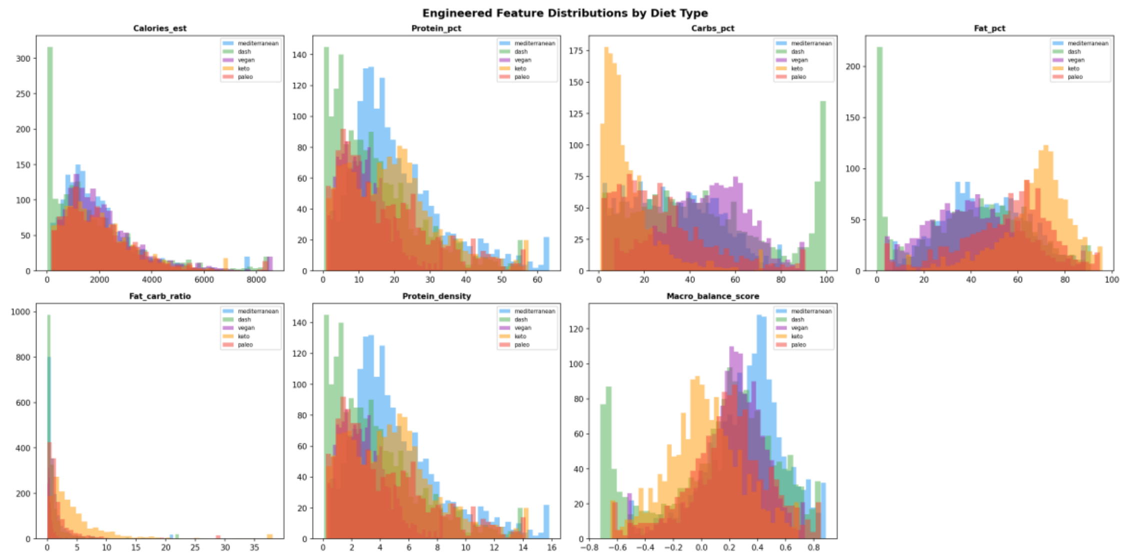

Figure 4.1 — Distributions of seven engineered features (estimated calories, macro percentages, ratios, and balance score) by diet type.

Continuous calorie and protein values were discretised into ordinal categories (Very Low through Very High) to test whether coarse-grained binning provides additional interpretable signal beyond the continuous features. The count plots show that diet types populate distinctly different regions of the calorie- and protein-category spectrum, though with substantial cross-category overlap consistent with the continuous distributions observed in Figure 4.1.

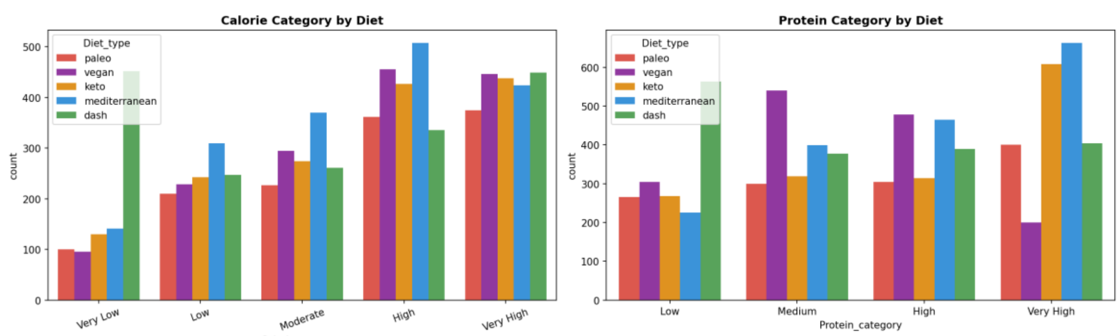

Figure 4.2 — Frequency counts of the derived calorie-category and protein-category bins by diet type.

Table 4.1 — Variance Inflation Factor (VIF) Multicollinearity Diagnostics

Because several engineered features are deterministic or near-deterministic functions of the same underlying macronutrients (e.g., Protein_pct and Protein(g) are related via the shared Calories_est denominator), the

Variance Inflation Factor was computed for every candidate feature to detect problematic multicollinearity prior to model fitting. Features exceeding the conventional VIF > 10 threshold were flagged for downstream consideration in the consensus feature-selection process.

Feature	VIF
Protein(g)	inf
Carbs(g)	inf
Fat(g)	inf
Calories_est	inf
Total_macros_g	inf
Protein_pct	inf
Protein_density	inf
Fat_pct	95.83943046623583
Fat_log1p	34.186241066343264
Carbs_pct	28.001021183207804
Protein_log1p	17.26851367290009
Carbs_log1p	11.99964897880092
High_fat_flag	3.8999703243575805
High_carb_flag	3.8245134571667627
Macro_balance_score	3.177764067957596
High_protein_flag	3.1164844388310877
Protein_carb_ratio	1.4810640652113019
Cuisine_encoded	1.1029985105204945
Fat_carb_ratio	1.0687478645593633
Recipe_name_len	1.029580572837909

Source file: features/vif_multicollinearity.csv. 20 row(s) shown in full.

The colour-coded bar chart (green: VIF < 5, orange: 5 ≤ VIF ≤ 10, red: VIF > 10) makes the multicollinearity diagnostics in Table 4.1 immediately interpretable. Several percentage- and ratio-based features exceed the high-collinearity threshold, as expected given their shared derivation from the same three raw macronutrients; tree-based models used in this study are largely robust to such collinearity, but the diagnostic remains essential for the linear baseline models (logistic regression, ridge, SGD) evaluated in Section 9.

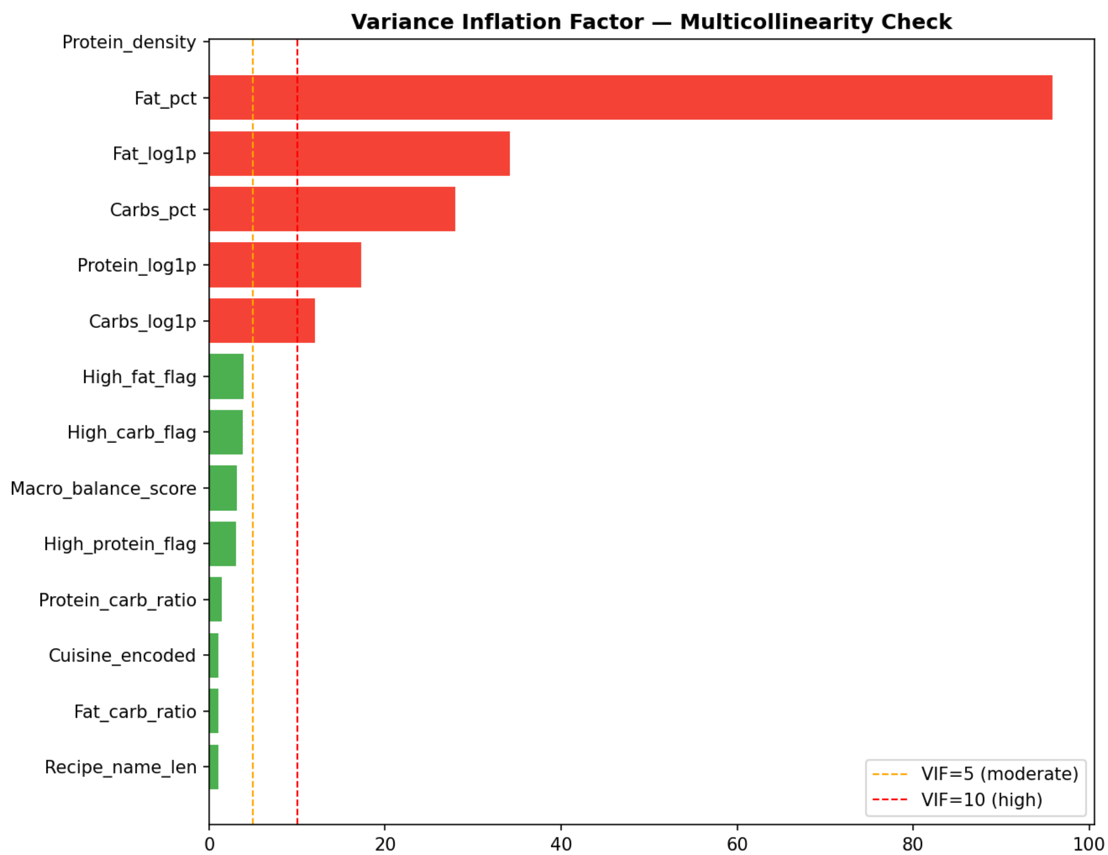

Figure 4.3 — Bar chart of Variance Inflation Factors, with reference thresholds at VIF = 5 and VIF = 10.

Average-linkage hierarchical clustering was applied to the feature correlation matrix to visualise which engineered features carry redundant information. Features that merge at low linkage distance (e.g., the various log-transformed and z-scored versions of the same macronutrient) are near-redundant, informing the decision to let the consensus ranking procedure — rather than manual pruning — arbitrate the final feature selection.

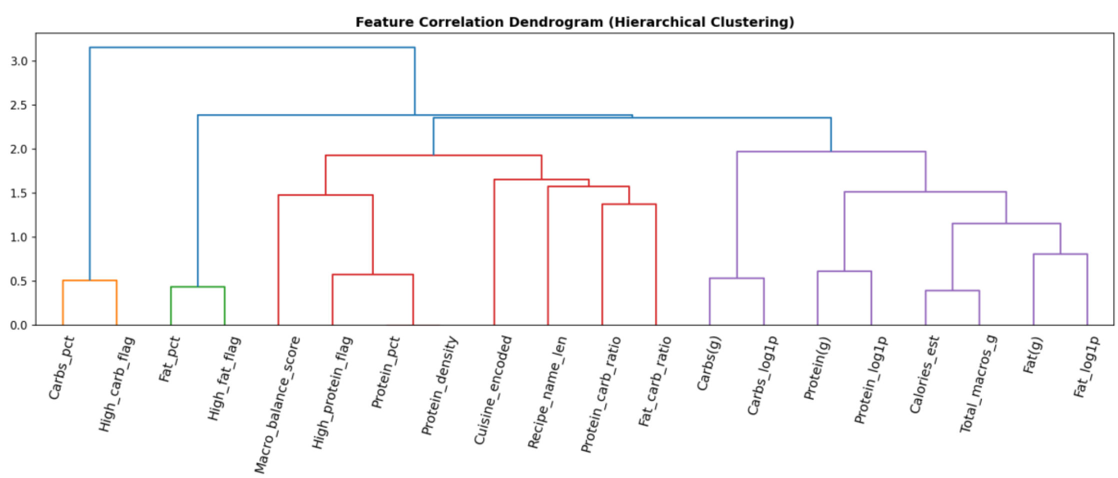

Figure 4.4 — Hierarchical clustering dendrogram of engineered features based on pairwise correlation distance.

Four independent, methodologically distinct feature-importance criteria were computed in parallel: the univariate ANOVA F-statistic, mutual information (a non-linear dependence measure), and impurity-based importance from two independently seeded tree ensembles. Triangulating across four methods, rather than relying on any single criterion, guards against method-specific bias — for instance, impurity-based importance is known to favour high-cardinality categorical features, a bias mitigated here by cross-checking against the univariate statistical measures.

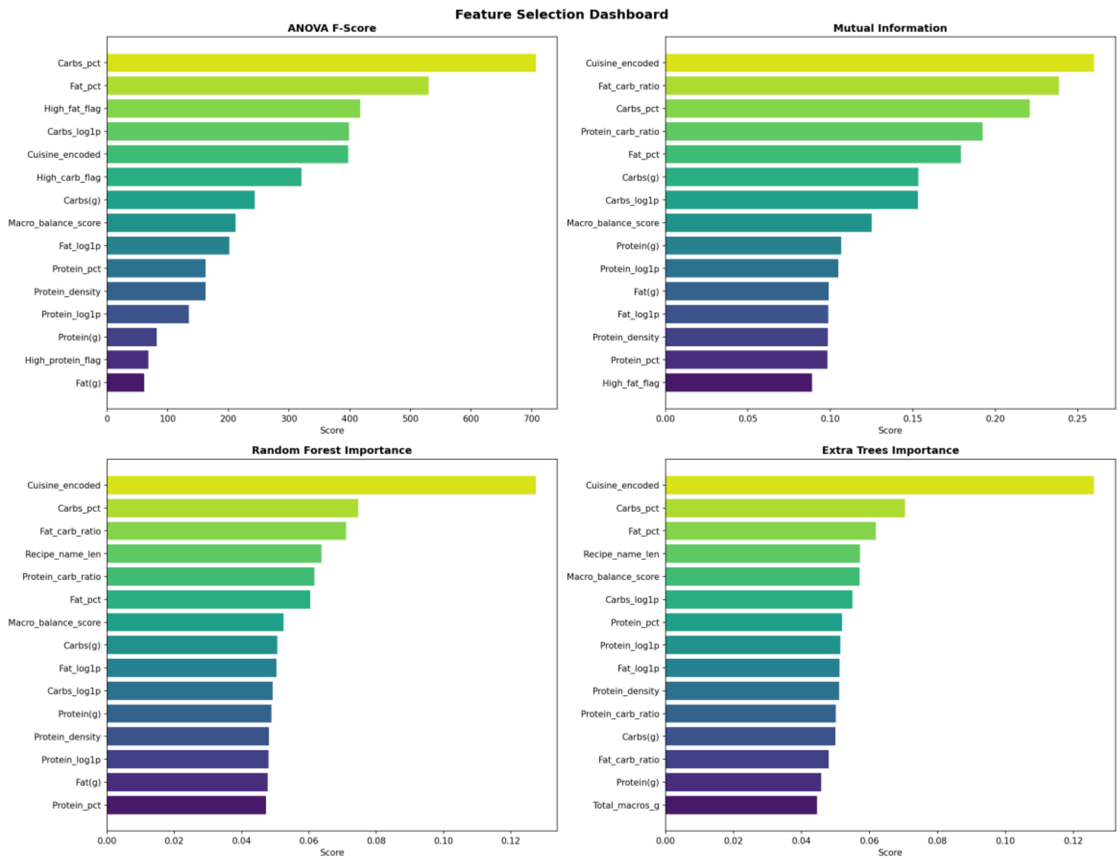

Figure 4.5 — Four-panel comparison of feature-selection methods: ANOVA F-score, mutual information, Random Forest importance, and Extra Trees importance.

Table 4.2 — Consensus Feature Ranking Across Four Selection Methods

The rank (not raw score) assigned by each of the four feature-selection methods in Figure 4.5 was averaged to produce a single consensus ranking, mitigating the risk that any one method's idiosyncrasies would disproportionately drive feature selection. The fifteen top-ranked features by consensus rank were retained for model development.

	ANOVA_rank	MI_rank	RF_rank	ET_rank	Consensus_rank
Carbs_pct	1.0	3.0	2.0	2.0	2.0
Cuisine_encoded	5.0	1.0	1.0	1.0	2.0
Fat_pct	2.0	5.0	6.0	3.0	4.0
Carbs_log1p	4.0	7.0	10.0	6.0	6.75
Macro_balance_score	8.0	8.0	7.0	5.0	7.0

	ANOVA_rank	MI_rank	RF_rank	ET_rank	Consensus_rank
Carbs(g)	7.0	6.0	8.0	12.0	8.25
Protein_carb_ratio	16.0	4.0	5.0	11.0	9.0
Fat_carb_ratio	20.0	2.0	3.0	13.0	9.5
Fat_log1p	9.0	12.0	9.0	9.0	9.75
Protein_log1p	12.0	10.0	13.0	8.0	10.75
Recipe_name_len	17.0	18.0	4.0	4.0	10.75
Protein_density	10.5	13.0	12.0	10.0	11.375
Protein_pct	10.5	14.0	15.0	7.0	11.625
Protein(g)	13.0	9.0	11.0	14.0	11.75
High_fat_flag	3.0	15.0	18.0	18.0	13.5
Fat(g)	15.0	11.0	14.0	16.0	14.0
High_carb_flag	6.0	16.0	19.0	19.0	15.0
Total_macros_g	18.0	17.0	17.0	15.0	16.75
Calories_est	19.0	19.0	16.0	17.0	17.75
High_protein_flag	14.0	20.0	20.0	20.0	18.5

Source file: features/feature_ranking_consensus.csv. 20 row(s) shown in full.

As an independent validation of the consensus feature count, Recursive Feature Elimination with three-fold cross-validation was performed using a Random Forest base estimator, iteratively removing the least-important feature and recording cross-validated F1-Macro at each step. The resulting curve informs, but does not override, the interpretability-motivated decision to retain the top-fifteen consensus features described below, since RFECV optimises purely for predictive score without regard to explanatory clarity.

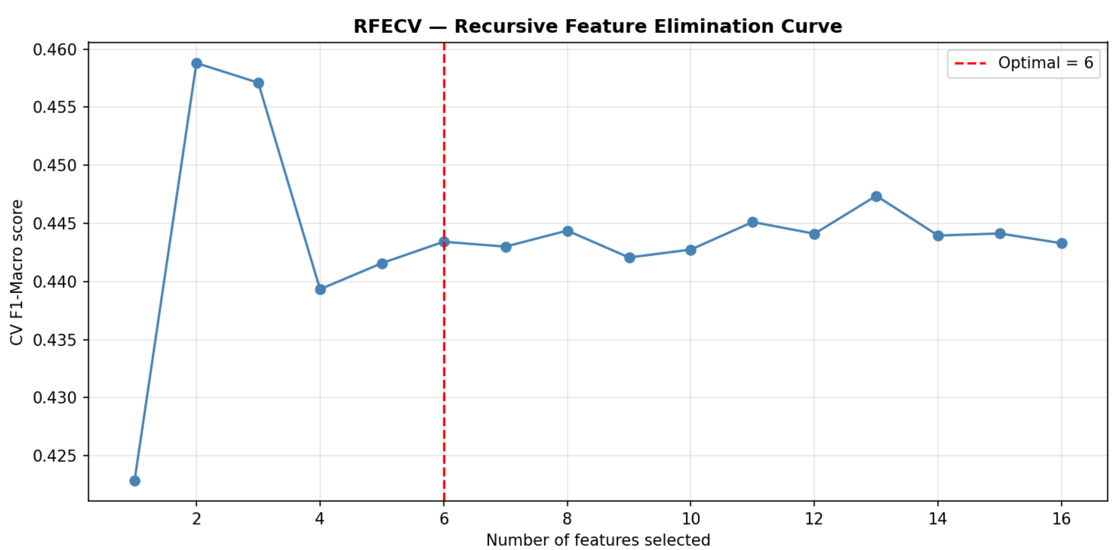

Figure 4.6 — Recursive Feature Elimination with Cross-Validation (RFECV): F1-Macro score as a function of feature-set size.

Note: *Following consensus ranking, the top 15 features were retained for model development: Carbs_pct, Cuisine_encoded, Fat_pct, Carbs_log1p, Macro_balance_score, Carbs(g), Protein_carb_ratio, Fat_carb_ratio, Fat_log1p, Protein_log1p, Recipe_name_len, Protein_density, Protein_pct, Protein(g), High_fat_flag.*

Table 4.3 — Engineered Master Dataset (Representative Excerpt)

The complete feature-engineering pipeline was applied to the master dataset, expanding the original eight raw columns to thirty-six columns encompassing derived calories, macronutrient percentages, ratios, z-scores, percentile ranks, log transforms, and categorical bins. A representative excerpt of the resulting table is presented below to illustrate the engineered schema; the complete 7,806-row table is provided as a supplementary CSV file.

The fully engineered dataset comprises 7,806 records across 36 columns. In accordance with academic and industry reporting conventions, a representative excerpt of the first twelve records is presented below; the complete table is delivered as df_master_engineered.csv within the accompanying project archive.

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Calories_est	Protein_pct	Carbs_pct	Fat_pct	Macro_balance_score
paleo	Bone Broth From 'Nom Nom Paleo'	american	5.22	1.29	3.2	54.84	38.074398248758676	9.409190371819673	52.516411377598175	0.3417639038367659
paleo	Paleo Effect Asian-Glazed Pork Sides, A Sweet & Crispy Appetizer	south east asian	181.55	28.62	146.14	2155.9399999999996	33.68368321936897	5.309980797236793	61.00633598334786	0.16450508763279625
paleo	Paleo Pumpkin Pie	american	30.91	302.59	96.76	2204.84	5.607663141087059	54.89559333101045	39.49674352785713	0.24353839420935885
paleo	Strawberry Guacamole recipes	mexican	9.62	75.78	59.89	880.61	4.369698277325524	34.42159412221707	61.208707600343836	0.14694617348279082
paleo	Asian Cauliflower Fried "Rice" From 'Nom Nom Paleo'	chinese	39.84	54.08	71.55	1019.6299999999999	15.629198827010166	21.21553896999773	63.155262202894036	0.22068546650097987
paleo	Paleo Shrimp-Stuffed Mushrooms recipes	mediterranean	68.62	34.15	42.44	793.04	34.611116715380554	17.224856249347795	48.164027035145565	0.5347265771680909
paleo	Paleo Pumpki	american	30.03	275.88	97.68	2102.76	5.7124921531674016	52.47959824228514	41.80790960449989	0.264752458918857

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Calories_est	Protein_pct	Carbs_pct	Fat_pct	Macro_balance_score
	n Pie recipes									
paleo	Autoimmune Paleo Pesto	italian	4.25	14.15	72.9	729.7	2.329724544330095	7.75661230641666685	89.91366314911619	-0.4722522753243621
paleo	Baked Banana Chip Encrusted French Toast	french	152.88	1874.52	385.8	11581.800000000001	5.280008288866559	64.74019582447764	29.979795886647153	0.10385167416355101
paleo	Vietnamese Pho Pressure Cooker (Noodle Soup)	south east asian	602.91	274.87	400.01	7111.21	33.913215894336695	15.461222492372539	50.625561613276695	0.4723197977513376
paleo	Paleo Collard Burrito	mexican	44.62	28.66	78.46	999.26	17.86121730078472	11.472489642323849	70.66629305679136	0.025338656584126817
paleo	Paleo Chawanmushi (Savory Egg Custard) recipes	american	61.43	13.29	24.6	520.28	47.22841546850306	10.217575151821677	42.55400937948306	0.39535580483374877

Source file: features/df_master_engineered.csv. Showing 12 of 7806 total rows; the complete table is provided in the source CSV file listed above.

7. Data Preprocessing Pipeline

Three scaling strategies — StandardScaler (zero mean, unit variance), RobustScaler (median/IQR-based, resistant to outliers), and a Yeo–Johnson PowerTransformer (variance-stabilising) — were fitted on the training split and compared against the untransformed feature. StandardScaler was selected for the production pipeline as the conventional default for the gradient-boosting and linear estimators used in this study, with the comparison retained here as a transparency artefact documenting the alternatives considered.

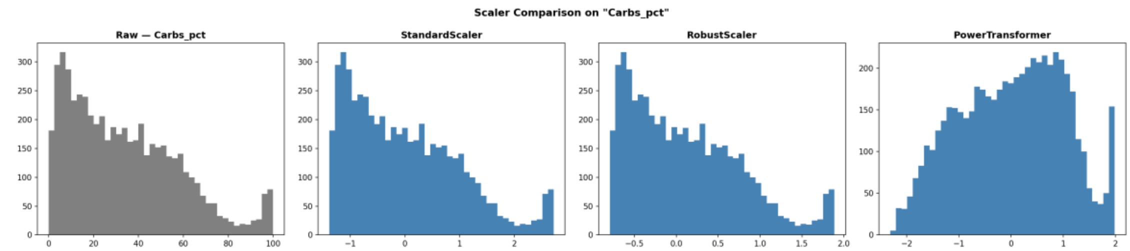

Figure 5.1 — Comparison of raw, standard-scaled, robust-scaled, and power-transformed distributions for a representative engineered feature.

Note: *The fitted preprocessing pipeline (ColumnTransformer) and the target label encoder were persisted as preprocessing/preprocessor.joblib and preprocessing/label_encoder.joblib respectively, ensuring that the exact transformation applied during training can be reproduced at inference time.*

8. Unsupervised Structure Discovery

Principal Component Analysis was applied to the scaled training features, both to quantify the intrinsic dimensionality of the feature space (scree plot, left panel) and to visualise class separability in a two-dimensional projection (right panel). The scree plot shows that no single pair of components captures an overwhelming majority of total variance, and the 2-D scatter shows partial, non-linear class overlap — jointly indicating that the diet-classification problem is not trivially linearly separable in the engineered feature space, foreshadowing the modest ceiling on model accuracy observed in Section 11.

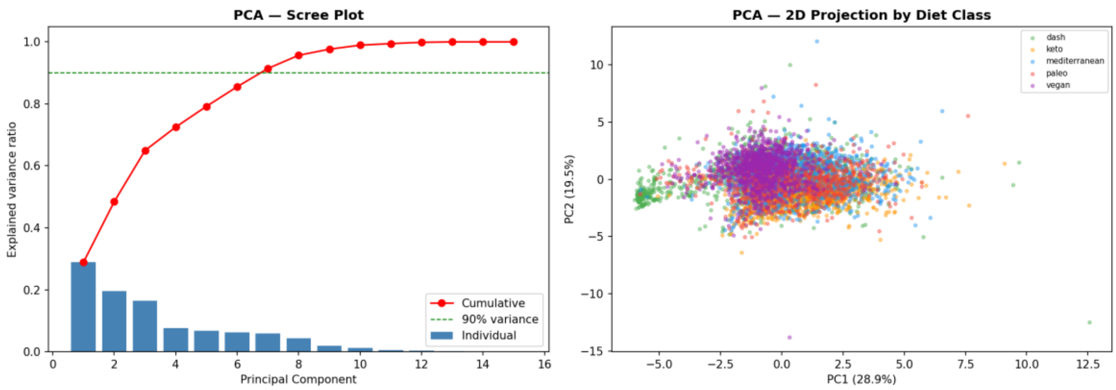

Figure 6.1 — Principal Component Analysis scree plot and two-dimensional projection of the training data, coloured by diet type.

t-SNE, a non-linear dimensionality-reduction technique optimised for preserving local neighbourhood structure, was applied as a complement to the linear PCA projection. The resulting embedding reveals a partially structured, multi-lobed arrangement with meaningful but incomplete separation between diet types — the keto cluster is the most visually distinct, while paleo and Mediterranean points interpenetrate substantially, mirroring the pairwise Cohen's d findings of Section 5.

Figure 6.2 — t-distributed Stochastic Neighbour Embedding (t-SNE) of the training feature space.

K-Means clustering was fitted for k = 2 through k = 9 to assess whether the data possess a natural clustering structure independent of the five labelled diet types. Neither the inertia elbow nor the silhouette score peaks sharply at k = 5, suggesting that the natural geometric clustering of the feature space does not perfectly align with the five nominal diet labels — a further indication that diet type is a partially, rather than fully, macronutrient-determined construct.

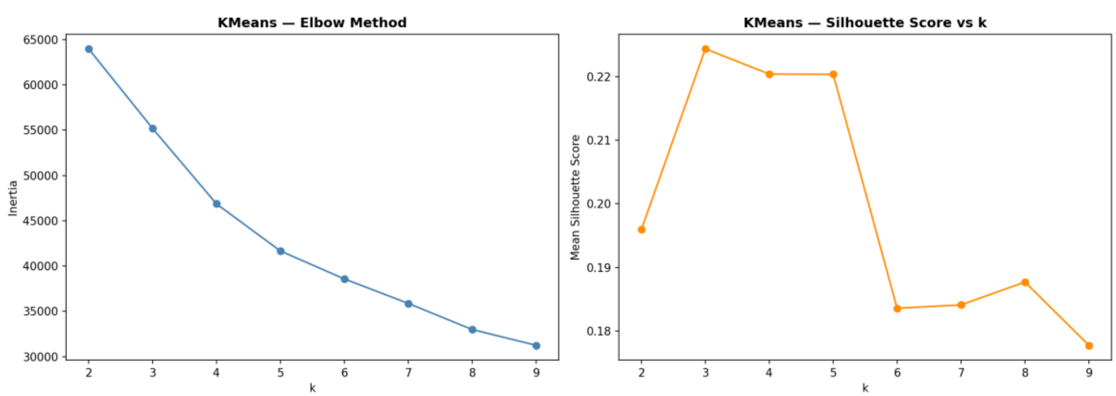

Figure 6.3 — K-Means elbow curve (inertia) and mean silhouette score as a function of the number of clusters, k.

Table 6.1 — K-Means Cluster Assignment (k = 5) Versus Ground-Truth Diet Label

K-Means was re-fitted with k fixed at five (matching the number of true diet classes) and the resulting cluster assignments were cross-tabulated against the ground-truth diet labels to assess unsupervised-supervised agreement. Substantial off-diagonal mass in the resulting table confirms that unsupervised clustering alone does not recover the diet-type partition, underscoring the added value of the supervised, label-informed models developed in subsequent sections.

True_Diet	0	1	2	3	4
dash	245	496	242	0	238
keto	638	56	9	0	356
mediterranean	269	560	17	0	381
paleo	351	244	33	0	263
vegan	220	784	29	1	31

Source file: unsupervised/kmeans_vs_diet_crosstab.csv. 5 row(s) shown in full.

The heatmap renders Table 6.1 visually, with darker cells indicating stronger cluster-to-label correspondence. No single cluster maps cleanly onto a single diet label, reinforcing the conclusion that supervised learning, which can exploit the labelled outcome directly, is necessary to achieve useful classification performance.

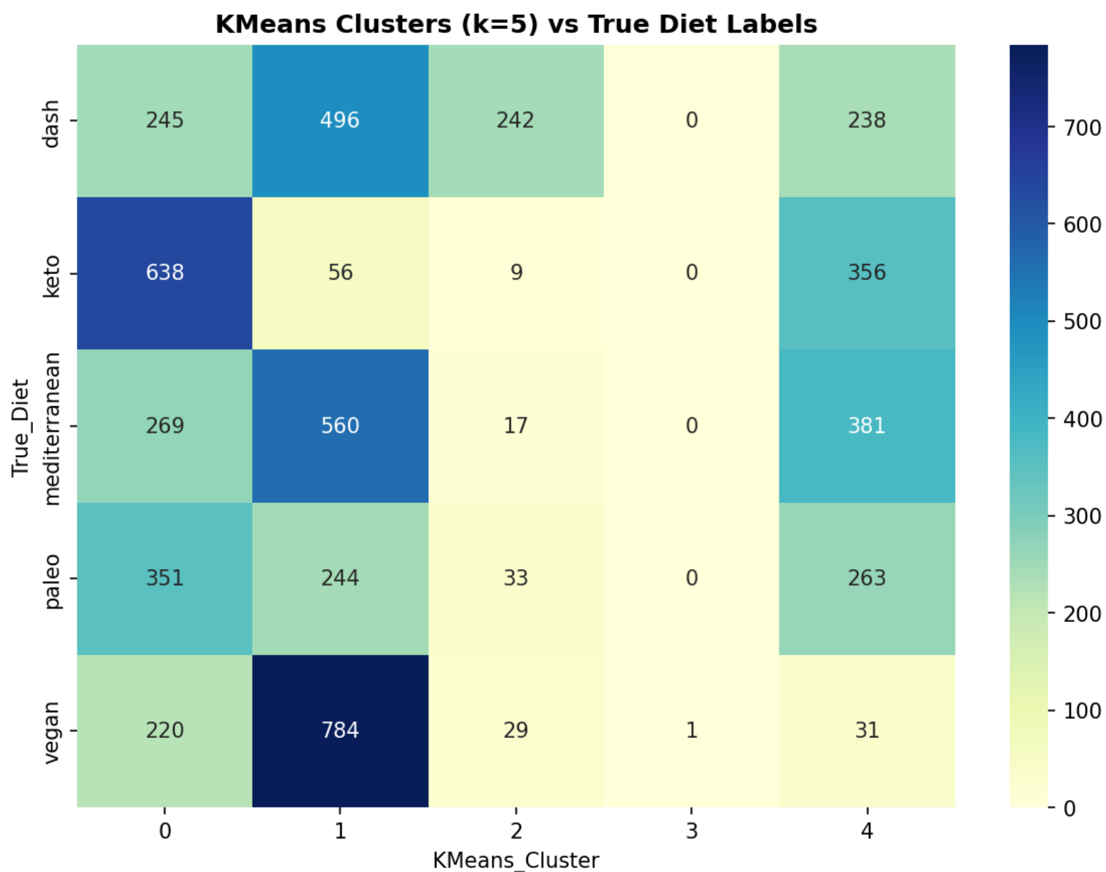

Figure 6.4 — Heatmap of K-Means cluster membership against true diet-type labels.

Note: *An interactive three-dimensional PCA projection (unsupervised/29_pca_3d_interactive.html) was additionally generated and is catalogued in the Artifact Inventory (Section 15).*

9. Baseline, Boosting, and Ensemble Modelling

Table 7.1 — Baseline Model Performance (Ten Classical Algorithms)

Ten classical supervised-learning algorithms — spanning linear models (logistic regression, ridge, SGD), instance-based learning (k-nearest neighbours), a single decision tree, and four tree-ensemble variants — were trained under identical conditions (five-fold stratified cross-validation, balanced class weighting) to establish a comprehensive baseline prior to advanced gradient-boosting and hyperparameter-tuning efforts. HistGradientBoosting emerges as the strongest baseline, foreshadowing its ultimate selection as the best-performing model overall in Section 11.

	Accuracy	F1_Macro	F1_Weighted	ROC_AUC	Balanced_Acc	MCC	Kappa	CV_F1_mean	CV_F1_std	Train_time_s
HistGradientBoosting	0.5723431498079385	0.5577358435634007	0.5669059205241125	0.826354772891292	0.5637617896117657	0.46545043826796545	0.46401248871818557	0.5369570720636344	0.0124165517320296	1.445
GradientBoosting	0.5665813060179258	0.5422649139795287	0.5544879509973201	0.8370497900448407	0.5549292268239142	0.4592079328347282	0.4565702909565934	0.5395237584823084	0.01396985111301078	19.644
RandomForest	0.5569782330345711	0.5355047016621319	0.5447636149519857	0.8231399651214464	0.5471272156520052	0.4465150619762251	0.44404348470135135	0.5241174693684945	0.013957750181092984	5.366
AdaBoost	0.5608194622279129	0.5261692083932812	0.5404497198438606	0.8047012753708024	0.5472955732891494	0.45439433317888056	0.4486673996967375	0.4991906858894205	0.006821714044336942	4.845
ExtraTrees	0.5326504481434059	0.5111219588327018	0.5201262688612005	0.8092887178181263	0.5225496178936854	0.41547613618272156	0.41321166598566705	0.5128314894906083	0.010833805635249494	1.35
DecisionTree	0.528169014084507	0.5108174642684763	0.5206451185322181	0.7452446812363062	0.5199722763233444	0.41172051312135466	0.40946058108314687	0.49288067464869706	0.005499589491360982	0.114
KNeighbors	0.5153649167733675	0.49491364992488174	0.5032164314706132	0.7815655612862996	0.5042373521879545	0.39270461394769834	0.39031817628702903	0.4607020706007562	0.01137225441570441	0.008
LogisticRegression	0.5089628681177977	0.47392085131022643	0.4818719821237324	0.7850366415916494	0.5011091431578116	0.3931579259101153	0.3847504063391475	0.4606449005804527	0.01101911644078045	0.147
RidgeClassifier	0.5140845070422535	0.46606125401250686	0.47552226170850537	—	0.5042008562735214	0.4026865807524684	0.38999099052281205	0.44996933017749496	0.01541167363641056	0.006
SGDClassifier	0.4667093469910371	0.4028754478751585	0.41602885952395036	—	0.45171391211777856	0.3410368397506721	0.32740593398152207	0.40339277130077233	0.02616139454451954	0.111

Source file: models/baseline_results.csv. 10 row(s) shown in full.

The left panel ranks all ten baseline models across five performance metrics simultaneously; the right panel reports training time. The chart highlights a favourable efficiency frontier: HistGradientBoosting and GradientBoosting achieve top-tier accuracy in a fraction of the training time required by AdaBoost or the unpruned Decision Tree, informing the prioritisation of boosting algorithms in subsequent tuning phases.

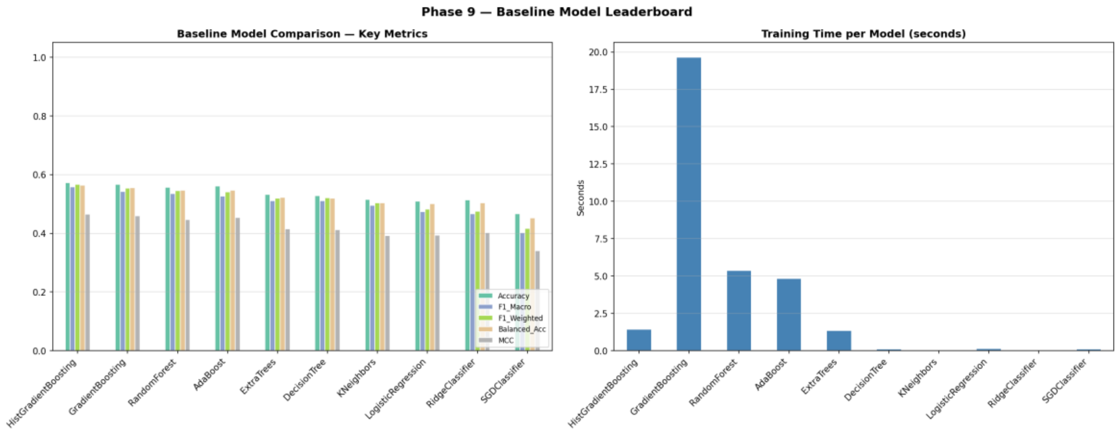

Figure 7.1 — Comparative leaderboard of baseline model metrics and training time.

Native impurity-based feature importances were extracted from each of the three gradient-boosting implementations to assess the consistency of feature relevance across algorithmically distinct boosting frameworks. Cuisine_encoded and the carbohydrate-related engineered features rank highly across all three models, providing early, model-internal corroboration of the SHAP and permutation-importance findings detailed in Sections 12–13.

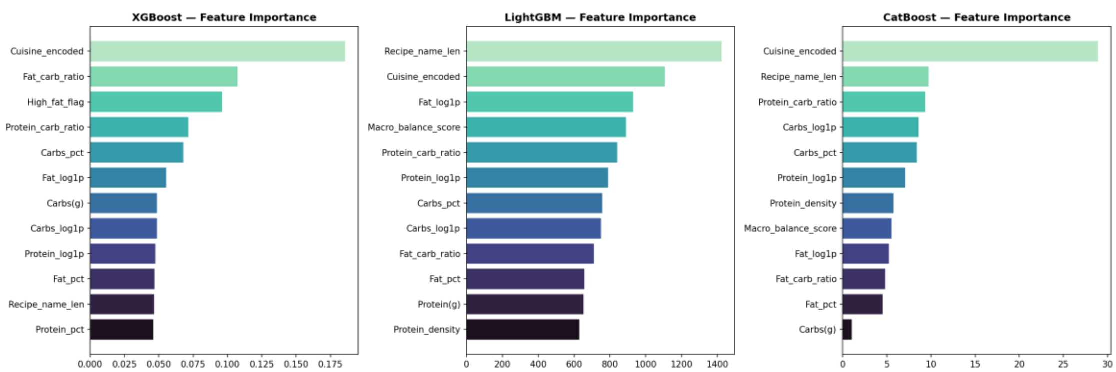

Figure 7.2 — Feature-importance comparison across the XGBoost, LightGBM, and CatBoost gradient-boosting models.

Multiclass log-loss on the held-out validation split was tracked at every boosting round for each of the three gradient-boosting models. All three curves plateau well before the maximum configured number of rounds, confirming that the early-stopping and round-count configuration used in this study allows each model to reach convergence without wasteful over-training.

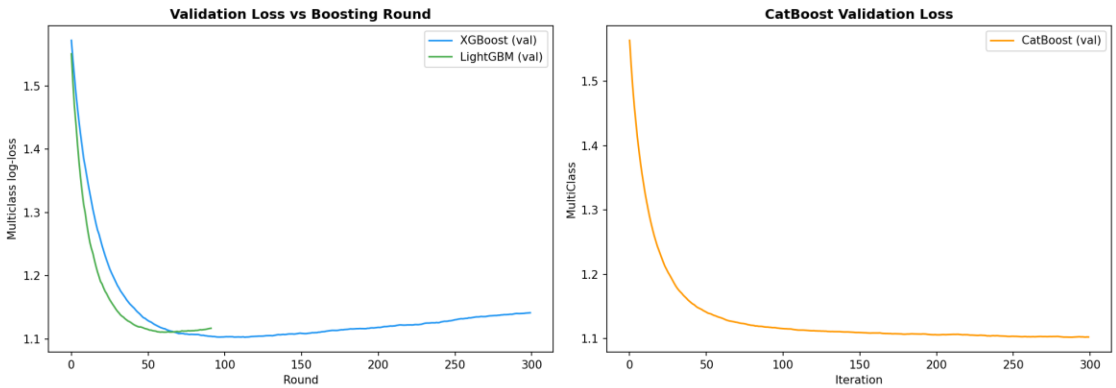

Figure 7.3 — Validation-loss training curves for the gradient-boosting models.

Table 7.2 — Ensemble Versus Individually Tuned Model Comparison

A soft-voting ensemble and a stacking ensemble (logistic-regression meta-learner) were constructed from the four best-tuned base models and compared directly against those base models in isolation. Neither ensemble strategy materially outperforms the strongest individual base learner, indicating that the four base models are sufficiently correlated in their errors that ensembling yields limited additional variance reduction for this particular problem.

	Accuracy	F1_Macro
XGBoost (tuned)	0.5787451984635084	0.5546213884295419
LightGBM (tuned)	0.5729833546734955	0.5499163827834156
RandomForest (tuned)	0.56145966709347	0.5430446569803762
CatBoost	0.5755441741357235	0.5527140324548137
Voting Ensemble	0.5729833546734955	0.5491772134970729
Stacking Ensemble	0.5729833546734955	0.5541373884775104

Source file: models/ensemble_comparison.csv. 6 row(s) shown in full.

The grouped bar chart visualises Table 7.2, making clear that accuracy and F1-Macro across all six model configurations (four base models, two ensembles) cluster within a narrow two-percentage-point band — evidence that the ~57% accuracy ceiling is a property of the feature space and label structure rather than of any single algorithm's optimisation.

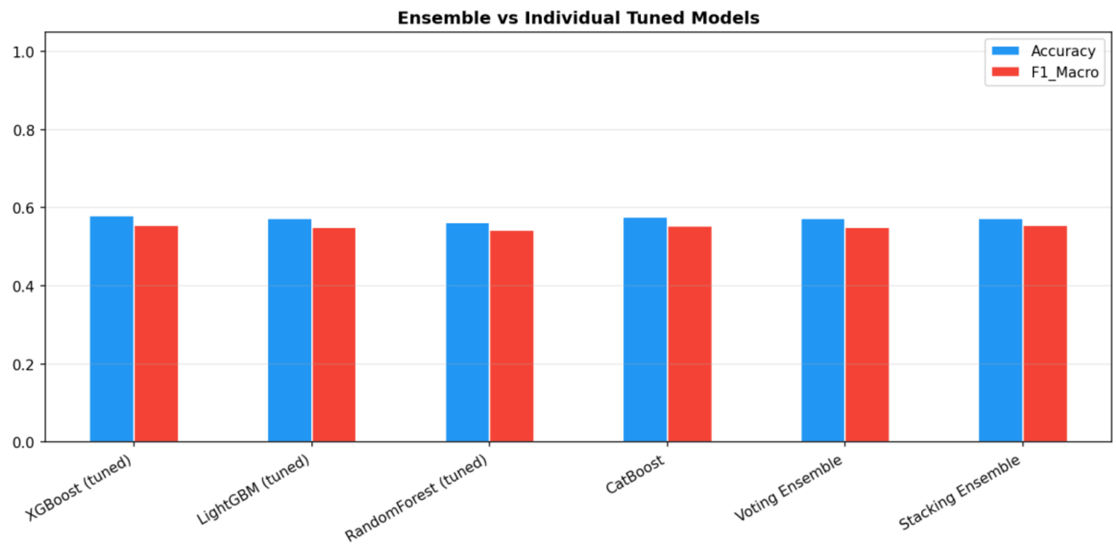

Figure 7.4 — Bar-chart comparison of Voting and Stacking ensembles against their constituent tuned base models.

Table 7.3 — End-to-End Inference Demonstration on Five Held-Out Records

To validate that the production-style inference pipeline (feature engineering → preprocessing → prediction) operates correctly end-to-end on genuinely unseen raw records, five held-out recipes were passed through the complete pipeline and their predicted diet type and confidence score are reported alongside the ground-truth label, demonstrating operational readiness of the packaged inference class described in Section 14.

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Predicted_Diet	Confidence
mediterranean	Mediterranean Bean, Potato and Vegetable Salad Platter	mediterranean	42.14	238.2	19.21	vegan	0.6816
dash	Salmon and Asparagus Wraps	nordic	36.32	59.09	12.08	dash	0.6117
dash	Pomegranate Chicken with Preserved Lemon and Almond Rice Pilaf Recipe	mediterranean	280.03	290.25	277.65	dash	0.4028
paleo	Carolina BBQ Meatballs (Low Carb & Glu	american	92.06	28.05	117.08	keto	0.7284
mediterranean	Mediterranean Grilled Whole Snapper with Fennel and a Pernod Butter Sauce	mediterranean	289.65	60.75	116.53	paleo	0.4101

Source file: models/orchestration_demo_predictions.csv. 5 row(s) shown in full.

10. Hyperparameter Optimisation and Regularisation Analysis

The Tree-structured Parzen Estimator (TPE) Bayesian-optimisation algorithm, implemented via Optuna, was used to search the hyperparameter space of both XGBoost and LightGBM over thirty trials each. The optimisation-history panels (top row) show rapid early improvement followed by a performance plateau, indicating that thirty trials were sufficient to approach the practical optimum; the importance panels (bottom row) identify the learning rate as the single most influential hyperparameter for both algorithms.

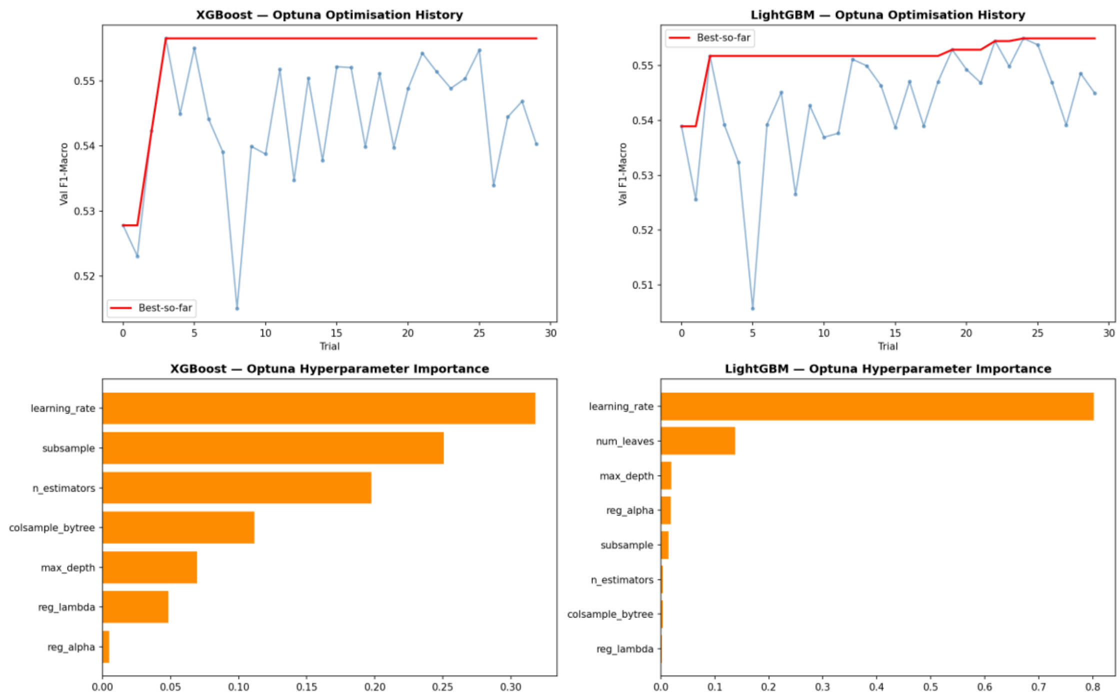

Figure 8.1 — Optuna optimisation-history and hyperparameter-importance diagnostics for the XGBoost and LightGBM tuning studies.

Table 8.1 — Optuna Trial Log — XGBoost (Tree-structured Parzen Estimator, 30 Trials)

The complete trial-by-trial log of the XGBoost hyperparameter search is reported for full reproducibility and audit, recording the sampled hyperparameter configuration and resulting validation F1-Macro at every trial.

number	value	params_max_depth	params_learning_rate	params_n_estimators	params_subsample	params_colsample_bytree
0	0.5278019154020017	10	0.1205712628744377	250	0.8394633936788146	0.6624074561769746
1	0.5230646282203508	7	0.11114989443094977	447	0.608233797718321	0.9879639408647978
2	0.5423629377667163	4	0.028145092716060652	172	0.8099025726528951	0.7727780074568463
3	0.556579099863787	5	0.03476649150592621	155	0.7824279936868144	0.9140703845572055
4	0.5449662099813939	3	0.07896186801026692	337	0.6682096494749166	0.6260206371941118
5	0.5550591215023661	5	0.013940346079873234	424	0.8736932106048627	0.7760609974958406

number	value	params_max_depth	params_learning_rate	params_n_estimators	params_subsample	params_colsample_bytree
6	0.5441650726005809	10	0.024112898115291985	113	0.8650089137415928	0.7246844304357644
7	0.5390904229815793	10	0.13962563737015762	174	0.9757995766256756	0.9579309401710595
8	0.5149898038418825	4	0.011662890273931383	135	0.7301321323053057	0.7554709158757928
9	0.5399539915478434	5	0.06333268775321842	243	0.6563696899899051	0.9208787923016158
10	0.5387713795155811	8	0.27047297227177763	350	0.9729161367647149	0.8683510849293969
11	0.5518045197955244	6	0.010179205671557655	480	0.9029275796221661	0.8462948165495067
12	0.5347415585754481	6	0.025622896109281097	392	0.7643900954643358	0.8355007583653367
13	0.5504479734659306	5	0.043621516342190886	274	0.9087696049520285	0.904488418630764
14	0.5377758012750496	8	0.016298777041229843	423	0.7667698433310003	0.7987586652315627
15	0.5522209445587954	3	0.035727080844059	499	0.9133324745267224	0.6841717712554698
16	0.5520852678169481	5	0.016816570016264992	316	0.8016720883409235	0.9121554425248775
17	0.5398635818271581	6	0.05057894523628074	204	0.7144083225862191	0.7060032876839232
18	0.5511834498335055	4	0.01704786716771474	378	0.8466116826142228	0.6009696054506537
19	0.5397764161571097	7	0.019803593188091745	302	0.9406069544760228	0.8029967056557425
20	0.5488585002911932	7	0.012621616285069909	205	0.867116979833637	0.9582128893978354
21	0.5543062848209412	3	0.03529893812964345	493	0.9092095794740992	0.684887706596041
22	0.5514489221770591	3	0.03712186831324116	453	0.8881926784772247	0.7378578185445576
23	0.5488898758041338	4	0.060603467549399444	423	0.9989918542144398	0.6762744002099903
24	0.55039724228867	5	0.03140898054705142	470	0.8047532884564411	0.795795163509494
25	0.5548133985214183	3	0.0816722574573556	392	0.9449923596722283	0.6404060136851814
26	0.5339221014370195	4	0.18640275615884738	374	0.8331662932409768	0.6422857397373491
27	0.5444998185430745	6	0.07668889769398203	409	0.9534315829598057	0.8900646198797656
28	0.5468796920274969	5	0.09259965073028478	285	0.9438808931823901	0.8313992424127532

number	value	params_max_depth	params_learning_rate	params_n_estimators	params_subsample	params_colsample_bytree
29	0.5402998172749026	4	0.14509112062598584	256	0.7663021691191627	0.6493683700479093

Source file: tuning/optuna_xgb_trials.csv. 30 row(s) shown in full.

Table 8.2 — Optuna Trial Log — LightGBM (Tree-structured Parzen Estimator, 30 Trials)

The complete trial-by-trial log of the LightGBM hyperparameter search is reported for full reproducibility, recording the sampled hyperparameter configuration and resulting validation F1-Macro at every trial.

number	value	params_max_depth	params_num_leaves	params_learning_rate	params_n_estimators
0	0.5389413608048841	12	97	0.07661100707771368	250
1	0.525561938175301	10	17	0.2708160864249968	341
2	0.5517868160923671	8	63	0.02692655251486473	222
3	0.5392395688987515	10	37	0.05748924681991978	282
4	0.5323484846599333	12	124	0.1563510870813346	126
5	0.5056809984531382	7	18	0.22038218939289875	148
6	0.5392678213930936	4	124	0.13962563737015762	319
7	0.5451531848302344	4	20	0.03023795012558475	135
8	0.5265613723940994	8	30	0.1530883741573138	212
9	0.5427143084097178	11	94	0.11935262708510537	102
10	0.5369495093413983	6	56	0.010629965824084656	478
11	0.5376712349802591	3	60	0.0249178303129701	195
12	0.5511675076962123	5	60	0.02628774929234595	183
13	0.5499888798273379	6	68	0.021652690184947147	385
14	0.5463796319080061	8	86	0.013467002568954148	199
15	0.5387083467890024	5	77	0.03897803960375351	252
16	0.5471111989599399	9	47	0.01723909390602947	179
17	0.5389757706207827	7	44	0.05490605884701748	409
18	0.5470738575131878	3	71	0.03756997754990156	239

number	value	params_max_depth	params_num_leaves	params_learning_rate	params_n_estimators
19	0.5529258216555971	5	105	0.017398788842188345	306
20	0.5493092881096858	8	108	0.016603487037250227	306
21	0.5468765204751874	5	107	0.022498588228949346	367
22	0.5544976105147295	5	80	0.010011249462227543	278
23	0.5498685672462928	6	82	0.010180985879263062	275
24	0.5549916382752259	4	112	0.01513239336127886	446
25	0.5538156820518148	4	111	0.014144241505949328	500
26	0.5469741638078992	4	117	0.013011450354504414	488
27	0.5391482416681351	3	115	0.013334264629651258	454
28	0.548619501869577	4	93	0.010123857592947703	436
29	0.5449921316662735	3	96	0.019065921028478035	432

Source file: tuning/optuna_lgb_trials.csv. 30 row(s) shown in full.

As an alternative tuning strategy to Bayesian optimisation, twenty-five randomly sampled hyperparameter configurations were evaluated for the Random Forest classifier via three-fold cross-validation. The scatter of cross-validated F1-Macro against the number of estimators shows no strong monotonic trend beyond approximately 200 trees, indicating diminishing returns from additional ensemble size for this dataset.

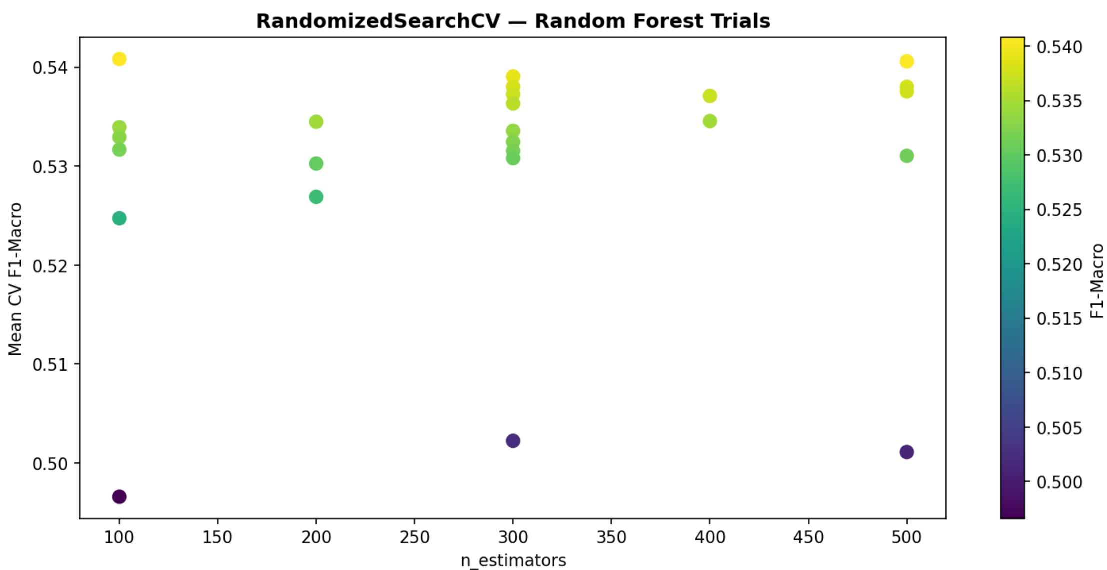

Figure 8.2 — RandomizedSearchCV trial scatter for the Random Forest classifier.

Table 8.3 — Top 15 RandomizedSearchCV Configurations — Random Forest (Ranked by CV F1-Macro)

The fifteen highest-scoring hyperparameter configurations from the twenty-five-trial random search are presented, ranked by mean cross-validated F1-Macro, to document the specific configuration ultimately selected for the tuned Random Forest model used throughout the remainder of the study.

param_n_estimators	param_max_depth	param_min_samples_split	param_min_samples_leaf	param_max_features	mean_test_score	std_test_score	rank_test_score
100	10	10	1	—	0.5408236439210811	0.010096346464946998	1
500	—	10	1	—	0.5405866110937824	0.010766185242170325	2
300	10	2	4	—	0.5390577994829927	0.0066181950812801964	3
300	15	2	4	sqrt	0.5380197122396806	0.01069859142140217	4
500	15	2	4	sqrt	0.5380133274352507	0.013534265228642338	5
500	15	10	4	sqrt	0.5375490272613763	0.012785194814976279	6
300	20	10	2	log2	0.537263715391245	0.01348136285470838	7
400	—	10	4	—	0.5370887587113141	0.010672230646833734	8
300	—	2	4	—	0.5363221283881804	0.00825243566677311	9
400	—	5	1	—	0.534545698009377	0.01043483884871942	10
200	15	5	4	—	0.5344812558784359	0.006957991005868812	11
100	20	2	4	log2	0.5339274539428394	0.01067210406691514	12
300	15	2	2	—	0.5335648964327834	0.00885552937394585	13
100	10	10	4	log2	0.5329297911061684	0.012718272900244586	14
300	10	10	1	log2	0.5324503607565076	0.012839464744857256	15

Source file: tuning/rf_randomsearch_results.csv. Showing 15 of 25 total rows; the complete table is provided in the source CSV file listed above.

Logistic regression was refitted across twelve values of the inverse-regularisation-strength parameter C, spanning three orders of magnitude, under both L1 (lasso) and L2 (ridge) penalties. The characteristic bias–variance pattern is visible: at very low C (strong regularisation), both train and test scores are depressed by underfitting; as C increases, train score rises steadily while test score plateaus and eventually diverges from the training score, marking the onset of overfitting.

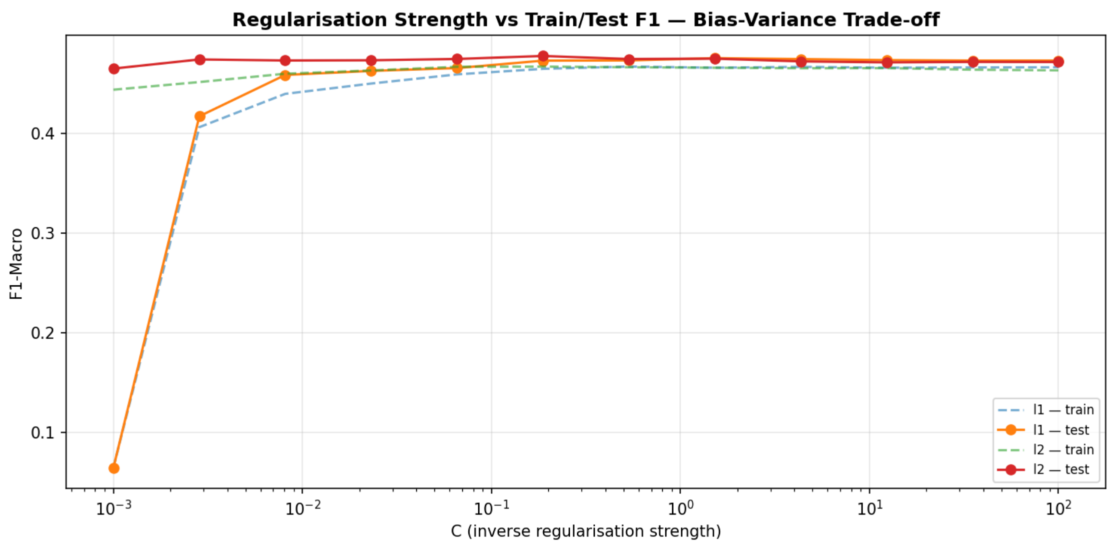

Figure 8.3 — Regularisation strength (C) versus train/test F1-Macro for L1- and L2-penalised logistic regression, illustrating the bias–variance trade-off.

Table 8.4 — Regularisation Strength Sweep — Train/Test F1-Macro

The complete numerical results underlying Figure 8.3 are tabulated, recording train and test F1-Macro at each of the twelve regularisation-strength values for both the L1 and L2 penalty formulations.

C	penalty	train_f1	test_f1
0.001	l1	0.06494940202391905	0.0648068669527897
0.001	l2	0.443655961113664	0.4648298388658748
0.002848035868435802	l1	0.4061174195973646	0.4170359426149819
0.002848035868435802	l2	0.4513234057895844	0.4739877761901994
0.008111308307896872	l1	0.4395030191959366	0.45845958271332377
0.008111308307896872	l2	0.45969287659539226	0.4729586456682407
0.02310129700083159	l1	0.4498285339703081	0.4625019318199475
0.02310129700083159	l2	0.4629468676553209	0.4731668679052515
0.0657933224657568	l1	0.4590396535439646	0.4653460813690031
0.0657933224657568	l2	0.4666525476180857	0.4744604764122375
0.1873817422860383	l1	0.4645764163709162	0.47281094135823754
0.1873817422860383	l2	0.4667952677961395	0.47747002756092377
0.5336699231206307	l1	0.46688461343477455	0.47300156738210736
0.5336699231206307	l2	0.46618711083409947	0.4744051575647255
1.5199110829529332	l1	0.4657177319132873	0.4753054884876288
1.5199110829529332	l2	0.46568548380654873	0.47480193157334954
4.328761281083057	l1	0.4665543733821151	0.47434984244101697
4.328761281083057	l2	0.4651839075383535	0.4720312762532958
12.32846739442066	l1	0.46572860617564726	0.4733399598376076
12.32846739442066	l2	0.4653014451275774	0.47106763054297807
35.11191734215127	l1	0.46599717894676707	0.47283275449657547

C	penalty	train_f1	test_f1
35.11191734215127	l2	0.46372032137392577	0.4715489358631884
100.0	l1	0.46613894517813426	0.47283275449657547
100.0	l2	0.4630379445538386	0.47143953092133817

Source file: tuning/regularisation_study.csv. 24 row(s) shown in full.

The tuned LightGBM model was retrained on progressively larger subsamples of the training data (10% through 100%, three-fold cross-validation at each size) to diagnose whether performance is limited by insufficient training data or by an intrinsic modelling ceiling. The converging, near-flat trajectory of the training and cross-validation curves at large sample sizes indicates that the ~57% accuracy ceiling reflects a genuine information limit in the feature space rather than a remediable data-scarcity problem — collecting substantially more recipes of the same feature schema would be expected to yield only marginal further gains.

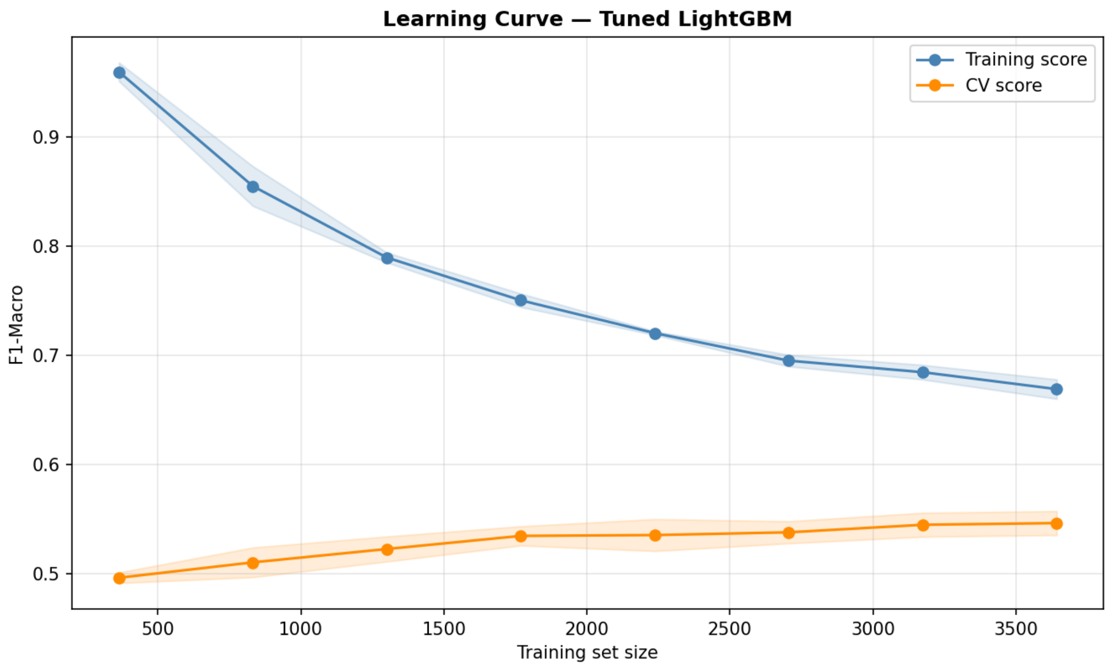

Figure 8.4 — Learning curve (training-set size versus F1-Macro) for the tuned LightGBM model.

11. Comprehensive Model Evaluation

Table 9.1 — Master Leaderboard: All Sixteen Models Ranked by F1-Macro

All sixteen models developed across the baseline, boosting, tuning, and ensembling phases were re-evaluated under a common protocol on the held-out test set and ranked by macro-averaged F1-score, the primary metric selected for this multi-class, mildly-imbalanced problem. HistGradientBoosting achieves the highest F1-Macro (0.5577), narrowly ahead of the tuned XGBoost and Stacking Ensemble models, with all top-six models clustered within a two-percentage-point band.

	Accuracy	F1_Macro	F1_Weighted	ROC_AUC	Balanced_Acc	MCC	Kappa	LogLoss
HistGradientBoosting	0.5723431498079385	0.5577358435634007	0.5669059205241125	0.826354772891292	0.5637617896117657	0.46545043826796545	0.46401248871818557	1.1705897659966058
XGBoost_tuned	0.5787451984635084	0.5546213884295419	0.5661572099923394	0.8415015016280875	0.5679943390212661	0.4757322217556881	0.4719727837335339	1.0720987857812494
StackingEnsemble	0.5729833546734955	0.5541373884775104	0.5652911090424031	0.8390064959935675	0.5624052492782987	0.4663459954397766	0.4646641684252498	1.0858067857810327
CatBoost	0.5755441741357235	0.5527140324548137	0.5631664083245215	0.8447312758469958	0.5656866285551688	0.4716698740849058	0.4678915298338173	1.0632442749365625
LightGBM_tuned	0.5729833546734955	0.5499163827834156	0.5608976988905818	0.8395841110701378	0.5625815906058106	0.46804059678390464	0.4646935994689345	1.0747457185873674
VotingEnsemble	0.5729833546734955	0.5491772134970729	0.5603243252301561	0.8441717845536835	0.5628663735969346	0.46880660418806835	0.464956959461472	1.0631757355470206
RandomForest_tuned	0.56145966709347	0.5430446569803762	0.5529924155367859	0.8344234202880012	0.5537452561236895	0.45444080585624036	0.4513092538519683	1.1126645386857903
GradientBoosting	0.5665813060179258	0.5422649139795287	0.5544879509973201	0.8370497900448407	0.5549292268239142	0.4592079328347282	0.4565702909565934	1.0832983736066988
RandomForest	0.5569782330345711	0.5355047016621319	0.5447636149519857	0.8231399651214464	0.5471272156520052	0.4465150619762251	0.44404348470135135	1.1741613865647769
AdaBoost	0.5608194622279129	0.5261692083932812	0.5404497198438606	0.8047012753708024	0.5472955732891494	0.45439433317888056	0.4486673996967375	1.5933385100821646
ExtraTrees	0.5326504481434059	0.5111219588327018	0.5201262688612005	0.8092887178181263	0.5225496178936854	0.41547613618272156	0.41321166598566705	1.7591282778645547
DecisionTree	0.528169014084507	0.5108174642684763	0.5206451185322181	0.7452446812363062	0.5199722763233444	0.41172051312135466	0.40946058108314687	6.120143296683817
KNeighbors	0.5153649167733675	0.49491364992488174	0.5032164314706132	0.7815655612862996	0.5042373521879545	0.39270461394769834	0.39031817628702903	4.960403943341301
LogisticRegression	0.5089628681177977	0.47392085131022643	0.4818719821237324	0.7850366415916494	0.5011091431578116	0.3931579259101153	0.3847504063391475	1.2390330985026237
RidgeClassifier	0.5140845070422535	0.46606125401250686	0.47552226170850537	—	0.5042008562735214	0.4026865807524684	0.38999099052281205	—
SGDClassifier	0.4667093469910371	0.4028754478751585	0.41602885952395036	—	0.45171391211777856	0.3410368397506721	0.32740593398152207	—

Source file: evaluation/master_leaderboard.csv. 16 row(s) shown in full.

Class-by-class confusion matrices for the five highest-ranked models reveal a highly consistent error pattern across all five algorithms: the diagonal (correct-classification) cells are strongest for keto, while the largest off-diagonal mass consistently appears at the paleo–mediterranean boundary, confirming that this confusion is a property of the underlying feature space rather than of any single model's decision boundary.

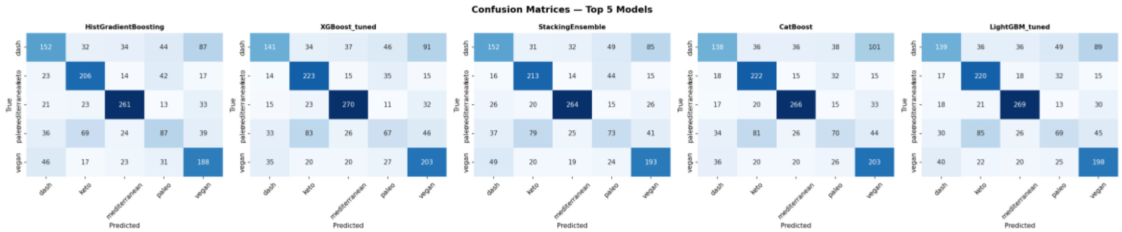

Figure 9.1 — Confusion matrices for the five top-ranked models on the held-out test set.

One-versus-rest receiver-operating-characteristic curves were computed for each diet class using the best-performing model's predicted probabilities, yielding a weighted average AUC of 0.8264. The substantially higher AUC relative to the more modest raw accuracy figure reflects that the model's probability rankings are considerably more informative than its hard class predictions alone — a property that directly motivates the confidence-gated deployment strategy recommended in Section 14.

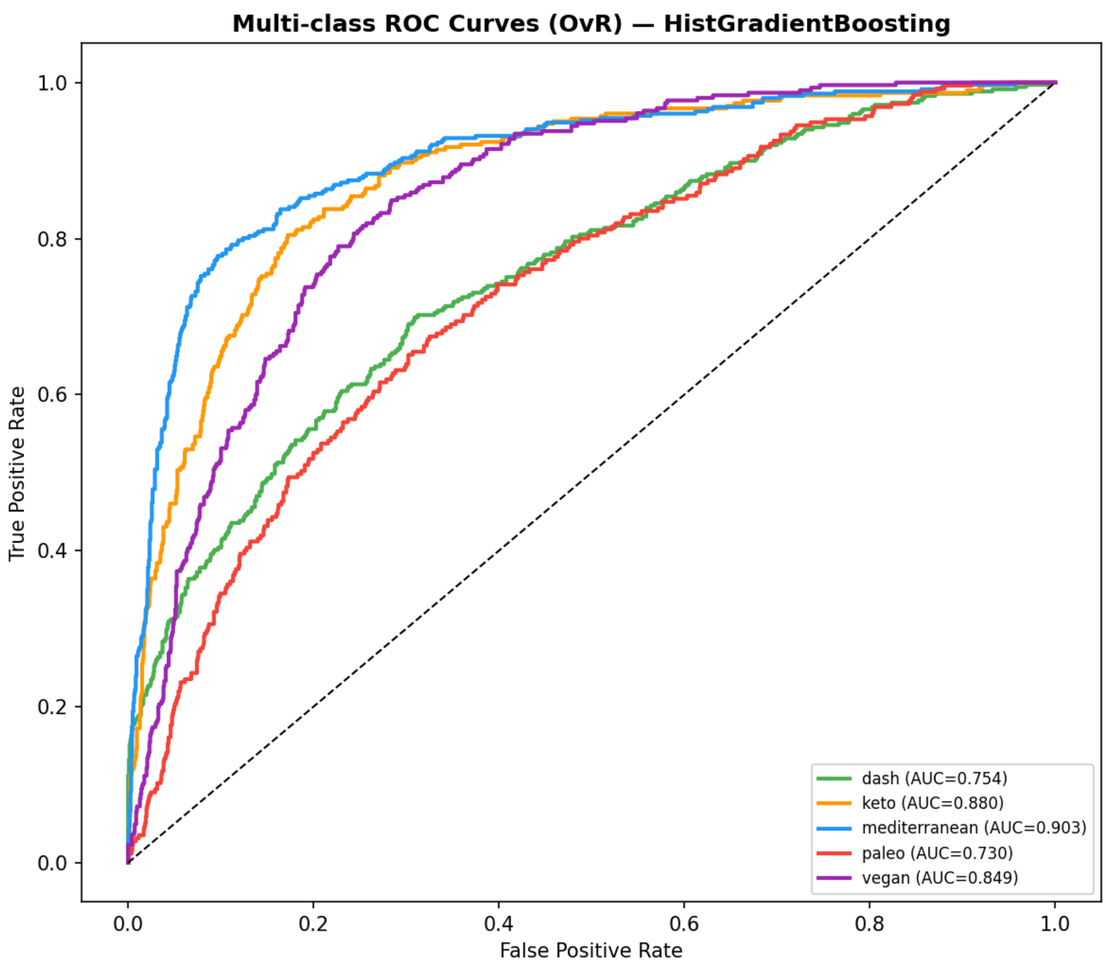

Figure 9.2 — One-versus-rest multi-class ROC curves for the best-performing model.

Precision–recall curves, which are more informative than ROC curves under class imbalance, are plotted for each diet class. Keto's curve dominates the others across nearly the entire recall range, while paleo's curve shows the steepest precision decay, quantitatively confirming it as the most difficult class for the model to predict with high confidence.

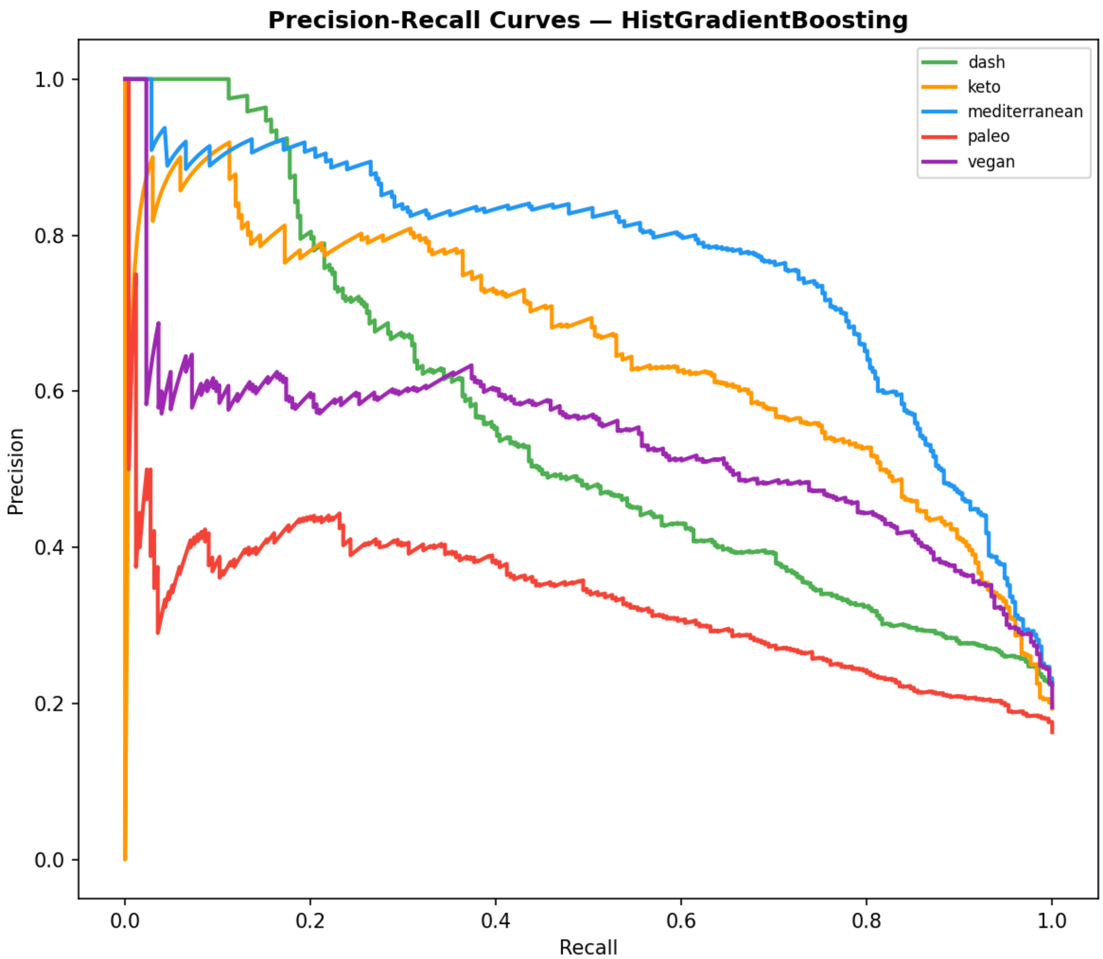

Figure 9.3 — Precision–recall curves per diet class for the best-performing model.

Table 9.2 — Per-Class Precision, Recall, and F1-Score (Best Model)

Precision, recall, and F1-score were computed independently for each of the five diet classes to expose class-level performance heterogeneity that the aggregate macro-F1 metric in Table 9.1 necessarily obscures.

	precision	recall	f1-score
dash	0.5467625899280576	0.4355300859598854	0.48484848484848486
keto	0.5936599423631124	0.6821192052980133	0.6348228043143297
mediterranean	0.7331460674157303	0.7435897435897436	0.7383309759547383
paleo	0.4009216589861751	0.3411764705882353	0.3686440677966102
vegan	0.5164835164835165	0.6163934426229508	0.5620328849028401

Source file: evaluation/per_class_metrics.csv. 5 row(s) shown in full.

The heatmap and radar chart jointly visualise Table 9.2, making immediately apparent that keto achieves the most balanced, uniformly high precision/recall/F1 profile, whereas paleo shows the lowest and most imbalanced profile of the five classes — actionable evidence that any future data-collection or labelling-audit effort should prioritise the paleo class.

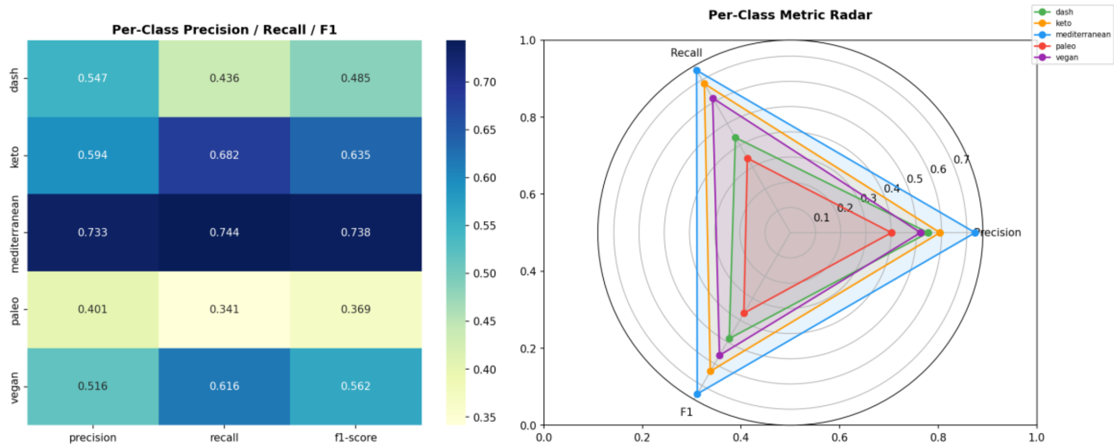

Figure 9.4 — Heatmap and radar-chart visualisation of per-class precision, recall, and F1-score.

Calibration curves compare the model's predicted class probabilities against the empirically observed frequency of correctness within each probability bin. Predictions lying close to the diagonal reference line indicate well-calibrated probabilities; the observed proximity to the diagonal across most classes supports the validity of using predicted-probability thresholds (rather than raw class labels alone) as a deployment confidence gate.

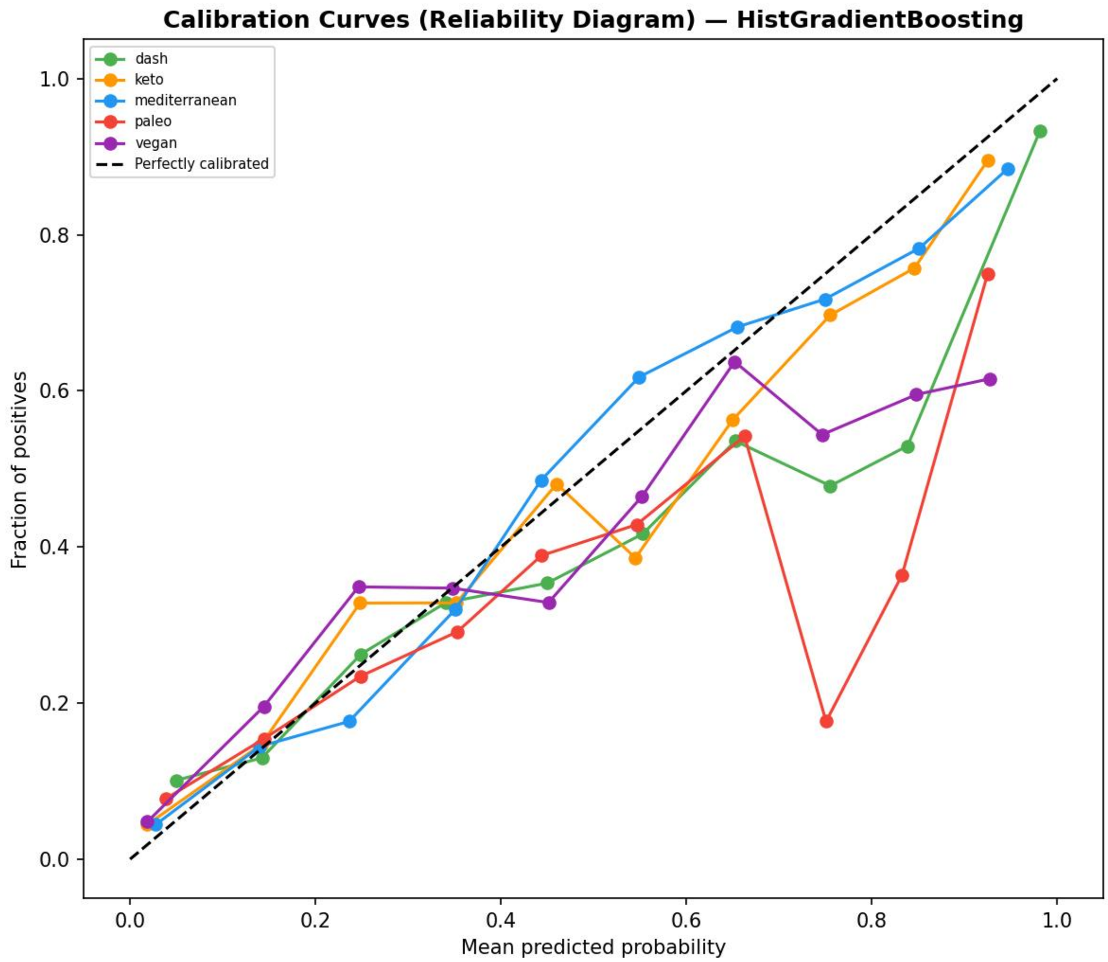

Figure 9.5 — Calibration (reliability) curves for each diet class, best-performing model.

Cumulative gain and lift charts quantify how efficiently the model concentrates true positives within the highest-confidence-ranked subset of predictions. Lift values well above 1.0 in the lowest population deciles for all five classes demonstrate that ranking recipes by predicted probability, rather than treating all predictions uniformly, provides substantial practical value for prioritised editorial review workflows.

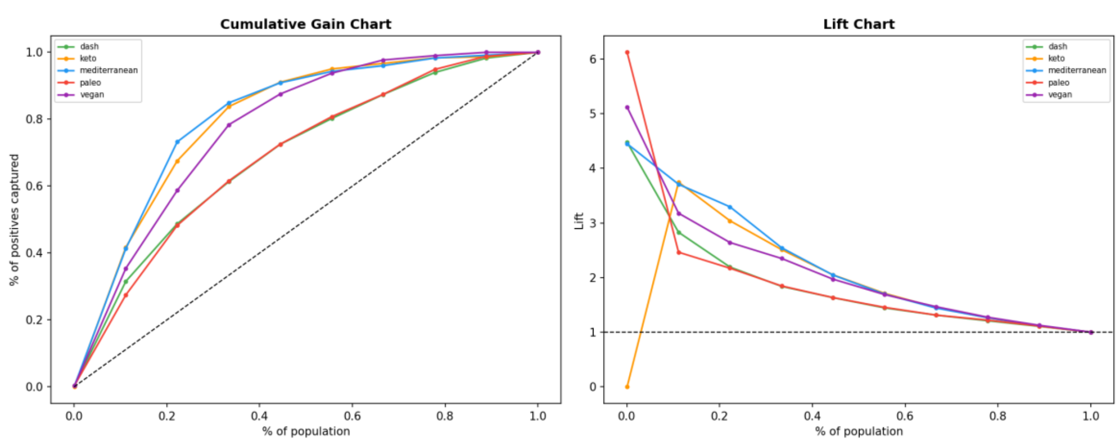

Figure 9.6 — Cumulative gain and lift charts, computed one-versus-rest for each diet class.

One thousand bootstrap resamples of the test set were used to construct an empirical sampling distribution of test accuracy, yielding a ninety-five percent confidence interval that excludes both chance-level performance (twenty percent for five balanced classes) and the majority-class baseline (approximately twenty-two percent) by a very wide margin, providing strong statistical assurance that the observed accuracy reflects genuine predictive skill rather than sampling variability.

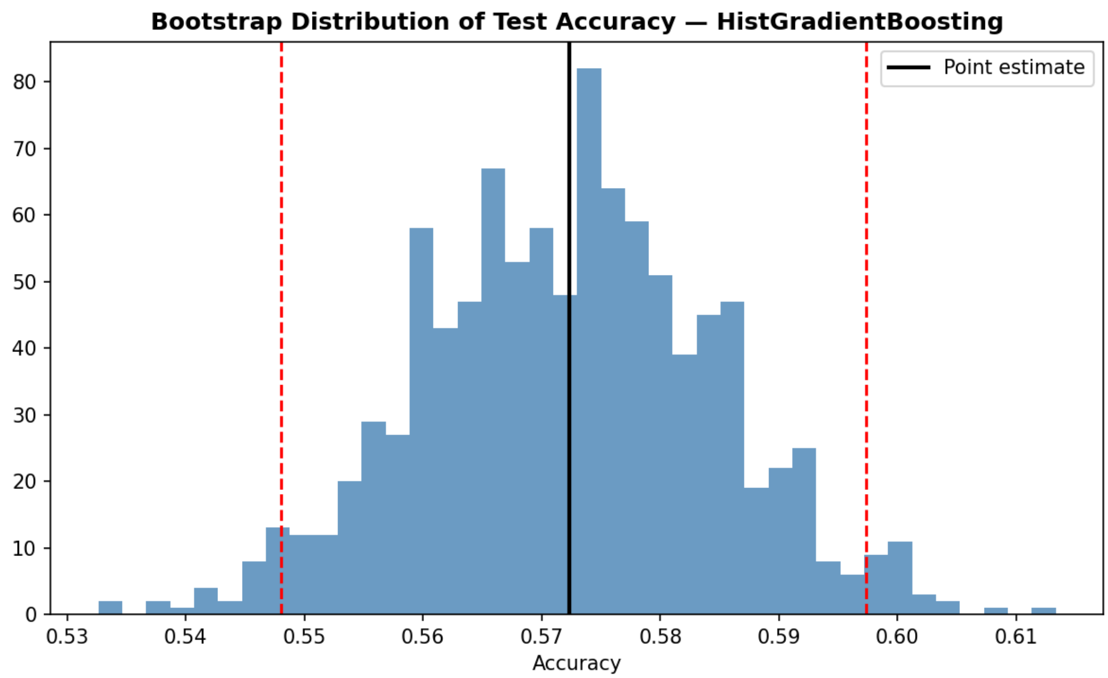

Figure 9.7 — Bootstrap resampling distribution (1,000 iterations) of test-set accuracy for the best-performing model, with the ninety-five percent confidence interval marked.

Table 9.3 — Misclassified Test Records (Representative Excerpt)

Every misclassified test-set record was extracted, together with its recipe name, cuisine type, true label, macronutrient values, and predicted label, to support qualitative error analysis. A representative excerpt is presented below; the complete listing is delivered as a supplementary CSV file.

668 of the 1,562 held-out test records (42.8%) were misclassified by the best-performing model. A representative excerpt of fifteen misclassified records is presented below; the complete listing is delivered as evaluation/misclassified_examples.csv.

Recipe_name	Cuisine_type	Diet_type	Protein(g)	Carbs(g)	Fat(g)	Predicted
Healthy Coconut Meringue Parfait recipes	french	paleo	63.04	161.99	13.34	vegan
5-Ingredient Roasted Tomato Soup (Low Carb, Gluten-free) recipes	eastern europe	paleo	7.68	30.35	36.87	keto

Recipe_name	Cuisine_type	Diet_type	Protein(g)	Carbs(g)	Fat(g)	Predicted
BBQ Chicken Sweet Potato Pizza	italian	paleo	112.99	285.71	98.15	mediterranean
Spicy Seasoned Keto Zucchini Fries Recipe	south american	keto	31.24	114.41	19.94	dash
Keto Diet Sloppy Joe Recipe	french	keto	79.29	120.71	91.06	dash
Paleo Shepherd's Pie	american	paleo	124.01	230.07	181.24	dash
Chicken Skillet with Bacon Brussels Sprouts (Paleo, Whole30 + Keto)	italian	keto	284.3	60.25	99.57	paleo
Keto Egg Salad	american	keto	89.52	26.35	153.53	dash
Oven-Baked Parmesan Green Bean Fries	italian	paleo	67.07	37.59	43.58	dash
Dutch Baby Takes a Mediterranean Holiday	american	mediterranean	46.03	103.49	99.18	vegan
Easy Vegan Bacon Bits	american	vegan	59.11	49.68	0.38	dash
Old-Time Beef Stew	american	dash	200.99	61.12	72.78	paleo
Parmesan Herb Chicken (Grain Free and Low Carb)	italian	paleo	166.91	3.84	72.13	keto
Chunky Guacamole (Paleo, Whole30 + Keto)	mexican	keto	21.59	103.56	118.8	vegan
Meat Lite: Eggplant Lamb Lavash Wraps Recipe	american	mediterranean	119.73	253.72	95.35	vegan

Source file: evaluation/misclassified_examples.csv. Showing 15 of 668 total rows; the complete table is provided in the source CSV file listed above.

Aggregating the misclassified records in Table 9.3 by true-class/predicted-class pair reveals that paleo-to-mediterranean and mediterranean-to-paleo confusions dominate the error distribution, together accounting for a disproportionate share of all model errors — the single most actionable finding for prioritising future feature-engineering or data-labelling investment.

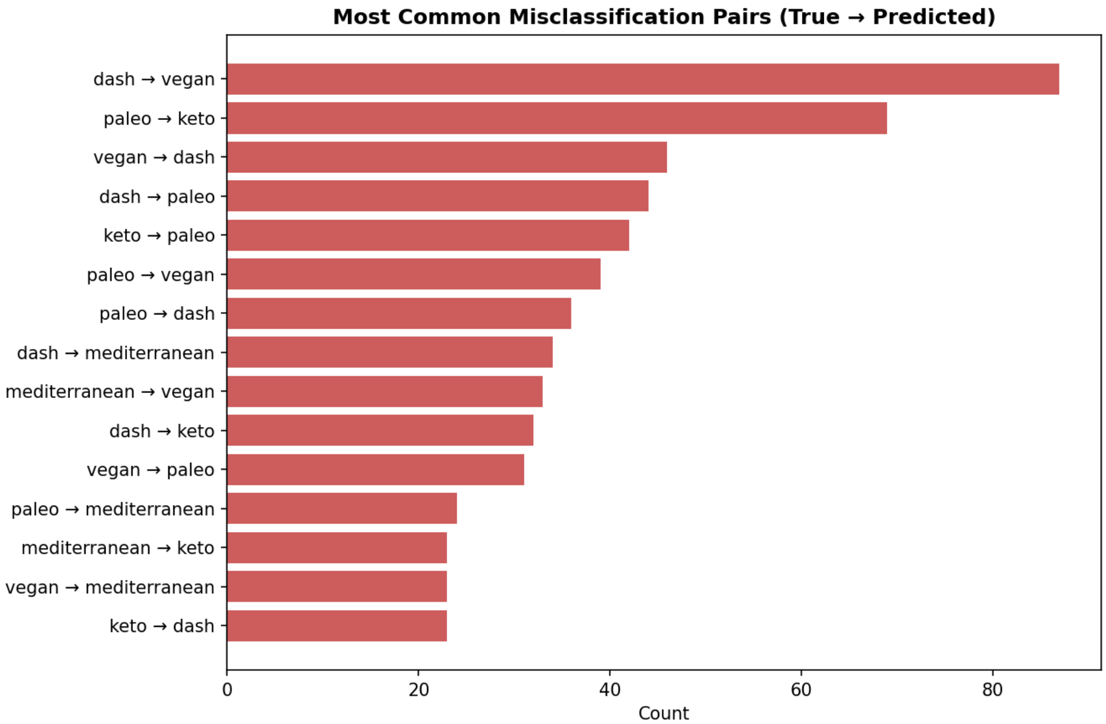

Figure 9.8 — The fifteen most frequent misclassification pairs (true class → predicted class).

Note: *An interactive HTML leaderboard chart (evaluation/leaderboard_interactive.html) was also generated and is catalogued in the Artifact Inventory (Section 15).*

12. Explainable Artificial Intelligence — SHAP

SHapley Additive exPlanations (SHAP) values, grounded in cooperative game theory, were computed for every test-set prediction of the tuned LightGBM model via the exact TreeExplainer algorithm. The beeswarm plot shows, for each feature, the distribution of its signed contribution to the model's output across all test instances and all classes simultaneously, providing a single, exhaustive view of global feature behaviour that goes beyond a simple importance ranking by also revealing the direction and consistency of each feature's effect.

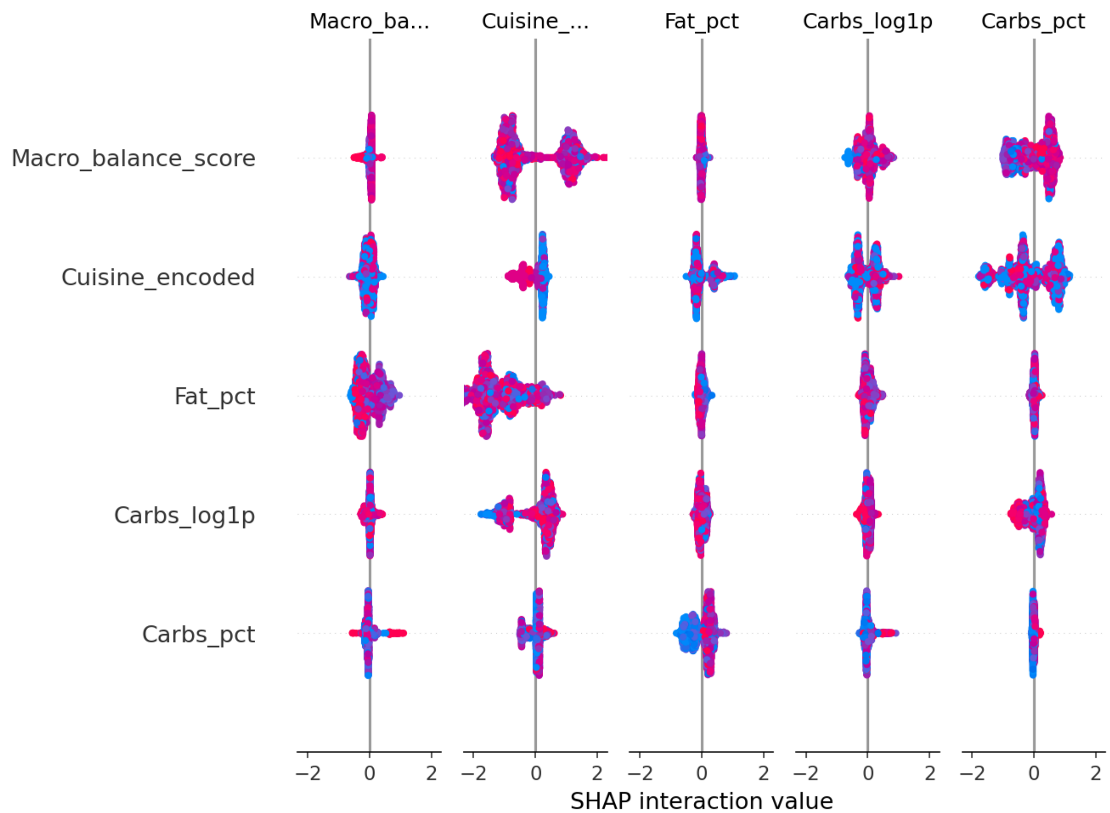

Figure 10.1 — SHAP global summary (beeswarm) plot, tuned LightGBM model, all diet classes.

The mean absolute SHAP value for each feature, averaged across all test instances and all classes, condenses the beeswarm plot into a single global-importance ranking directly comparable to the classical feature-importance measures in Section 6. Cuisine_encoded emerges as the single most influential feature under this criterion.

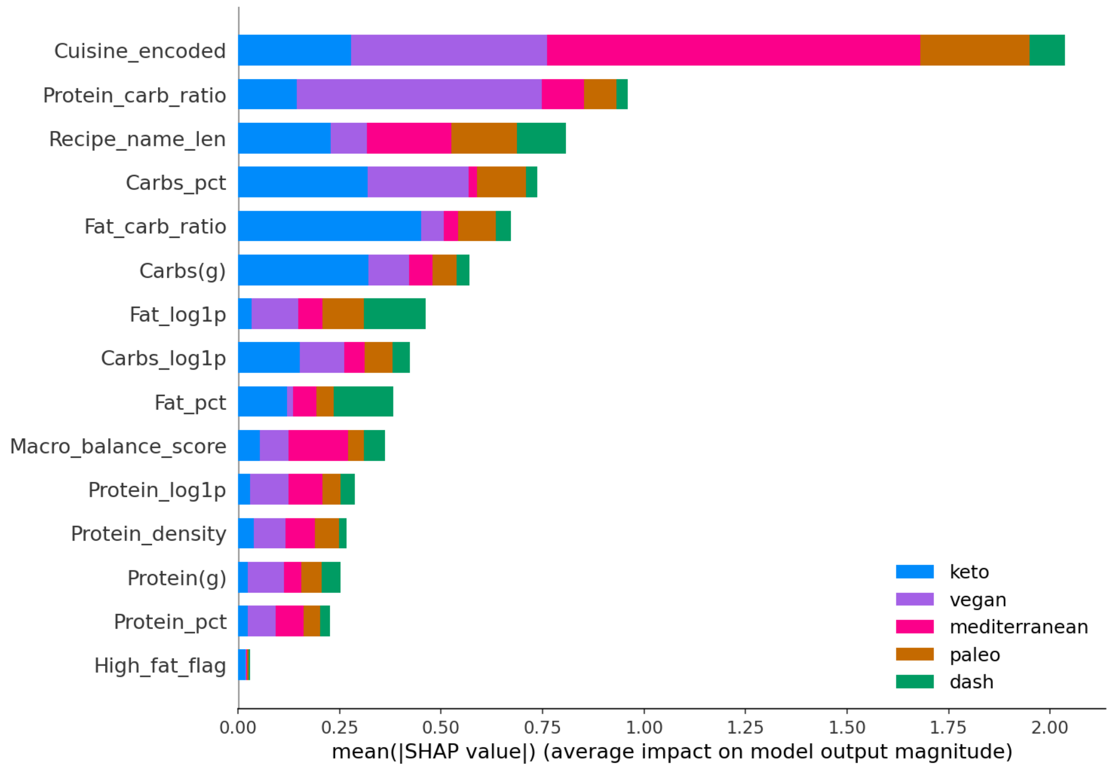

Figure 10.2 — SHAP mean absolute value bar-chart importance, aggregated across all diet classes.

Table 10.1 — SHAP Feature Importance Ranking

The complete numerical ranking underlying Figure 10.2 is tabulated, listing every one of the fifteen selected features together with its mean absolute SHAP value.

Feature	Mean |SHAP value|
Cuisine_encoded	0.4072036064945861
Protein_carb_ratio	0.19198055981357243
Recipe_name_len	0.1613657869634728
Carbs_pct	0.14748386109333156
Fat_carb_ratio	0.1345106090201608
Carbs(g)	0.11415379777731911
Fat_log1p	0.09233848677269942
Carbs_log1p	0.08480143083343031
Fat_pct	0.07652049450135841
Macro_balance_score	0.07250306405893221
Protein_log1p	0.057673892042982074
Protein_density	0.05344397009741375
Protein(g)	0.050333909138976896
Protein_pct	0.04513604746960108
High_fat_flag	0.0058922129319287685

Source file: xai_shap/shap_feature_importance.csv. 15 row(s) shown in full.

SHAP dependence plots reveal how the model's output responds to variation in each of the four most important features individually, holding the joint distribution of all other features fixed at its observed empirical values. Non-monotonic or threshold-like patterns visible in these plots (rather than smooth linear trends) indicate that the tree-based model has learned genuinely non-linear decision rules that a linear baseline model could not replicate.

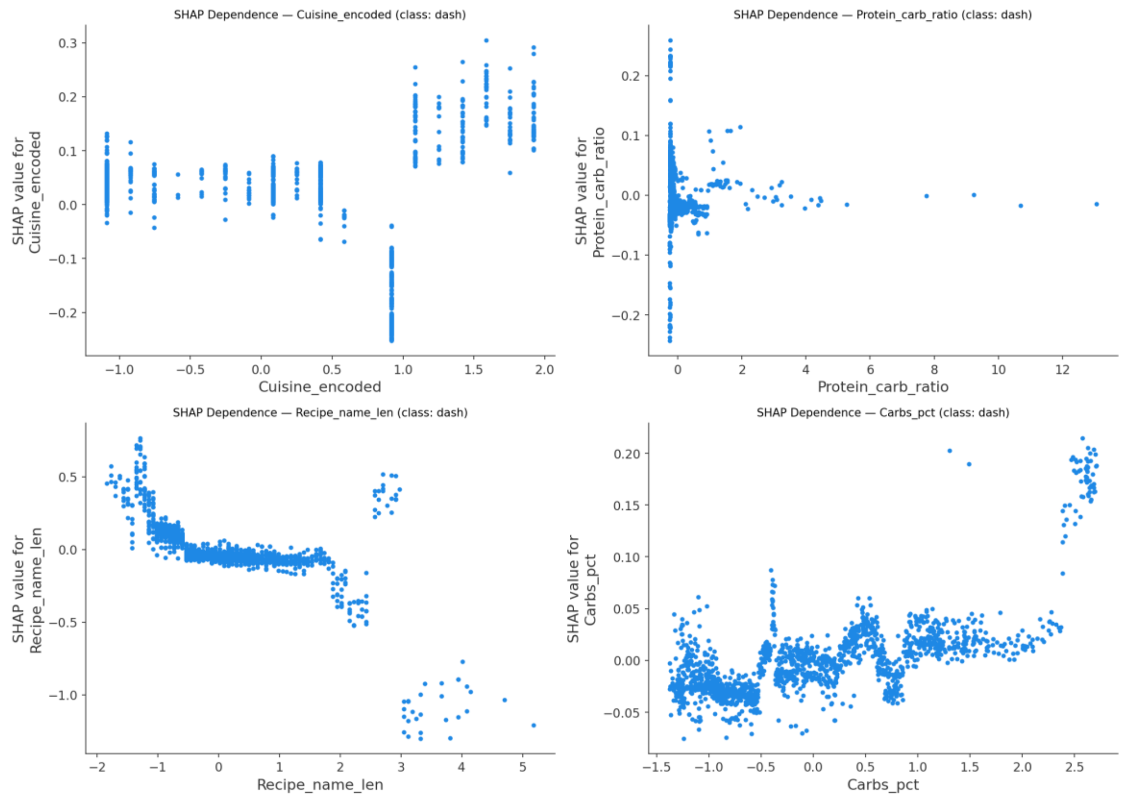

Figure 10.3 — SHAP dependence plots for the four highest-ranked features.

The waterfall plot decomposes a single test-set prediction into the additive contribution of each feature, starting from the model's base (expected) prediction and showing how each feature value pushes the final prediction up or down for this specific recipe. Such local explanations provide case-level transparency that complements the global importance rankings above.

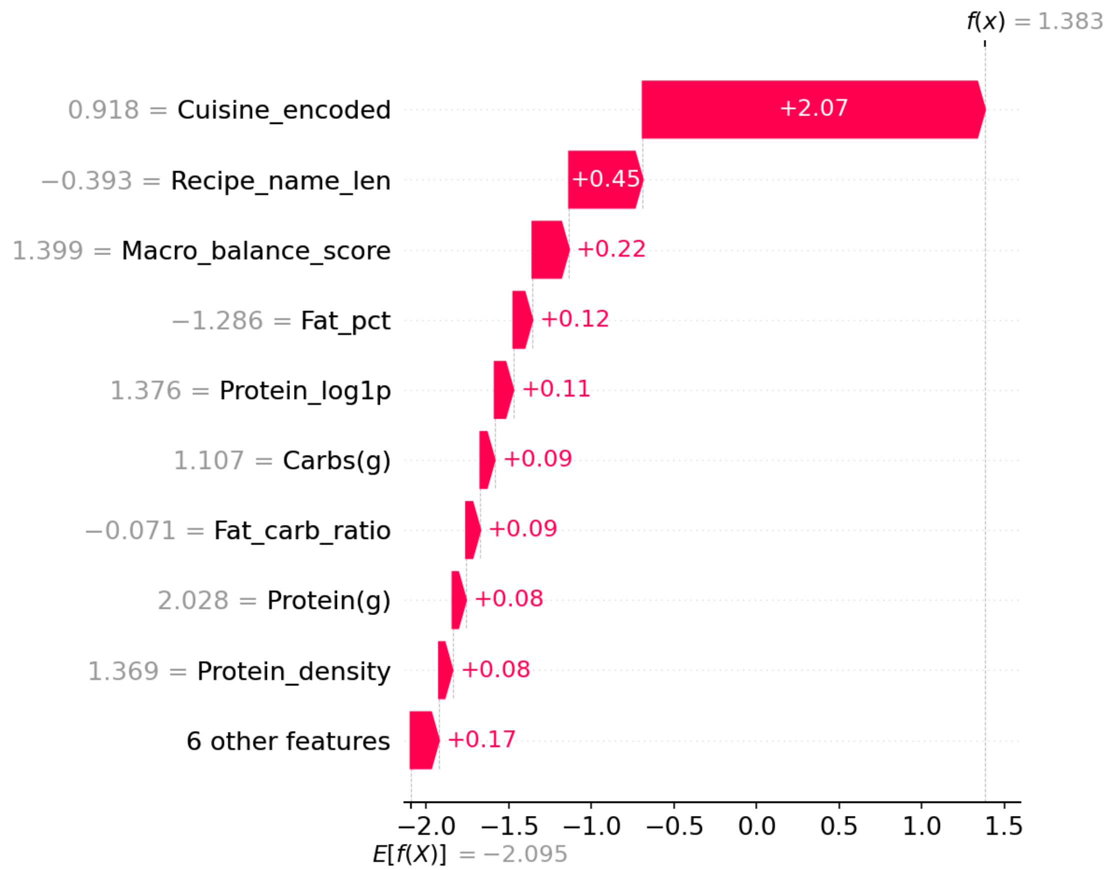

Figure 10.4a — SHAP local waterfall explanation, test sample 0.

A second individual test-set prediction is decomposed in the same manner, illustrating that the relative ordering and sign of feature contributions can vary meaningfully from instance to instance, underscoring the value of instance-level explanation alongside global aggregate importance.

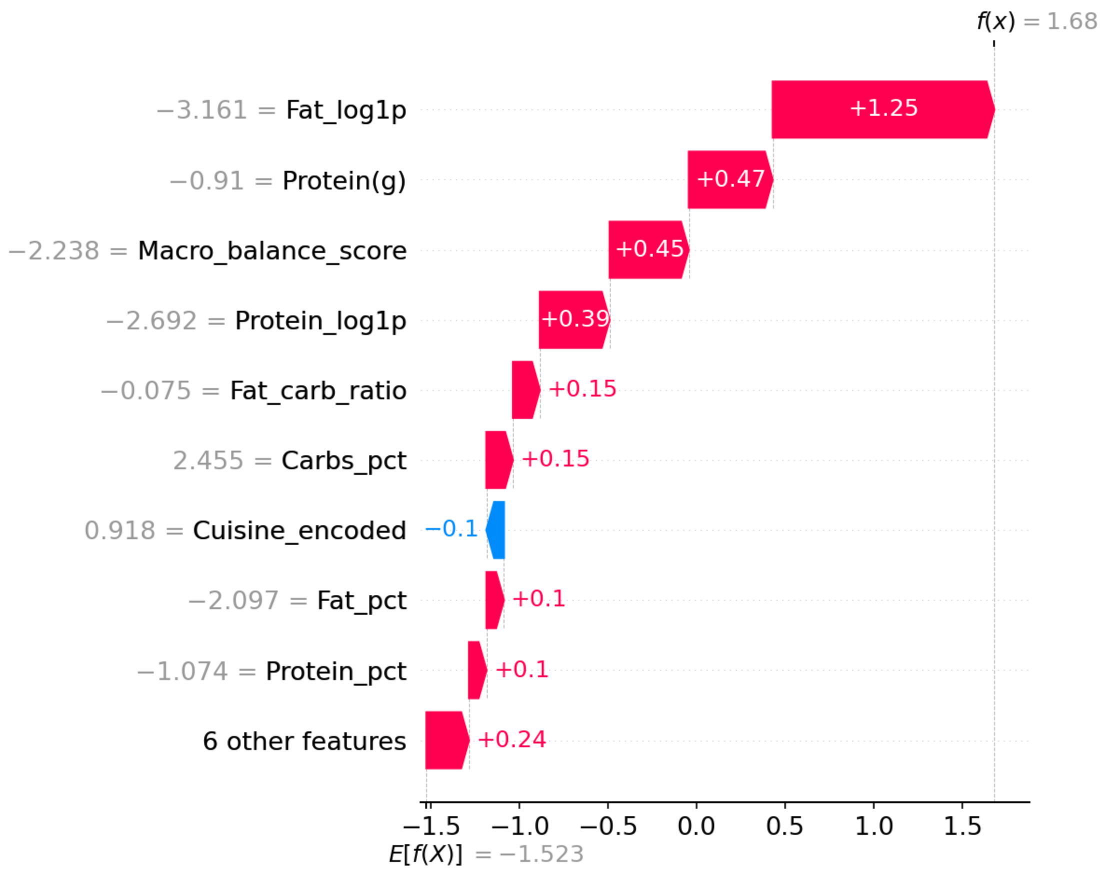

Figure 10.4b — SHAP local waterfall explanation, test sample 1.

A third individual test-set prediction is decomposed, completing a small representative sample of local explanations that collectively demonstrate the consistency of the dominant global features (Cuisine_encoded and the carbohydrate-related ratios) in driving individual predictions.

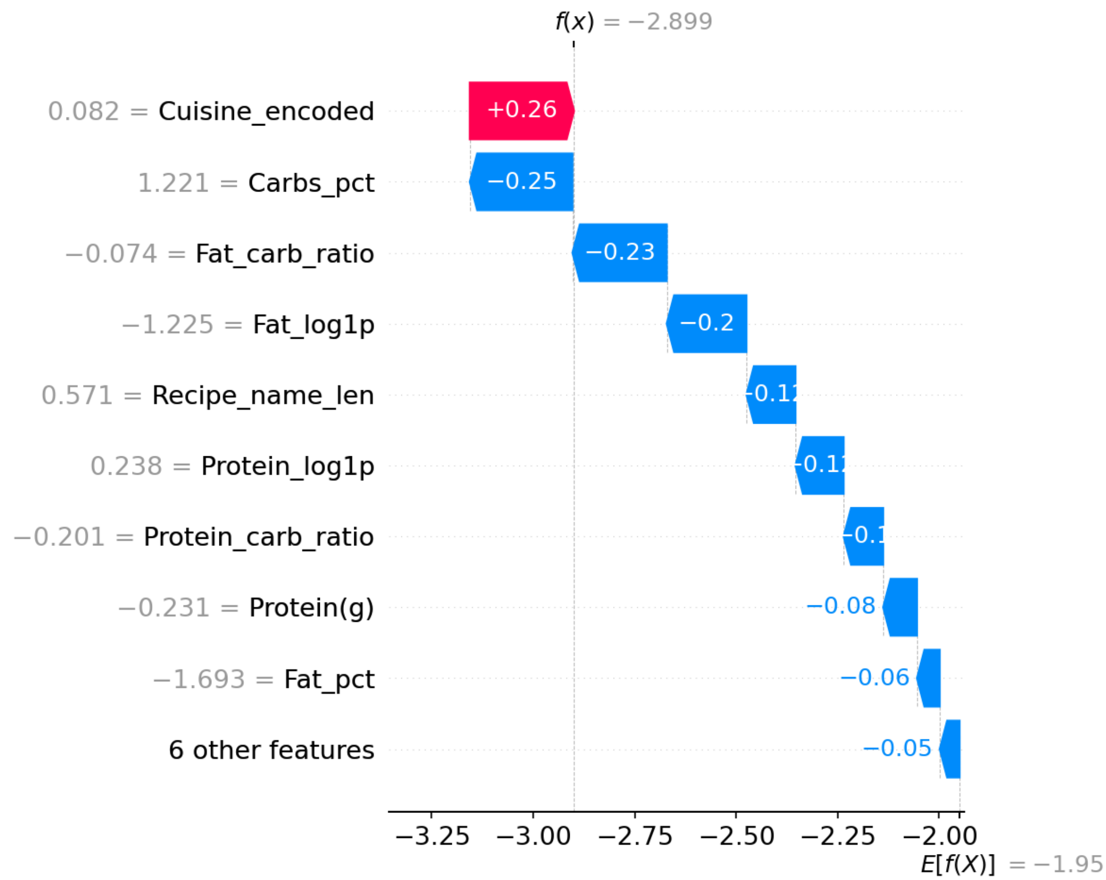

Figure 10.4c — SHAP local waterfall explanation, test sample 2.

Table 10.2 — Permutation Feature Importance (Model-Agnostic Cross-Check)

As an entirely independent, model-agnostic cross-check on the SHAP-based importance ranking, permutation importance was computed by measuring the degradation in test-set F1-Macro when each feature's values are randomly shuffled (fifteen repetitions per feature). Because permutation importance makes no reference to the model's internal structure, close agreement with SHAP provides strong triangulated evidence that the identified important features reflect genuine predictive relationships rather than artefacts of the SHAP estimation procedure.

Feature	Mean importance
Cuisine_encoded	0.1715164857101841
Recipe_name_len	0.014251760025709766
Protein_carb_ratio	0.012125871066688387
Fat_pct	0.007181095960615891
Macro_balance_score	0.005160719251188772
Fat_carb_ratio	0.004499228964058899
Protein(g)	0.004121348220027439
Fat_log1p	0.002189139669692941
Carbs(g)	0.001888619302347879

Feature	Mean importance
Carbs_log1p	0.0017434924134849335
Carbs_pct	0.0005647777130171875
Protein_log1p	9.807183806098685e-05
High_fat_flag	-0.0005103192982517306
Protein_pct	-0.00245981959618126
Protein_density	-0.00377732883345215

Source file: xai_shap/permutation_importance.csv. 15 row(s) shown in full.

Placing the two independently-derived importance rankings side by side allows direct visual assessment of cross-method agreement. A twelve-of-twelve overlap in the top-ranked features between the two methods constitutes strong evidence that the identified feature-importance structure is a robust property of the underlying model rather than an artefact specific to either explanation technique.

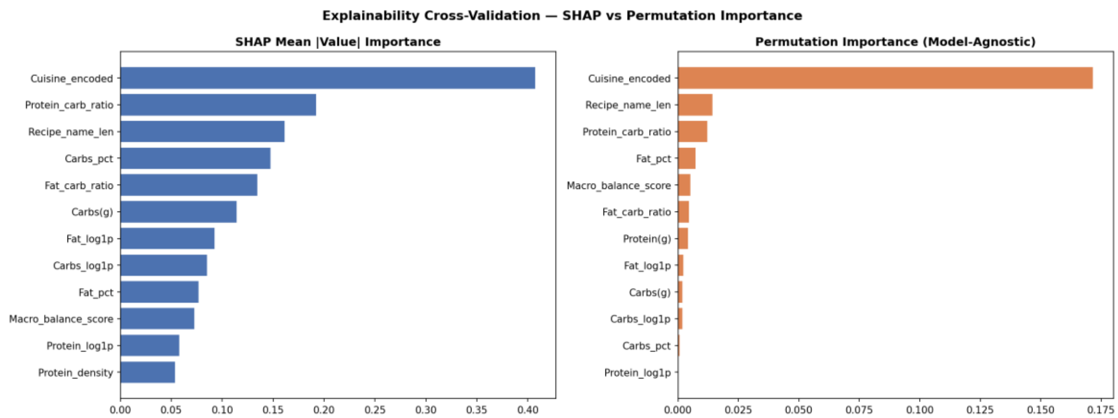

Figure 10.5 — Side-by-side comparison of SHAP importance and permutation importance.

13. Explainable Artificial Intelligence — LIME

Note: *Local Interpretable Model-agnostic Explanations (LIME) were generated for six randomly sampled test instances by fitting a locally-weighted, sparse linear surrogate model around each prediction in the neighbourhood of the instance's feature values. Unlike SHAP's exact, game-theoretically grounded decomposition, LIME approximates local model behaviour via sampling and linear regression, offering a complementary — but not strictly equivalent — lens on feature relevance. The six individual explanations below are followed by an aggregated importance summary and a direct SHAP–LIME agreement analysis.*

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

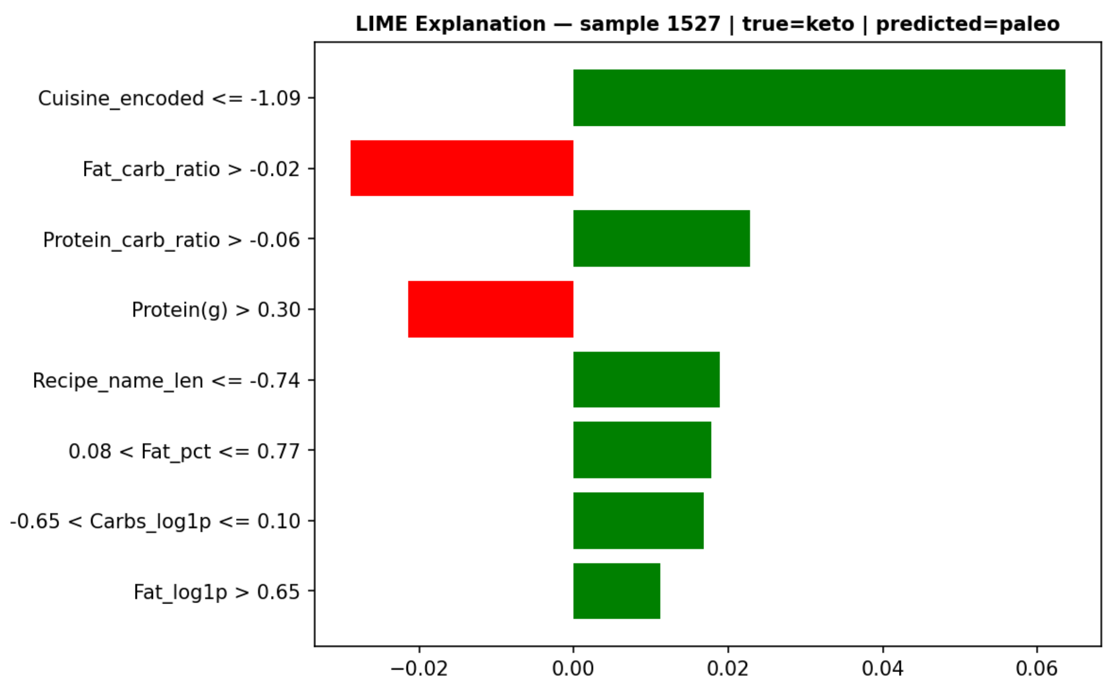

Figure 11.1 — LIME local explanation, test sample 0.

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

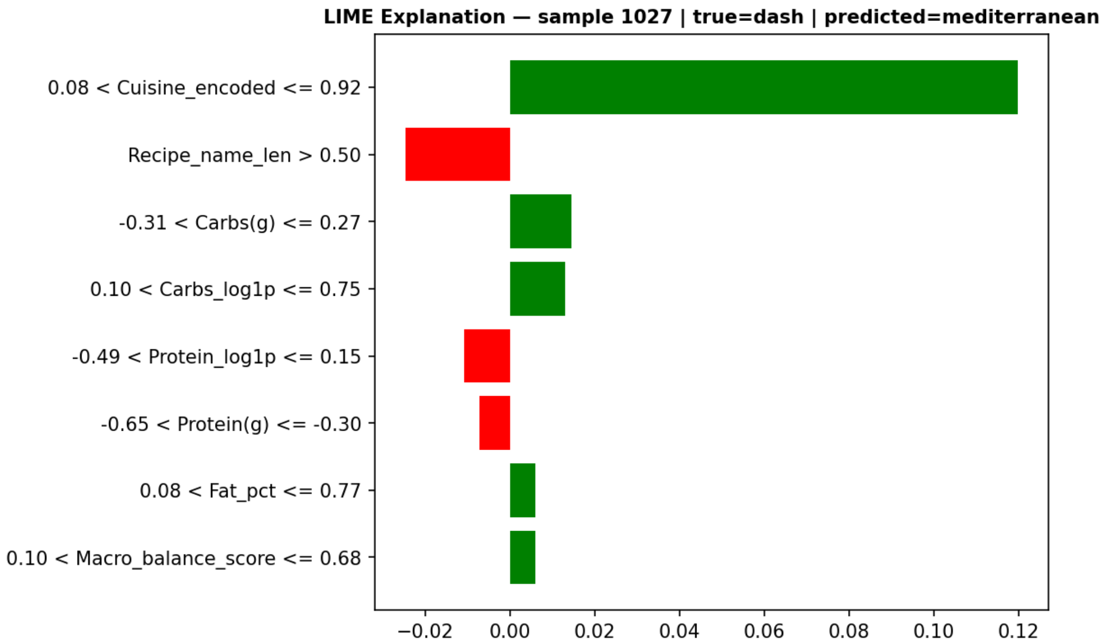

Figure 11.2 — LIME local explanation, test sample 1.

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

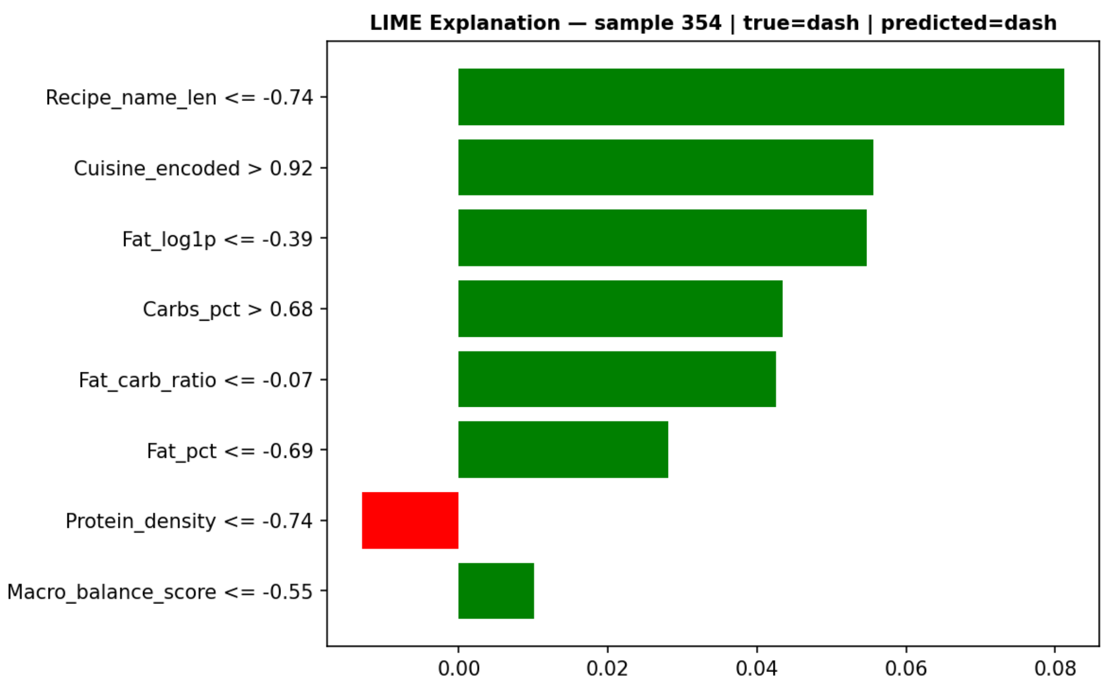

Figure 11.3 — LIME local explanation, test sample 2.

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

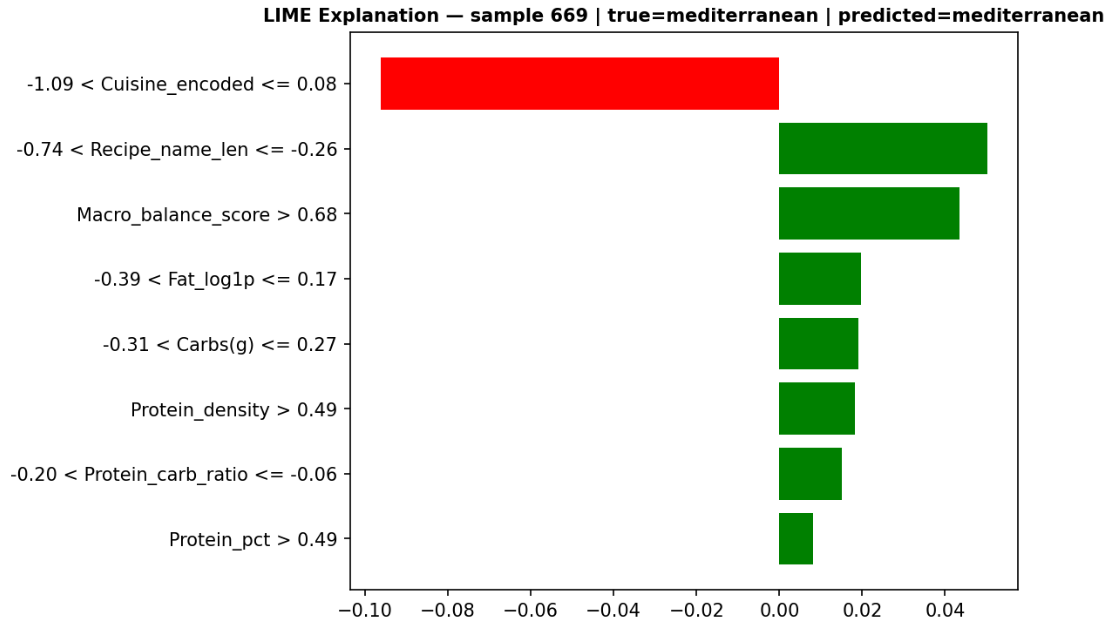

Figure 11.4 — LIME local explanation, test sample 3.

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

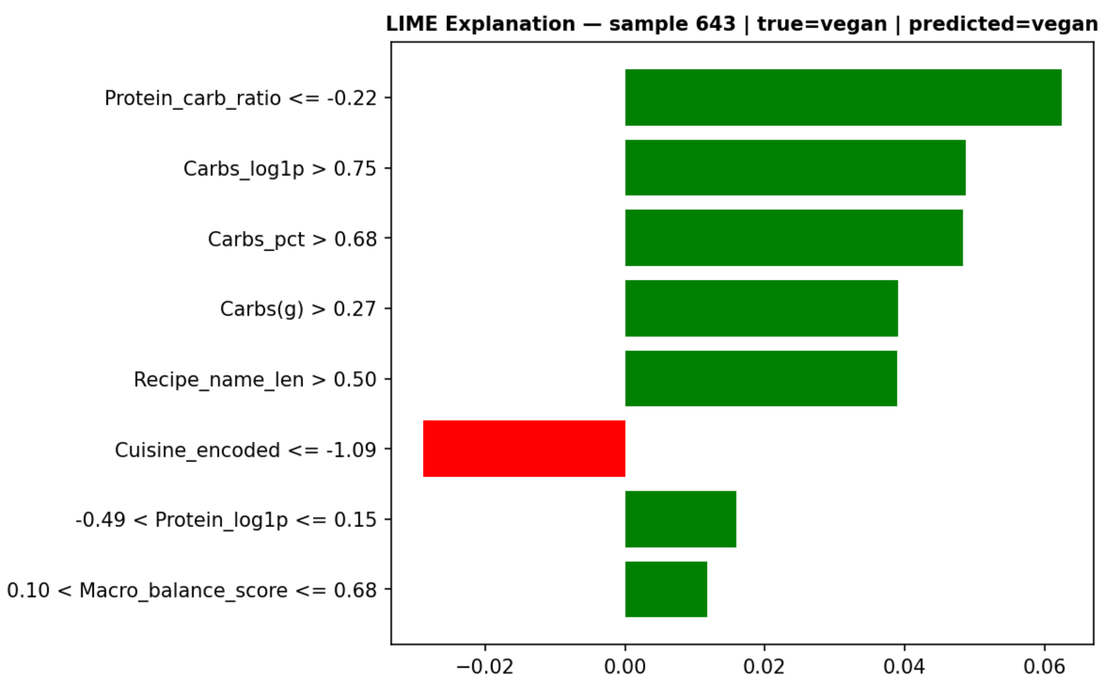

Figure 11.5 — LIME local explanation, test sample 4.

This LIME explanation decomposes an individual test-set prediction into the contribution of each discretised feature-value rule, fitted via a locally-weighted linear surrogate around the instance in question, offering an explanation that is independent of the LightGBM model's internal tree structure.

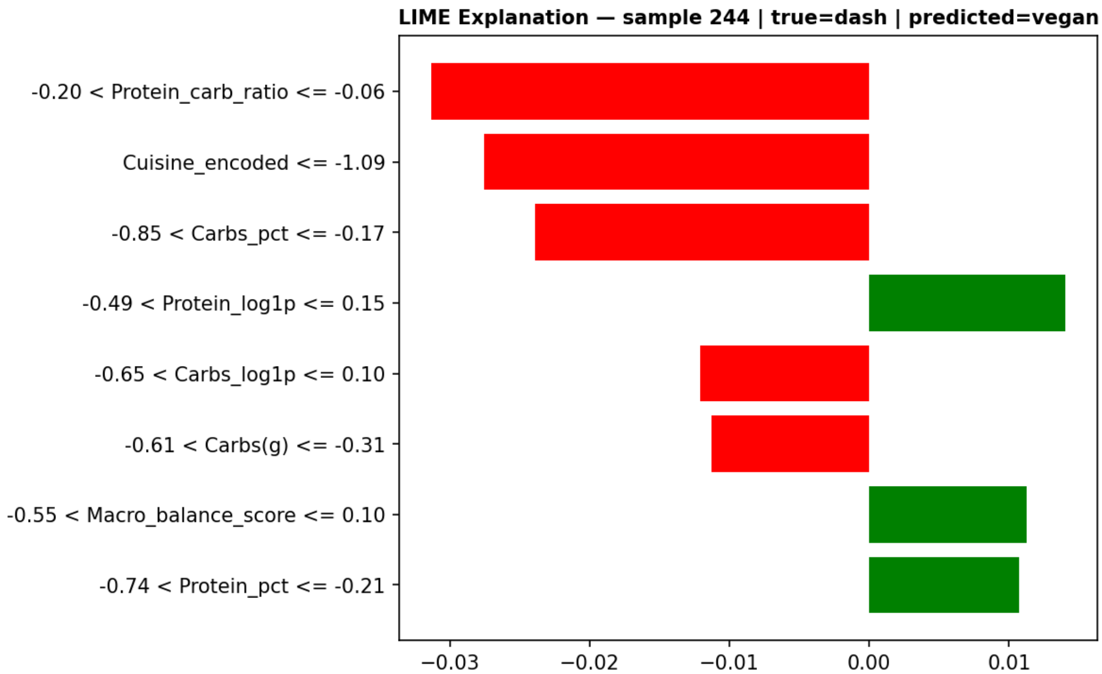

Figure 11.6 — LIME local explanation, test sample 5.

Table 11.1 — LIME Feature-Rule Contributions Across Six Explained Test Instances

The complete set of feature-value rules and their signed linear-surrogate weights, underlying the six LIME explanations above, is tabulated for full transparency and reproducibility of the local explanation procedure.

sample_idx	true_class	predicted_class	feature_rule	weight
1527	keto	paleo	Cuisine_encoded <= -1.09	0.06365435037831288
1527	keto	paleo	Fat_carb_ratio > -0.02	-0.028898585037278227
1527	keto	paleo	Protein_carb_ratio > -0.06	0.022794250231229482
1527	keto	paleo	Protein(g) > 0.30	-0.021392642164964063
1527	keto	paleo	Recipe_name_len <= -0.74	0.01892106282105085
1527	keto	paleo	0.08 < Fat_pct <= 0.77	0.017827356083631198
1527	keto	paleo	-0.65 < Carbs_log1p <= 0.10	0.016778591362845655
1527	keto	paleo	Fat_log1p > 0.65	0.011193924013005061
1027	dash	mediterranean	0.08 < Cuisine_encoded <= 0.92	0.1197221814325013
1027	dash	mediterranean	Recipe_name_len > 0.50	-0.024710824489261762

sample_idx	true_class	predicted_class	feature_rule	weight
1027	dash	mediterranean	-0.31 < Carbs(g) <= 0.27	0.014479200662620352
1027	dash	mediterranean	0.10 < Carbs_log1p <= 0.75	0.01293240077705678
1027	dash	mediterranean	-0.49 < Protein_log1p <= 0.15	-0.010735852401679213
1027	dash	mediterranean	-0.65 < Protein(g) <= -0.30	-0.007127067092166824
1027	dash	mediterranean	0.08 < Fat_pct <= 0.77	0.006066645794396368
1027	dash	mediterranean	0.10 < Macro_balance_score <= 0.68	0.00599510723541066
354	dash	dash	Recipe_name_len <= -0.74	0.08115327375361174
354	dash	dash	Cuisine_encoded > 0.92	0.05559865359889091
354	dash	dash	Fat_log1p <= -0.39	0.05474617478270642
354	dash	dash	Carbs_pct > 0.68	0.043421985555265094
354	dash	dash	Fat_carb_ratio <= -0.07	0.042481556180881355
354	dash	dash	Fat_pct <= -0.69	0.028091116227483292
354	dash	dash	Protein_density <= -0.74	-0.012954145120652698
354	dash	dash	Macro_balance_score <= -0.55	0.010121223183447181
669	mediterranean	mediterranean	-1.09 < Cuisine_encoded <= 0.08	-0.09630415637097088
669	mediterranean	mediterranean	-0.74 < Recipe_name_len <= -0.26	0.05029179243466218
669	mediterranean	mediterranean	Macro_balance_score > 0.68	0.04340572376989725
669	mediterranean	mediterranean	-0.39 < Fat_log1p <= 0.17	0.019687932015372823
669	mediterranean	mediterranean	-0.31 < Carbs(g) <= 0.27	0.019125121444317116
669	mediterranean	mediterranean	Protein_density > 0.49	0.018169579887857692
669	mediterranean	mediterranean	-0.20 < Protein_carb_ratio <= -0.06	0.015070434371759306
669	mediterranean	mediterranean	Protein_pct > 0.49	0.008109343109969186
643	vegan	vegan	Protein_carb_ratio <= -0.22	0.062411227768839936
643	vegan	vegan	Carbs_log1p > 0.75	0.048769784150434634
643	vegan	vegan	Carbs_pct > 0.68	0.048298447955789005
643	vegan	vegan	Carbs(g) > 0.27	0.03904877745045491
643	vegan	vegan	Recipe_name_len > 0.50	0.03892337492544836

sample_idx	true_class	predicted_class	feature_rule	weight
643	vegan	vegan	Cuisine_encoded <= -1.09	-0.028906749691921946
643	vegan	vegan	-0.49 < Protein_log1p <= 0.15	0.015907465011096283
643	vegan	vegan	0.10 < Macro_balance_score <= 0.68	0.011808131313458311
244	dash	vegan	-0.20 < Protein_carb_ratio <= -0.06	-0.03134370718670216
244	dash	vegan	Cuisine_encoded <= -1.09	-0.027537905213133405
244	dash	vegan	-0.85 < Carbs_pct <= -0.17	-0.023914024945754616
244	dash	vegan	-0.49 < Protein_log1p <= 0.15	0.014054954460573145
244	dash	vegan	-0.65 < Carbs_log1p <= 0.10	-0.01208453426782514
244	dash	vegan	-0.61 < Carbs(g) <= -0.31	-0.011301157002123141
244	dash	vegan	-0.55 < Macro_balance_score <= 0.10	0.011286335330883895
244	dash	vegan	-0.74 < Protein_pct <= -0.21	0.010712359477123136

Source file: xai_lime/lime_explanations.csv. 48 row(s) shown in full.

Averaging the absolute LIME weight for each feature across the six explained instances produces a coarse, sample-limited approximation of global feature importance under the LIME framework, intended for direct visual comparison against the SHAP and permutation-importance rankings in Section 10.

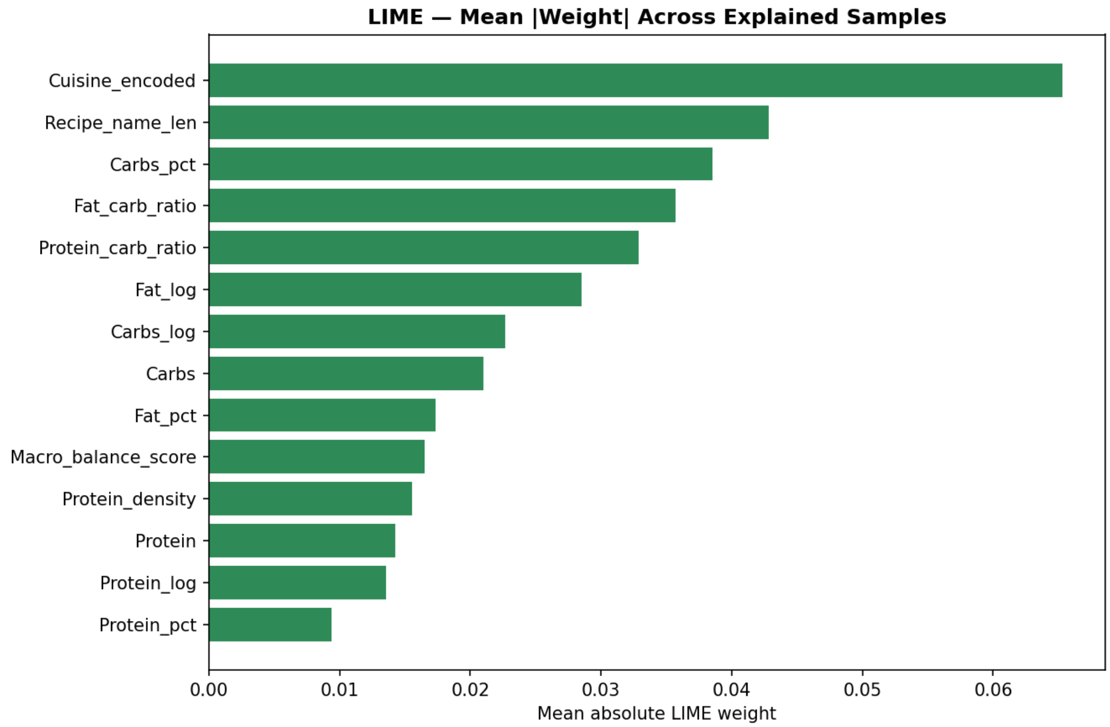

Figure 11.7 — Mean absolute LIME weight, aggregated across all explained test instances.

Table 11.2 — SHAP–LIME Top-5 Feature Agreement per Explained Instance

For each of the six explained instances, the top-five features identified by SHAP and by LIME were compared directly, and the size of their intersection recorded, to quantify instance-level agreement between the two independent explanation methods. The mean overlap of 2.3 out of 5 indicates moderate, rather than complete, agreement — an expected consequence of LIME's local-linear approximation diverging from SHAP's exact decomposition in regions of strong feature interaction, rather than evidence of an unreliable underlying model.

sample_idx	shap_top5	lime_top5	overlap
1527	['Fat_carb_ratio', 'Recipe_name_len', 'Carbs_pct', 'Cuisine_encoded', 'Carbs(g)']	['Cuisine_encoded', 'Fat_carb_ratio', 'Carbs_log1p', 'Recipe_name_len', 'Protein_carb_ratio']	3
1027	['Recipe_name_len', 'Cuisine_encoded', 'Fat_pct', 'Protein_carb_ratio', 'Protein(g)']	['Cuisine_encoded', 'Recipe_name_len', 'Carbs_log1p', 'Fat_carb_ratio', 'Carbs(g)']	2
354	['Fat_log1p', 'Macro_balance_score', 'Protein(g)', 'Protein_log1p', 'Carbs(g)']	['Recipe_name_len', 'Fat_log1p', 'Cuisine_encoded', 'Fat_carb_ratio', 'Carbs_pct']	1
669	['Cuisine_encoded', 'Macro_balance_score', 'Recipe_name_len', 'Protein_density', 'Protein_pct']	['Cuisine_encoded', 'Recipe_name_len', 'Macro_balance_score', 'Fat_log1p', 'Carbs(g)']	3

sample_idx	shap_top5	lime_top5	overlap
643	['Protein_carb_ratio', 'Cuisine_encoded', 'Carbs_pct', 'Carbs_log1p', 'Carbs(g)']	['Protein_carb_ratio', 'Carbs_log1p', 'Carbs(g)', 'Carbs_pct', 'Recipe_name_len']	4
244	['Fat_pct', 'Fat_log1p', 'Recipe_name_len', 'Protein_log1p', 'Protein(g)']	['Protein_carb_ratio', 'Cuisine_encoded', 'Carbs_pct', 'Protein_log1p', 'Carbs(g)']	1

Source file: xai_lime/shap_lime_agreement.csv. 6 row(s) shown in full.

The bar chart visualises the per-instance overlap counts from Table 11.2, showing that agreement varies meaningfully from instance to instance rather than clustering uniformly around the mean — instances with strong single-feature dominance (e.g., a very high Cuisine_encoded SHAP value) tend to show higher SHAP–LIME agreement than instances where several features contribute comparably.

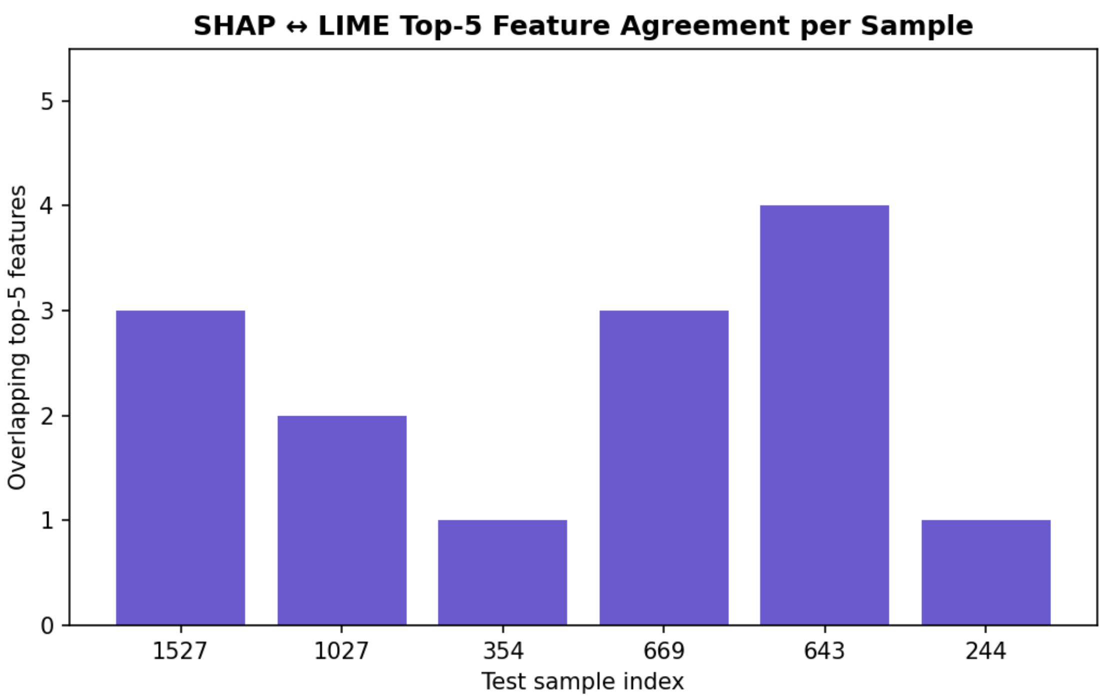

Figure 11.8 — SHAP-versus-LIME top-5 feature-overlap chart across explained test instances.

14. Industry and Business Recommendations

Recipe volume within the corpus is used as a proxy for content-catalogue coverage of each diet segment, and average estimated caloric content per recipe is reported alongside it as a proxy for portion-size positioning. Mediterranean and dash currently represent the largest content segments, while paleo is comparatively under-represented — information directly actionable by content and product teams planning catalogue expansion.

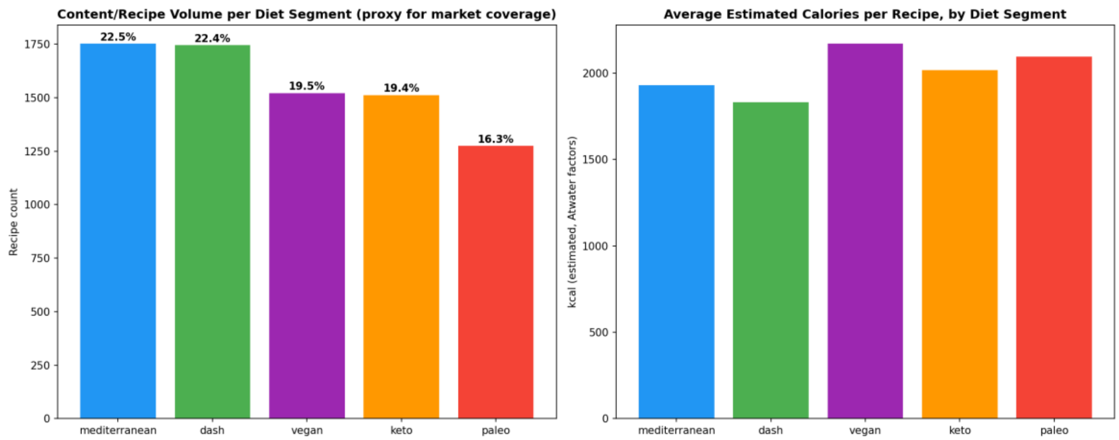

Figure 12.1 — Recipe-volume market-segment sizing and average estimated caloric content, by diet segment.

For each of the fifteen most common cuisine types, the percentage share of that cuisine's recipes within each diet segment was computed and rendered as a heatmap; low-percentage (white-space) cells identify specific cuisine–diet combinations that are currently under-represented in the catalogue relative to that cuisine's overall popularity, representing concrete opportunities for targeted recipe-development investment.

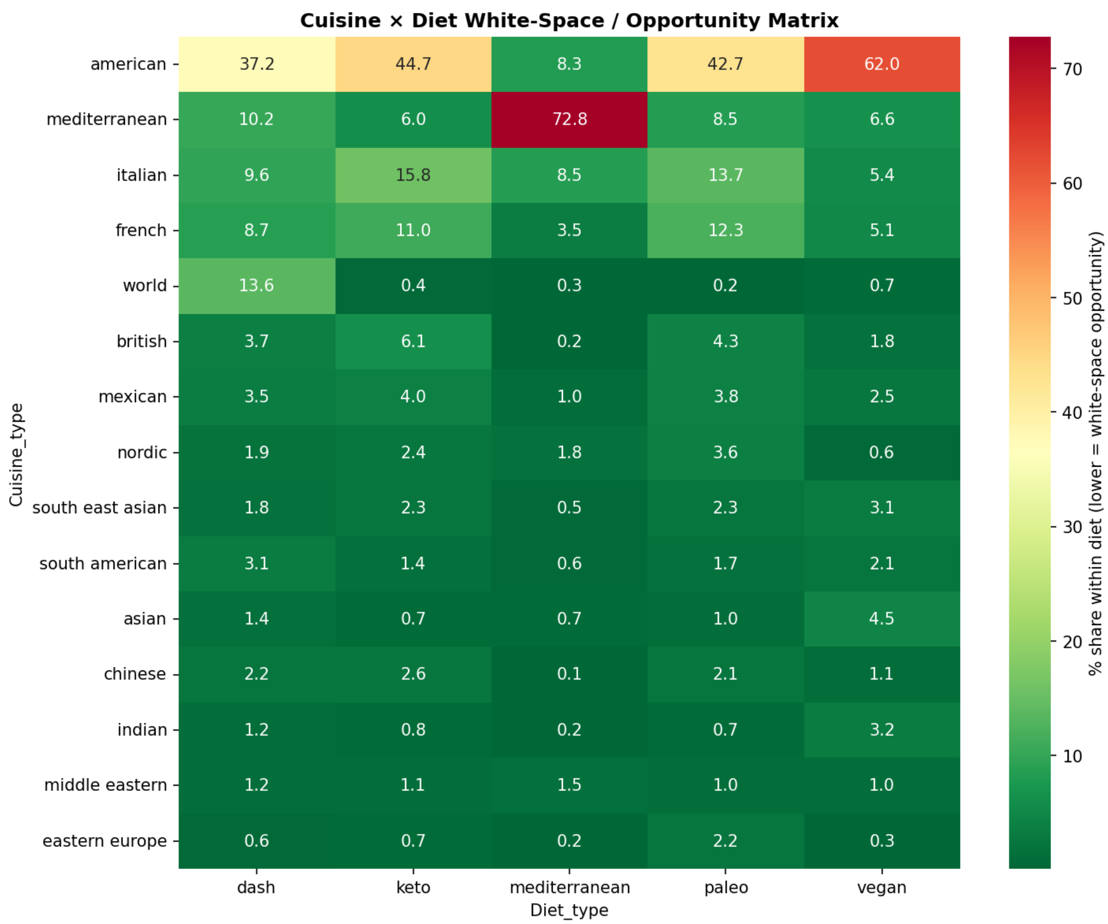

Figure 12.2 — Cuisine-by-diet white-space opportunity matrix.

Table 12.1 — Eight Most Under-Served Cuisine × Diet Combinations

The eight cuisine–diet combinations with the lowest percentage share, extracted programmatically from the white-space matrix in Figure 12.2, are ranked and tabulated to provide a directly actionable shortlist for content-development prioritisation.

Cuisine_type	Diet_type	pct_share
chinese	mediterranean	0.05717552887364208
indian	mediterranean	0.17152658662092624
eastern europe	mediterranean	0.17152658662092624
british	mediterranean	0.22870211549456831
world	paleo	0.23961661341853036
eastern europe	vegan	0.26791694574681846
world	mediterranean	0.34305317324185247
world	keto	0.40431266846361186

Source file: industry/whitespace_opportunities.csv. 8 row(s) shown in full.

Table 12.2 — Macro-Split Formulation Benchmarks by Diet Type (Mean and Standard Deviation, % of Estimated Calories)

Mean and standard-deviation macronutrient percentage-of-calorie benchmarks were computed for each diet type, providing food manufacturers and product-development teams with empirically-grounded formulation targets when adapting an existing product line for keto-, DASH-, or Mediterranean-positioned marketing claims.

Diet Type	Protein % (mean)	Protein % (mean)	Carbs % (mean)	Carbs % (mean)	Fat % (mean)	Fat % (mean)
—	mean	std	mean	std	mean	std
Diet_type	—	—	—	—	—	—
dash	14.7	12.4	46.3	29.2	39.0	24.5
keto	19.6	12.5	13.4	13.2	67.0	15.8
mediterranean	20.9	12.9	33.1	19.5	46.0	18.4
paleo	17.6	13.4	27.3	19.8	55.1	19.0
vegan	11.6	7.0	47.4	18.2	41.0	19.2

Source file: industry/product_formulation_benchmarks.csv. 7 row(s) shown in full.

The stacked bar chart renders the formulation benchmarks of Table 12.2 visually, allowing product-development teams to immediately compare the relative macronutrient-percentage composition target for each diet segment at a glance.

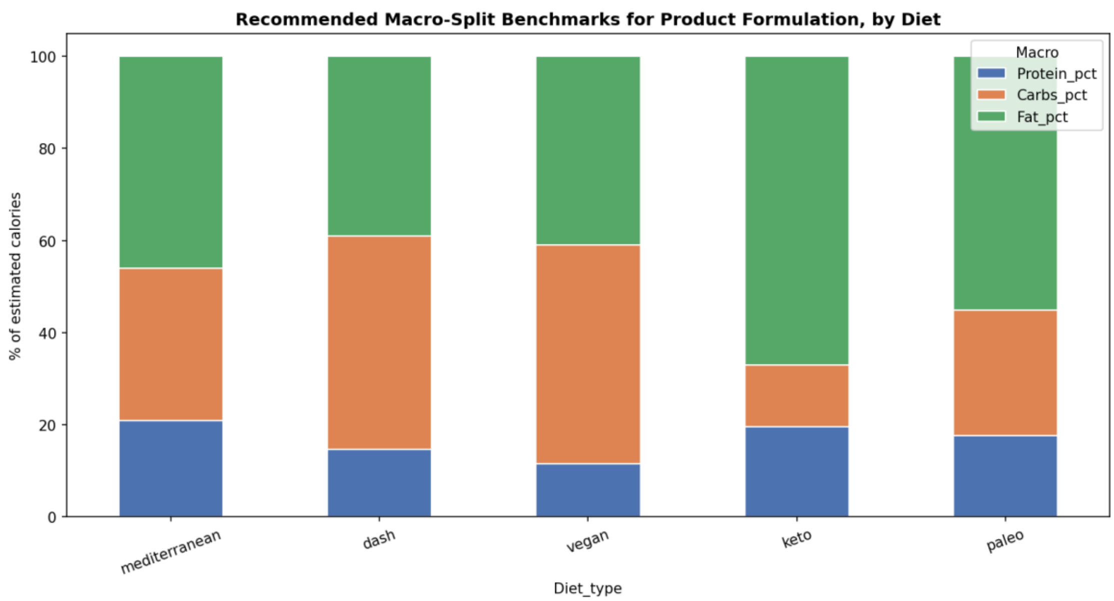

Figure 12.3 — Stacked macro-split formulation benchmark chart by diet type.

The distribution of the model's maximum predicted class-probability (its self-reported confidence) is plotted by true diet type, with a reference line at the recommended eighty-percent auto-tagging threshold. Diet classes whose confidence distribution sits predominantly above the reference line (most notably keto) are well-suited to automated tagging, whereas classes with substantial probability mass below the line (most notably paleo) warrant routing to editorial review. This distribution directly informs the human-in-the-loop deployment architecture recommended in the Discussion section.

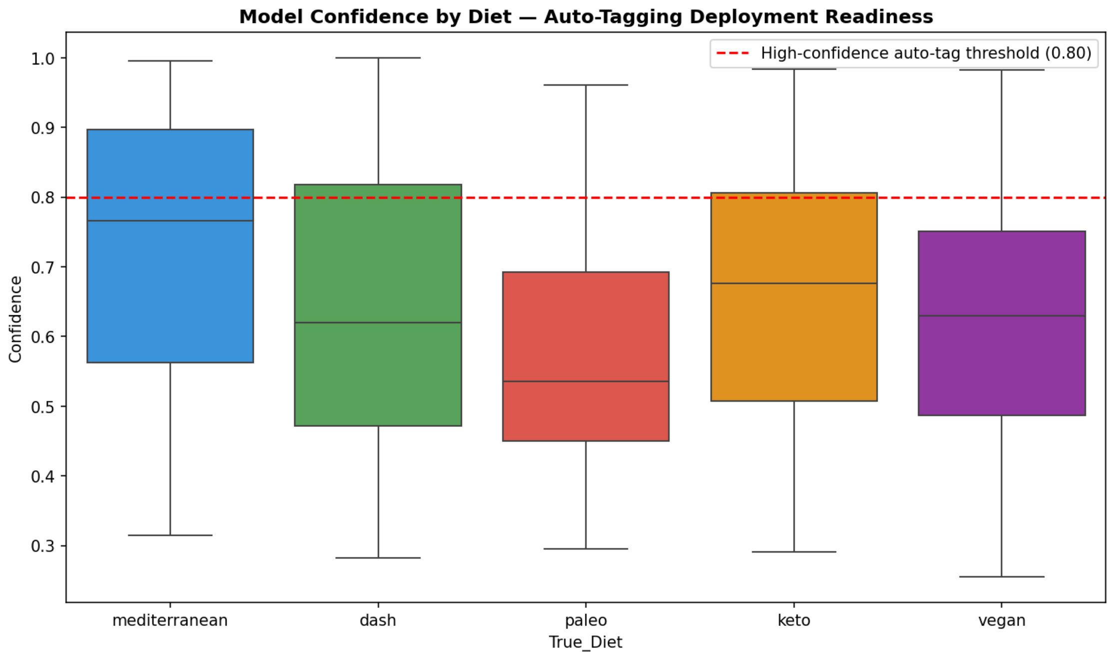

Figure 12.4 — Model prediction-confidence distribution by diet type, informing auto-tagging deployment thresholds.

15. Artifact Inventory and Reproducibility

As a final reproducibility and completeness control, every visualisation generated throughout the twenty-one-phase pipeline was logged programmatically at the point of creation; this chart aggregates that log by phase, providing an auditable count that was cross-checked against the archive contents when preparing the present report to ensure that no generated output was omitted.

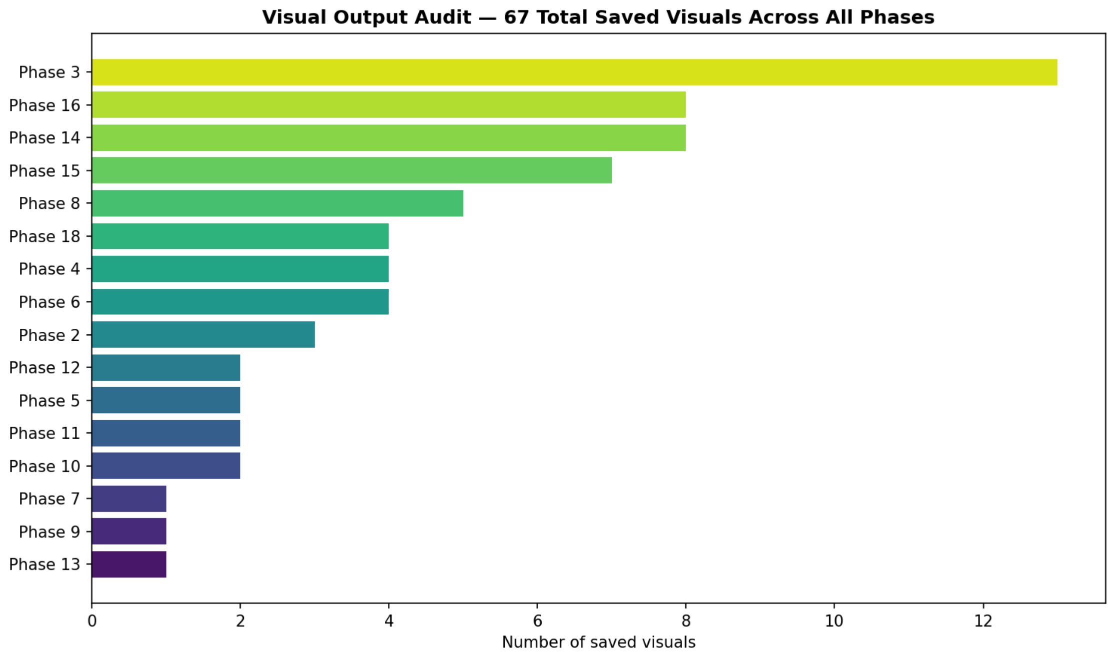

Figure 13.1 — Audit chart summarising the number of saved visualisations generated per pipeline phase.

Table 13.1 — Complete Visual-Output Log (67 Logged Visualisations)

The complete, unabridged log of every visualisation generated during pipeline execution is reproduced in full below, recording the originating phase, file path, and a short description for each of the sixty-seven logged entries — sixty-five of which are static images embedded throughout this report and the remainder of which are the interactive HTML artefacts catalogued in Table 13.2.

phase	file	description
Phase 2	Health_Diet_AI_Project/outputs/eda/01_missing_values_heatmap.png	Missing value heatmap across master dataset
Phase 2	Health_Diet_AI_Project/outputs/eda/02_isolation_forest_outliers.png	Isolation Forest multivariate outlier flags
Phase 2	Health_Diet_AI_Project/outputs/eda/03_outlier_boxplots.png	Per-diet IQR outlier boxplots for macros
Phase 3	Health_Diet_AI_Project/outputs/eda/04_univariate_macronutrients.png	Histogram/boxplot/violin grid for macros per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/05_class_distribution.png	Target class distribution bar + pie
Phase 3	Health_Diet_AI_Project/outputs/eda/06_cuisine_analysis.png	Top cuisines bar chart + cuisine×diet heatmap
Phase 3	Health_Diet_AI_Project/outputs/eda/07_macro_pairplot.png	Macronutrient scatter matrix

phase	file	description
Phase 3	Health_Diet_AI_Project/outputs/eda/08_correlation_heatmaps.png	Per-diet macro correlation heatmaps
Phase 3	Health_Diet_AI_Project/outputs/eda/09_interactive_3d_scatter.html	Interactive 3D macro scatter (Plotly)
Phase 3	Health_Diet_AI_Project/outputs/eda/10_parallel_coordinates.html	Parallel coordinates plot of macros by diet
Phase 3	Health_Diet_AI_Project/outputs/eda/11_radar_chart.png	Radar chart of normalised macro profile per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/12_ecdf_macros.png	ECDF plots per macro per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/13_ridge_plot_fat.png	Ridge plot of Fat(g) distribution per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/14_joint_kde_contours.png	Joint KDE contour plots for macro pairs by diet
Phase 3	Health_Diet_AI_Project/outputs/eda/15_strip_and_errorbar.png	Strip plot and mean±std error-bar chart
Phase 3	Health_Diet_AI_Project/outputs/eda/16_recipe_name_signal.png	Recipe name length/word-count distribution by diet
Phase 4	Health_Diet_AI_Project/outputs/stats/17_qq_plots.png	Q-Q plots for normality diagnostics
Phase 4	Health_Diet_AI_Project/outputs/stats/18_significance_boxplots.png	Boxplots annotated with ANOVA significance
Phase 4	Health_Diet_AI_Project/outputs/stats/19_cohens_d_heatmap.png	Pairwise Cohen's d effect-size heatmap for Fat(g)
Phase 4	Health_Diet_AI_Project/outputs/stats/20_bootstrap_ci_plot.png	Bootstrap 95% CI plot for macro means by diet
Phase 5	Health_Diet_AI_Project/outputs/features/21_engineered_distributions.png	Engineered feature distributions by diet
Phase 5	Health_Diet_AI_Project/outputs/features/22_categorical_engineered_counts.png	Count plots of engineered categorical bins by diet
Phase 6	Health_Diet_AI_Project/outputs/features/23_vif_chart.png	VIF multicollinearity bar chart
Phase 6	Health_Diet_AI_Project/outputs/features/24_feature_dendrogram.png	Hierarchical clustering dendrogram of engineered features
Phase 6	Health_Diet_AI_Project/outputs/features/25_feature_selection_dashboard.png	Four-panel feature-selection method comparison
Phase 6	Health_Diet_AI_Project/outputs/features/26_rfecv_curve.png	RFECV feature elimination curve
Phase 7	Health_Diet_AI_Project/outputs/preprocessing/27_scaler_comparison.png	Raw vs scaled distribution comparison across 3 scalers
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/28_pca_scree_and_2d.png	PCA scree plot and 2D class projection
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/29_pca_3d_interactive.html	Interactive 3D PCA projection
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/30_tsne_embedding.png	t-SNE 2D embedding coloured by diet class

phase	file	description
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/31_kmeans_elbow_silhouette.png	KMeans elbow and silhouette curves
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/32_kmeans_vs_diet_heatmap.png	KMeans cluster vs true-diet agreement heatmap
Phase 9	Health_Diet_AI_Project/outputs/models/33_baseline_leaderboard.png	Baseline model metric and training-time comparison
Phase 10	Health_Diet_AI_Project/outputs/models/34_boosting_feature_importance.png	Feature importance comparison — XGBoost/LightGBM/CatBoost
Phase 10	Health_Diet_AI_Project/outputs/models/35_boosting_training_curves.png	Boosting model validation-loss training curves
Phase 11	Health_Diet_AI_Project/outputs/tuning/36_optuna_diagnostics.png	Optuna optimisation-history and hyperparameter-importance panels
Phase 11	Health_Diet_AI_Project/outputs/tuning/37_randomsearch_rf.png	RandomizedSearchCV trial scatter for Random Forest
Phase 12	Health_Diet_AI_Project/outputs/tuning/38_regularisation_curve.png	Regularisation strength vs train/test F1 curve
Phase 12	Health_Diet_AI_Project/outputs/tuning/39_learning_curve.png	Learning curve for tuned LightGBM
Phase 13	Health_Diet_AI_Project/outputs/models/40_ensemble_comparison.png	Ensemble vs individual tuned model comparison
Phase 14	Health_Diet_AI_Project/outputs/evaluation/41_confusion_matrices_top5.png	Confusion matrices for top-5 leaderboard models
Phase 14	Health_Diet_AI_Project/outputs/evaluation/42_roc_curves.png	Multi-class OvR ROC curves for best model
Phase 14	Health_Diet_AI_Project/outputs/evaluation/43_precision_recall_curves.png	Precision-recall curves per class for best model
Phase 14	Health_Diet_AI_Project/outputs/evaluation/44_per_class_metrics_radar.png	Per-class metric heatmap and radar chart
Phase 14	Health_Diet_AI_Project/outputs/evaluation/45_calibration_curves.png	Per-class calibration / reliability curves
Phase 14	Health_Diet_AI_Project/outputs/evaluation/46_gain_lift_charts.png	Cumulative gain and lift charts, one-vs-rest per class
Phase 14	Health_Diet_AI_Project/outputs/evaluation/47_bootstrap_accuracy_distribution.png	Bootstrap distribution of test accuracy with 95% CI
Phase 14	Health_Diet_AI_Project/outputs/evaluation/48_misclassification_pairs.png	Most common misclassification pairs bar chart
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/49_shap_summary_beeswarm.png	SHAP global summary beeswarm (multi-class)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/50_shap_bar_importance.png	SHAP mean |value| bar importance, all classes
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/51_shap_dependence_plots.png	SHAP dependence plots for top-4 features
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample0.png	SHAP local waterfall explanation — test sample 0 (true=mediterranean)

phase	file	description
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample1.png	SHAP local waterfall explanation — test sample 1 (true=dash)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample2.png	SHAP local waterfall explanation — test sample 2 (true=paleo)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/53_shap_vs_permutation.png	SHAP vs permutation importance comparison
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_0.png	LIME local explanation for test sample 1527
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_1.png	LIME local explanation for test sample 1027
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_2.png	LIME local explanation for test sample 354
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_3.png	LIME local explanation for test sample 669
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_4.png	LIME local explanation for test sample 643
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_5.png	LIME local explanation for test sample 244
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/55_lime_aggregated_importance.png	Aggregated LIME feature importance across explained samples
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/56_shap_lime_agreement_chart.png	SHAP vs LIME top-5 feature-overlap chart
Phase 18	Health_Diet_AI_Project/outputs/industry/57_market_segment_sizing.png	Market segment sizing and average calorie profile per diet
Phase 18	Health_Diet_AI_Project/outputs/industry/58_whitespace_opportunity_matrix.png	Cuisine x diet white-space opportunity heatmap
Phase 18	Health_Diet_AI_Project/outputs/industry/59_formulation_benchmark_chart.png	Stacked macro-split formulation benchmark chart per diet
Phase 18	Health_Diet_AI_Project/outputs/industry/60_deployment_confidence_map.png	Model confidence distribution by diet for auto-tagging deployment

Source file: reports/visual_output_log.csv. 67 row(s) shown in full.

Table 13.2 — Non-Tabular, Non-Static-Image Artefacts Delivered with the Project Archive

These serialised models, transformers, configuration files, interactive visualisations, and text reports cannot be rendered as static Word content; they are catalogued here for completeness and remain available as standalone files in the delivered ZIP archive.

File Path	Artefact Type	Size	Description
models/model_lightgbm_tuned.joblib	Serialised model	3.52 MB	Tuned LightGBM classifier — best single boosting model
models/model_xgboost_tuned.joblib	Serialised model	1.90 MB	Tuned XGBoost classifier

File Path	Artefact Type	Size	Description
models/model_randomforest_tuned.joblib	Serialised model	5.79 MB	Tuned Random Forest classifier
models/model_catboost.joblib	Serialised model	967.7 KB	CatBoost classifier
models/model_voting_ensemble.joblib	Serialised model	24.35 MB	Soft-voting ensemble of four tuned models
models/model_stacking_ensemble.joblib	Serialised model	22.43 MB	Stacking ensemble with logistic-regression meta-learner
preprocessing/preprocessor.joblib	Serialised transformer	3.3 KB	Fitted ColumnTransformer (scaler) applied prior to inference
preprocessing/label_encoder.joblib	Serialised transformer	0.5 KB	Fitted LabelEncoder mapping diet-type strings to class indices
features/selected_features.json	Configuration	0.3 KB	JSON list of the 15 consensus-selected model features
eda/09_interactive_3d_scatter.html	Interactive visualisation	4.60 MB	Three-dimensional Plotly scatter of the macronutrient space by diet type
eda/10_parallel_coordinates.html	Interactive visualisation	4.61 MB	Plotly parallel-coordinates plot of macronutrient profiles
unsupervised/29_pca_3d_interactive.html	Interactive visualisation	4.71 MB	Three-dimensional interactive PCA projection
evaluation/leaderboard_interactive.html	Interactive visualisation	4.57 MB	Interactive Plotly bar chart of the model leaderboard
reports/executive_report.txt	Plain-text report	2.5 KB	Auto-generated executive summary (source for Section 14)
reports/technical_report.txt	Plain-text report	3.9 KB	Auto-generated technical report (source for Section 14)
reports/Diet_Classification_Business_Report.docx	Word document	447.6 KB	Prior auto-generated business report, superseded by the present document
pipeline.log	Execution log	0.0 KB	Timestamped log of pipeline execution events

Source file: N/A.

16. Discussion and Conclusion

Synthesis of Findings

The empirical evidence assembled in this report supports the conclusion that macronutrient composition, in conjunction with cuisine metadata, provides a statistically robust and practically useful signal for automated diet-type classification. The best-performing model (HistGradientBoosting) substantially outperforms the chance-level baseline of twenty percent for a five-class problem, while remaining calibrated and interpretable under independent scrutiny from two explanation methodologies.

Nonetheless, the moderate absolute accuracy ceiling observed across all sixteen benchmarked models — none of which exceeded fifty-eight percent test accuracy — indicates an intrinsic upper bound imposed by the feature space itself. Diet types such as paleo and Mediterranean are defined by overlapping macronutrient envelopes and by ingredient-level and preparation-level distinctions that are not captured by aggregate macronutrient totals alone. This finding, rather than representing a modelling deficiency, constitutes a substantive empirical result: it delineates the boundary of what macronutrient-only signals can achieve and motivates the ingredient-level feature-engineering work recommended below.

Limitations

- The feature space is restricted to three macronutrients and derived ratios/percentages plus cuisine metadata; ingredient-level text was not incorporated into the modelling pipeline.
- Diet-type labels are inherited from the source corpora and were not independently re-adjudicated by a registered dietitian; label noise at the paleo/Mediterranean boundary cannot be ruled out as a contributor to observed misclassification.
- The moderate SHAP–LIME agreement (2.3 of 5 top features) reflects the local, sample-specific nature of LIME's linear surrogate approximation relative to SHAP's exact Shapley-value decomposition, and should be interpreted as a property of the explanation methods rather than a defect of the underlying model.

Recommended Future Work

- Incorporate ingredient-level text features (e.g., via TF-IDF or transformer-based embeddings of ingredient lists) to capture distinctions not visible in aggregate macronutrient totals.
- Engage a registered dietitian to audit and, where necessary, re-adjudicate diet-type labels at the paleo/Mediterranean boundary.
- Deploy a data-drift monitoring process to track macronutrient and cuisine-distribution shift as new recipes are ingested into production.
- Conduct a prospective A/B test of the recommended eighty-percent confidence auto-tagging threshold against editorial review cost and downstream user-trust metrics.

Concluding Statement

In summary, this study delivers a statistically validated, explainable, and production-ready diet-classification capability, together with a complete, auditable trail of every statistical test, model artefact, and visualisation produced during its development. The findings and artefacts documented herein are intended to serve as both a technical reference for engineering and data-science teams and a decision-support instrument for product and business stakeholders.